

Answers to Student's Book

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1 Data representation

Answers to activities

1.1

- a 51
- b 127
- c 153
- d 116
- e 255
- f 15
- g 143
- h 179
- i 112
- j 238
- k 487
- l 1364
- m 3855
- n 1992
- o 2047
- p 31984
- q 16141
- r 49983
- s 34952
- t 32767

1.2

- a** 00101001
- b** 01000011
- c** 01010110
- d** 01100100
- e** 01101111
- f** 01111111
- g** 10010000
- h** 10111101
- i** 11001000
- j** 11111111
- k** 1000000011101000
- l** 0000001101111000
- m** 0000111111111111
- n** 0100000000010000
- o** 1111001101100011

1.3

- a** C3
- b** F7
- c** 27F
- d** 4EE
- e** 1E1
- f** 89E
- g** (0)4FE
- h** (0)E9C
- i** FF7D
- j** 67AE

1.4

- a** 0110 1100
- b** 0101 1001
- c** 1010 1010
- d** 1010 0000 0000
- e** 0100 0000 1110
- f** 1011 1010 0110
- g** 1001 1100 1100
- h** 0100 0000 1010 1010
- i** 1101 1010 0100 0111
- j** 0001 1010 1011 0000

1.5

- a** 107
- b** 156
- c** 74
- d** 255
- e** 511
- f** 2561
- g** 2996
- h** 3240
- i** 4782
- j** 44425

1.6

- a** 62
- b** E3
- c** 1EA
- d** 1FF
- e** 33A
- f** 3E8
- g** A4A
- h** E9F
- i** FA7
- j** 1388

1.7

- 1** Student investigation

- 2 a** 35, 37, 8B
- b** C9, 7A, CC
- c** C, 6F, 51

1.8

- a** 10000011
- b** 01100110
- c** 01111011
- d** 10110110
- e** 01101111
- f** 10110111
- g** 01111110
- h** 01110000
- i** 11111110
- j** 11011101

1.9

- a** $98 = 01100010$ and $15 = 00001111$; sum = 01110001 (= 113)
- b** $29 = 00011101$ and $88 = 01011000$; sum = 01110101 (= 117)
- c** $49 = 00110001$ and $100 = 01100100$; sum = 10010101 (= 149)
- d** $51 = 00110011$ and $171 = 10101011$; sum = 11011110 (= 222)
- e** $82 = 01010010$ and $69 = 01000101$; sum = 10010111 (= 151)
- f** $100 = 01100100$ and $140 = 10001100$; sum = 11110000 (= 240)
- g** $19 = 00010011$ and $139 = 10001011$; sum = 10011110 (= 158)
- h** $203 = 11001011$ and $30 = 00011110$; sum = 11101001 (= 233)
- i** $66 = 01000010$ and $166 = 10100110$; sum = 11101000 (= 232)
- j** $211 = 11010011$ and $35 = 00100011$; sum = 11110110 (= 246)

1.10

- 1 a** $89 = 01011001$ and $175 = 10101111$; sum = $1]00001000$ (= 8)
overflow error has occurred (9 bits)
- b** $168 = 10101000$ and $99 = 01100011$; sum = $1]00001011$ (= 11)
overflow error has occurred (9 bits)
- c** $88 = 01011000$ and $215 = 11010111$; sum = $1]00101111$ (= 47)
overflow error has occurred (9 bits)
- 2 a** 1101 1111 0010 1010 – result is correct
- b** $1]1110\ 1011\ 1110\ 0100$ – overflow error has occurred

1.11

- 1 a** denary value = $64 + 32 + 8 = 104$

$$0\ 0\ 0\ 0\ 1\ 1\ 0\ 1$$
- b** denary value is 13 ($104 \div 8$) therefore shift three places right is the same as division by 2^3 (i.e. 8)
- c** denary value = 15

$$1\ 1\ 1\ 1\ 0\ 0\ 0\ 0$$
- d** denary value is 240 (15×16) therefore shift four places left is the same as multiplication by 2^4 (i.e. 16)
- 2 a** $29 = 00011101$ and $51 = 00110011$
- b** sum = 01010000
- c** 00001010
- d** $75 = 01001011$
- e** 01010101
- f** 10101010

1.12

- 1**
- a** 00100111
 - b** 01000010
 - c** 01011000
 - d** 01100110
 - e** 01101111
 - f** 01111101
 - g** 01001101
 - h** 00010100
 - i** 00110001
 - j** 00111000

- 2**
- a** 85
 - b** 51
 - c** 76
 - d** 126
 - e** 15
 - f** 125
 - g** 65
 - h** 30
 - i** 113
 - j** 120

1.13

- a** 11101110
- b** 11100001
- c** 11010001
- d** 11000001
- e** 10101000
- f** 10100100
- g** 10011100
- h** 11111111
- i** 11110000
- j** 10000001

1.14

- a** -51
- b** -66
- c** -17
- d** -121
- e** -96
- f** -7
- g** -81
- h** -1
- i** -127
- j** -10

1.15

- 1 i** The number of possible colours of a group of pixels (usually three or four) based on the bit depth.
- ii** The coding system for all the characters on a standard keyboard and control codes; extended ASCII, gives another 128 codes to allow for characters in non-English alphabets and for some graphical characters to be included.
- iii** The coding system which represents all the languages of the world (the first 128 characters are the same as ASCII code).
- iv** The number of sound samples taken per second.
- v** Images which are made up of pixels.
- 2 i, ii, iii** each colour is $2^8 = 256$
- iv** $256 \times 256 \times 256 = 16\,777\,216$ possible colours
- 3** They both increase the size of the sound file.

1.16

- 1** $1920 \times 1536 = 2\,949\,120$ pixels
 colour depth = 16 $\Rightarrow 16 \times 2\,949\,120 = 47\,185\,920$ bits
 bits ($\div 8$) $\Rightarrow 5\,898\,240$ bytes
 ($\div (1024 \times 1024)$) $\Rightarrow 5.625$ MiB
- 2** $1024 \times 1536 = 1\,572\,864$ pixels
 colour depth = 24 $\Rightarrow 24 \times 1\,572\,864 = 37\,748\,736$ bits
 bits ($\div 8$) $\Rightarrow 4\,718\,592$ bytes
 16 GiB = 17 179 869 bytes
 $\Rightarrow$ number of photos = $(17\,179\,869)/(4\,718\,592) = 3640$ photos
- 3 a** sample size = $44\,100 \times 8 \times 2$ (two tracks) = 705 600 bits
- b** file size = $44\,100 \times 8 \times 30 \times 2$ (two tracks) = 21 168 000 bits
 bytes ($\div 8$) $\Rightarrow 2\,646\,000$ bytes = 2.5 MiB

- 4** sample size = $44\,100 \times 16 \times 210 \times 2$ (two songs) = 296 352 000 bits
 bytes ($\div 8$) $\Rightarrow$ 37 044 000 bytes $\Rightarrow$ 35.33 MiB
 $\Rightarrow$ 740 MiB CD can store $740/35.33 = 20.94 \Rightarrow$ 20 full songs

Answers to exam-style questions

Answers to questions 6 and 7 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

- 1 a i** The number of bits used to represent sound amplitude.
ii Better sound quality
 Less sound distortion
- b i** The number of pixels that make up an image; for example, an image could contain 4096×3192 pixels (i.e. 12 738 656 pixels in total).
ii 4 bits 16 colours
iii $16\,384 \times 512 = 8\,388\,608$ pixels
 Colour depth = 256 (8 bits per pixel)
 $\Rightarrow 8\,388\,608 \times 8 = 67\,108\,864$ bits
 $\Rightarrow (67\,108\,864 / (1024 \times 1024 \times 1024)) = 0.0625$ GiB
- iv** The developer may wish to use lossy compression and use the JPEG format to save space on their camera and/or computer.
- 2 a** The editor decides how many bits to represent the sound amplitude. Increasing the number of bits to represent sound amplitude increases accuracy; this is sampling resolution.
 The editor also decides how many samples per second should be taken; this is the sampling rate – the higher the number of sound samples per second, the better the quality of the sound recording.
- b** The editor will use lossy format, for example, MP3 format, which uses perceptual music shaping, which removes sounds the ear cannot hear properly, therefore the sound quality is essentially retained.
 Lossless files would be larger than lossy files for little improvement in sound quality.
- c i** RLE is a form of lossless compression. It reduces the size of a string of adjacent, identical data, for example, repeated colours of an image. The repeating string is encoded into two values; one value represents the number of identical items and the second value represents the code of the data item (ASCII code).
ii 3G 2W 4G 9W 4G 2W 1G 2W 2G 2W 1G (W = 1 and G = 0)
 $\Rightarrow 30\,21\,40\,91\,40\,21\,10\,21\,20\,21\,10$
 8×4 grid = 32 bytes
 RLE code = $11 \times 2 = 22$ bytes
 $\Rightarrow$ RLE saves 10 bytes of memory/storage

- 3 a** $32 + 16 + 4 + 2 = 54$ (base 10)
- b i** 0 0 0 1 1 0 1 1
- ii** $16 + 8 + 2 + 1 = 27$ (base 10)
- c i** 1 1 0 1 1 0 0 0
- ii** $128 + 64 + 16 + 8 = 216$ (base 10)
- Therefore two places shift to the left = multiply by 4 (2^2)
- 4 a** Parts **i** to **iii**
- | | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
- b i** $123 + 55 = 0\ 1\ 1\ 1\ 1\ 0\ 1\ 1 + 0\ 0\ 1\ 1\ 0\ 1\ 1\ 1 = 1\ 0\ 1\ 1\ 0\ 0\ 1\ 0$ (178)
- ii** $123 + 180 = 0\ 1\ 1\ 1\ 1\ 0\ 1\ 1 + 1\ 0\ 1\ 1\ 0\ 1\ 0\ 0 = 1\ 0\ 0\ 1\ 0\ 1\ 1\ 1\ 1$ (47), which is incorrect; overflow has occurred
- c i** 0 1 1 1 0 1 0 0
- two's complement gives: 1 0 0 0 1 1 0 0 (-116)
- ii** $-112 = 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0$
- iii** $1\ 0\ 1\ 1\ 1\ 0\ 0\ 1 = -128 + (32 + 16 + 8 + 1) = -128 + 57 = -71$
- d i** 0 1 1 0 1 0 0 0
- ii** 1 0 0 1 1 0 0 0
- 5 a** A picture element is the smallest element used to make up an image; each pixel is represented by a minimum of 8 bits and 3 pixels are needed to make up the RGB colours.
- b** The number of possible colours of a group of pixels (usually three or four) based on the bit depth.
- c** $1140 \times 1080 = 1\ 231\ 200$ pixels
- $\Rightarrow 1\ 231\ 200 \times 24 = 29\ 548\ 800$ bits
- $\Rightarrow (29\ 548\ 800)/8 = 3\ 693\ 600$ bytes
- Memory card is $32 \times (1024 \times 1024 \times 1024)$ bytes
- $\Rightarrow$ number of photos $= (32 \times 1024 \times 1024 \times 1024)/(3\ 693\ 600) = 9302$ photos
- d** Use file compression (e.g. JPEG) which reduces the size of the files/images being stored. Therefore more files/images can be stored on a memory stick.

- 6 a**
- image converted from analogue to digital
 - image turned into pixels
 - each pixel is given a binary value
 - pixels form a grid/pattern
 - each pixel has associated colour
 - pixels are stored in sequence
 - meta data stored (dimensions, colour depth, etc.)
 - for example, JPEG, raw, TIFF, PSD, PNG

b Lossy:

- lossy would reduce file size more than lossless
- unnecessary data removed from the file
- images can be of similar quality/won't notice any difference
- no requirement for files to be the same as the original
- can send photos faster/faster to upload

7 a

Hours	0	0	0	0	0	0	1	0
Minutes	0	0	0	1	1	1	1	1
Seconds	0	0	1	1	1	0	1	0

- b**
- 0 0 0 0 0 1 0 1 = 5
- 0 0 0 1 1 0 1 0 = 26
- 0 0 1 1 0 1 1 1 = 55

Time shown as:

05 : 26 : 55

- 8 a** $64 \text{ GiB} = 64 \times 1024 \times 1024 \text{ KiB}$
 $\Rightarrow (64 \times 1024 \times 1024)/10 = 6\,710\,886 \text{ photos}$

b i Use JPEG lossy file compression

- ii**
- saves storage space
 - faster to send photos by email
 - quicker to download/upload photos

iii loss of quality

c i Each primary colour intensity needs to be stored as a byte to allow for sufficient different colour variations.

ii $2^8 \times 2^8 \times 2^8 = 256 \times 256 \times 256 = 16\,777\,216$

9

An 8-bit register uses two's complement notation. What is the denary value of: 0 0 1 0 1 1 0 1 ?	10
Convert the following into GiB: 59 055 800 320 bytes	16
If $2^x = 1\,048\,576$ bytes What is the value of x?	20
Give the denary equivalent of the following hexadecimal number: 3F	28
What is the denary result of the following binary addition: 0 0 0 1 0 0 1 1 + 0 0 0 1 1 0 1 1	45
Find the hexadecimal value of the following denary number: 40	46
	55
	57
	60
	63
	80

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2 Data transmission

Answers to activities

2.1

1 Parts i and ii

- picture and sound out of synchronisation
 - packets arriving at different times
- pauses in the video
 - time delay in assembling packets
- degraded video and sound
 - caused by competing traffic on the communications link
- possible 'drop out'
 - packets take different routes, so some may get lost

2 • web page divided into data packets

- each packet has an IP address of the destination
- routers check header against IP address values to determine next router to send data packet
- hop value checked to see if it is zero
- destination computer reassembles data packets to build up the web page

3 a This takes all the 1-values in the payload and adds them together to generate a hex number.

- this hex value is stored in the trailer
- receiving computer carries out the same calculation on received payload
- if values match, data packets has been received error-free
- if values do not match, data packet needs to be re-sent

b 29 1-bits $\Rightarrow$ 1D in hex. If recalculated value doesn't equal 1D_{hex} then it needs to be re-sent.

- ##### 4 a
- packets keep 'bouncing around' from router to router and never reach the destination
 - use hop numbers; if hop number = 0 then the packet is deleted and is therefore lost

- b
 - apply a hop number ...
 - ... each time a data packet reaches a router, the hop number is decreased by 1
 - ... once the hop number = 0, and the destination hasn't been reached, then the data packet is deleted
- c
 - each packet takes a different route ...
 - ... therefore if a packet is intercepted, hacker still won't have the whole message

2.2

- 1
 - i data sent one bit at a time over a single channel in both directions, but **not** at the same time
 - ii several bits of data sent down multiple channels in both directions, at the same time
 - iii data sent one bit at a time over a single channel in one direction only
- 2
 - i serial, simplex data transmission
 - ii parallel, simplex data transmission
 - iii parallel, full-duplex data transmission
 - iv serial, full-duplex data transmission
 - v parallel, simplex data transmission

2.3

Statement	True	False
Packets have a header which contains the IP address of the sender and the receiver	✓	
Packets don't require any form of error checking		✓
USBs use a protocol that allows for error-free data transmission between device and computer	✓	
Serial data transmission suffers from data skewing		✓
The longest cable length supported by USB is 5 metres or less	✓	
Simplex data transmission occurs when data is transmitted one bit at a time		✓
Full-duplex data transmission involves sending 8 bits of data at a time		✓
USB uses serial data transfer	✓	
Packet switching prevents loss of any data packets		✓
USB connections can transfer data using half-duplex or full-duplex	✓	

2.4

Parity bits: 1, 0, 1, 1, 0, 0, 1, 0, 1, 0

2.5

- 1 no error, error, error, no error, error, error, no error, no error, no error, no error

- 2 It isn't possible to determine where the error occurred.

2.6

- 1 error occurred in: byte 4 (four 1-bits)
and in: bit 6 (four 1-bits)
byte 4 should be: 0 1 1 0 1 1 1 0
- 2 error occurred in: byte 8 (three 1-bits)
and in: bit 7 (three 1-bits)
byte 8 should be: 0 1 0 1 0 0 1 1

2.7

- 1 $9 + 8 + 5 + 0 + 5 + 5 = 32$
 $3 \times (7 + 1 + 1 + 4 + 7 + 9) = 3 \times 29 = 87$
 $(32 + 87)/10 = 119/10 = 11$ remainder 9
 $10 - 9 = 1$, the check digit
- 2 i ISBN 13: $2 + 3 + 1 + 0 + 0 + 2 = 8$
 $3 \times (1 + 1 + 1 + 0 + 4 + 8) = 3 \times 15 = 45$
 $(8 + 45)/10 = 53/10 = 5$ remainder 3
 $10 - 3 = 7$, the check digit
Modulo-11: $\text{sum} = (13 \times 2) + (12 \times 1) + (11 \times 3) + (10 \times 1) + (9 \times 1) + (8 \times 1) + (7 \times 0)$
 $+ (6 \times 0) + (5 \times 0) + (4 \times 4) + (3 \times 2) + (2 \times 8)$
 $\Rightarrow 26 + 12 + 33 + 10 + 9 + 8 + 0 + 0 + 0 + 0 + 16 + 6 + 16 = 136$
 $136/11 = 12$ remainder 4
 $11 - 4 = 7$, the check digit
- ii ISBN 13: $9 + 9 + 1 + 1 + 3 + 4 = 27$
 $3 \times (0 + 8 + 2 + 2 + 5 + 4) = 3 \times 21 = 63$
 $(27 + 63)/10 = 90/10 = 9$ remainder 0
check digit = 0
Modulo-11: $\text{sum} = (13 \times 9) + (12 \times 0) + (11 \times 9) + (10 \times 8) + (9 \times 1) + (8 \times 2) + (7 \times 1)$
 $+ (6 \times 2) + (5 \times 3) + (4 \times 5) + (3 \times 4) + (2 \times 4)$
 $\Rightarrow 117 + 0 + 99 + 80 + 9 + 16 + 7 + 12 + 15 + 20 + 12 + 8 = 395$
 $395/11 = 35$ remainder 10
 $11 - 10 = 1$, the check digit

2.8

- 1 Tom would need to send Jane his public key.
- 2 Two keys are required; one of the keys is only known to the recipient of the document (i.e. the private key).

2.9

- a E – text following an encryption algorithm
- b B – the receiver
- c B – plaintext
- d E – there is a security problem when transmitting the encryption key
- e A – symmetric key encryption
- f E – check digit
- g A – automatic repeat request
- h B – parity check
- i D – checksum
- j A – echo check

Answers to exam-style questions

Answers to questions 7 and 8 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

- 1 a i Data is returned to sender. Sender's computer compares the two sets of data. If any errors found, then the data is re-sent.

This doesn't make it a very suitable method for this application.

- ii ARQ uses acknowledgement and timeout. When a receiving device detects an error following data transmission, it asks for the data to be re-sent. If there are no errors, a positive acknowledgement is sent to the sender.

The sending device will re-send data if:

- it receives a request to re-send data
- a timeout has occurred.

The whole process is continuous until all data packets received are error-free or until the timeout is reached.

- b Three differences from:

Checksum

- calculated from the block of data
- sent with block of data
- re-calculated at the receiving station
- if different, an error has occurred
- used to check errors following data transmission

Check digit

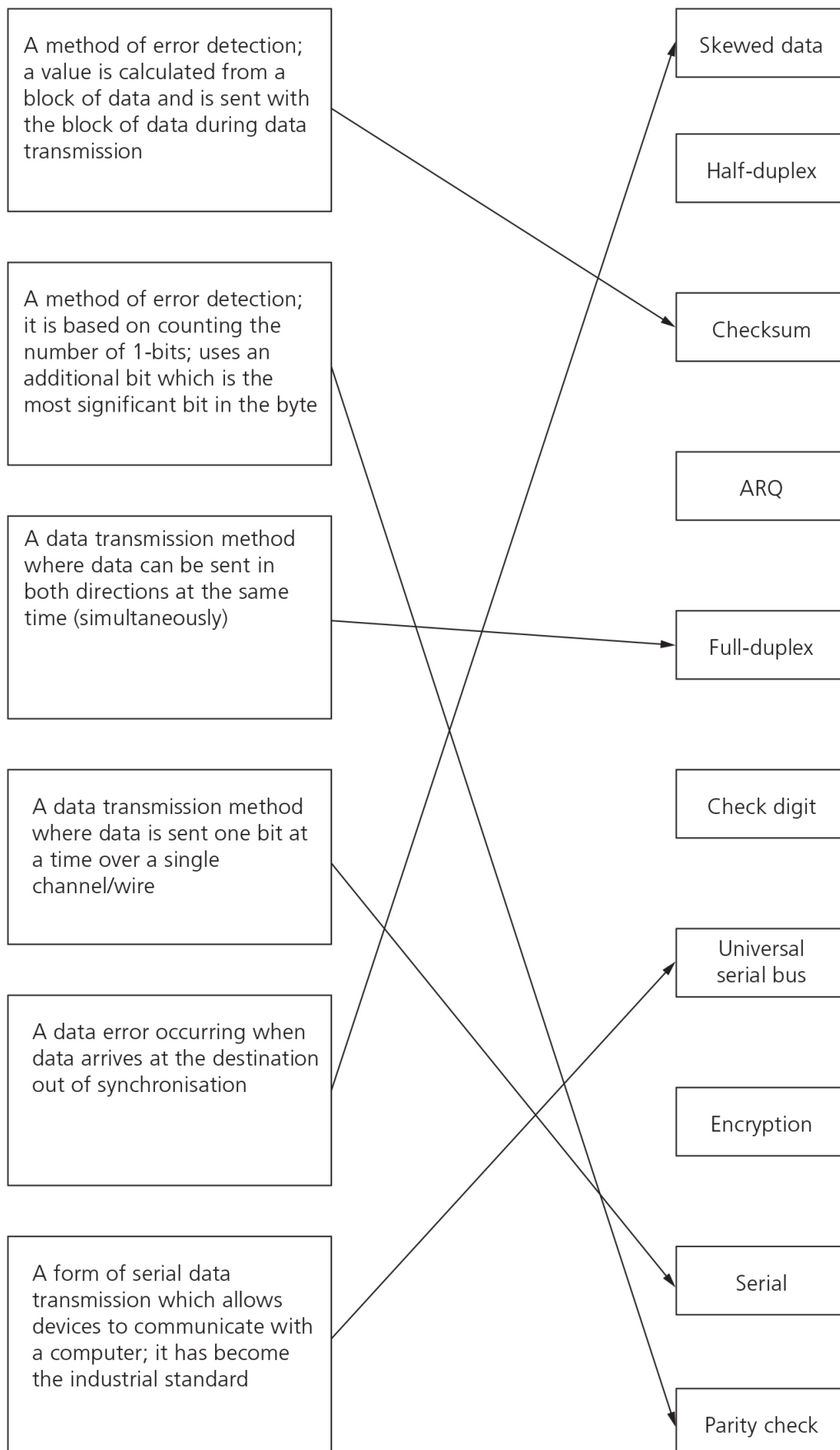
- final digit included in the code
- calculated from the other digits in the code
- used to check for errors during data entry
- detects errors such as transposition, missing/extra digits, and so on

- 2 i Packet switching is a method of data transmission where a message is broken into many data packets, which can then be sent along different pathways independently of each other.
- ii Cyclic redundancy check is an error-checking method which adds up all the 1-bits in the data packet payload and stores this hex number in the trailer; the same calculation is done at the receiving station.
- iii Data skewing occurs when data arrives at the destination with the bits no longer synchronised; this usually occurs during parallel data transmission.
- iv Universal serial bus is a type of serial data transmission which has become the industry standard for connecting computers to devices using the USB port.
- v Parity bit (0 or 1) is added to a byte of data in the most significant bit position (left-most bit); this ensures that the byte follows the correct even parity or odd parity protocol.

3 Description	Checksum ✓	Parity check ✓	ARQ ✓
Extra bit sent with each byte of data		✓	
Makes use of timeout and acknowledgement			✓
If an error is found, a request is made to re-send the data			✓
Check on whether a data packet has been changed following transmission	✓		✓
Re-calculation made on any additional data values sent to the recipient	✓	✓	
Data is transmitted in blocks or packets	✓	(✓)	✓
A method that can determine which bit in a data stream has been changed		✓	
Additional value sent at the end of a block of data to be used to check if any data transmission errors occurred	✓		

- 4 a Sequence: iii, i, ii, iv
- b Sequence: v, ii, iii, iv, i
- c Sequence: ii, iii, i, iv, v
- d Sequence: vi, v, ii, iv, iii, i
- 5 a Symmetric encryption occurs when the same encryption key is used to encrypt and decrypt a message.
- b Asymmetric encryption uses public keys (known to everyone) and private keys (known to a single user only); the recipient sends a public key to each sender who uses it to encrypt the message. The encrypted message is then decrypted using a matching private key at the recipient's computer.
- c If data is intercepted it cannot be understood by a hacker/eavesdropper without the correct decryption key.

6



7 a i

Method 1	Tick ✓	Method 2	Tick ✓
Serial	✓	Simplex	
Parallel		Half-duplex	
		Full-duplex	✓

ii Serial has lower/less interference.

Serial is more reliable over long distances.

In serial, bits won't be skewed/out of synchronisation.

In serial, it is easier to collate bits together again after transmission.

Full-duplex transmits data in both directions at the same time.

Simplex and half-duplex won't allow read and write operations at the same time.

b Any two from:

Checksum:

- value calculated from the data block
- this value is transmitted along with the data
- checksum recalculated after transmission
- if the values match, data has been transmitted error-free

Parity check:

- parity bit is transmitted with each byte of data
- odd or even parity protocol is being used
- number of 1-bits are added together to determine if parity bit is 0 or 1
- byte checked at destination to see if it has the correct parity protocol (odd or even)
- if the parity has changed, an error has occurred

ARQ:

- uses acknowledgement and timeout
- when device detects an error in the data transmission, it requests the packet to be re-sent; if no error, a positive acknowledgement is sent
- sending device re-sends packet after request to re-send is made/timeout reached
- process is continuous until the data packet received is correct/until ARQ time limit is reached

Echo check:

- copy of data sent back to sender
- data is compared to see if they match
- if it does not match, error has been detected

8 a

Statement	True ✓	False ✓
Full-duplex data transmission can be either serial or parallel	✓	
Full-duplex data transmission is when data is transmitted both ways, but only one way at a time		✓
Full-duplex transmission is always used to connect a device to a computer		✓
Full-duplex data transmission is when data is transmitted both ways at the same time	✓	
Full-duplex data transmission automatically detects any errors in data		✓

b parallel data transmission

c Two from:

- universal standard, so it is likely to be compatible with the computer
- can only be inserted one way, so there is less chance of connecting a device incorrectly
- high-speed data connection, so data transmission is fast
- uses serial transmission, so less chance of skewing errors
- does not require (wireless) network, therefore can be used even if network is down
- backwards compatible, so no additional technology required
- no need for a separate power source, because USB supplies (5V) power
- drivers are automatically identified/loaded, so no need to find them online/installs drivers automatically

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3 Hardware

Answers to activities

3.1

- 1
 - a address bus, data bus, control bus
 - b
 - address bus: carries addresses throughout the computer; it is unidirectional
 - data bus: bidirectional bus allowing data to be sent throughout the computer
 - control bus: carries signals from the control unit to all other computer components; it is bidirectional
 - c
 - wider address and data buses would increase computer performance
 - wider control bus would have little, if any, effect on performance
 - higher clock speed would increase processor speed, (overclocking could have serious consequences however, such as unsynchronised operations or overheating)
- 2
 - fetches
 - current instruction register (CIR)
 - program counter (PC)
 - memory address register (MAR)
 - address bus
 - memory data register (MDR)
 - decoded
 - executed
 - control signals
 - control bus
 - arithmetic and logic unit (ALU)
 - accumulator (ACC)

3.2

- 1 a
- increasing the width of the address bus and width of the data bus
 - increasing clock speed but care needed to avoid over-clocking
 - use of larger cache memories (uses static RAM) which have faster data access time than normal RAM (which is dynamic)
 - using dual core and quad core processors; but doubling number of cores doesn't double processor performance since the CPU needs to communicate with each core which uses up time
- b
- instructions are a set of operations which are decoded in sequence
 - each operation is made up of an opcode (which informs the CPU about what operation needs to be done) and an operand (data which needs to be acted on/register)
 - limited number of opcodes exist and this is known as an instruction set
 - instruction sets are low level language instructions which instruct the microprocessor how to carry out an operation
- 2 a
- destination address input
 - other factors (e.g. avoiding toll roads) are input
 - data from satellites, which gives the location of three or four satellites ...
 - ... together with very accurate timings
 - Sat Nav computer uses satellite data to calculate the position of a car on the stored maps
 - this combination allows accurate tracking of the vehicle on stored roads
- b
- use of satellite or cellular network used to download the latest map updates/software updates
 - if the Sat Nav is a portable unit, it can be connected to a computer via USB port and new maps/software will be downloaded

3.3

- 1 a || wwwbwbb/wwwbbwb/wwwbbwb || bwwwwbw/bwbbbww/bbbwwbw ||
- b || wwbbwwb/wwbwbb/wbbwwb/wbbbwb ||
bwbwww/bwbwww/bwbbbww/bwbwww ||
- c || wwwbbwb/wbbwwb/wbbbbb/wbbbbb/wwwbwb ||
bbbwwb/bbwbww/bbbwwb/bbwbbw/bwbwww ||
- 2 a
- hold more information
 - fewer errors due to built-in error-checking in QR codes
 - easier to read – QR codes can be read by a smartphone/tablet camera and app
 - QR codes can be encrypted
- b $40 \times 40 = 1600 \text{ pixels} = 1600 \text{ bits} = 200 \text{ bytes}$
- c
- they are used as a form of alignment when reading the QR code
 - the remaining corner square is used to ensure the correct size and correct angle of QR camera shot
- d
- more than one QR format exists

- QR codes can contain malicious codes (called attagging); QR code software is free; therefore, it is easy to embed malicious software or fake websites

3.4

1 a Laser printer:

- uses dry powdered ink (toner)
- makes use of the properties of static electricity to produce text and images on paper
- prints whole page in one go
- ink is fused onto paper using heated rollers

Inkjet printer:

- uses liquid ink in four or five cartridges
- printing is done line by line as paper advances forward after each printed line
- ink droplets are ejected onto paper using thermal bubble or piezoelectric technology

b Parts i and ii

Laser printer:

- high-quality flyers and posters (in large quantities)
- any large volume printing job
- equipped with large capacity toner cartridges
- have large paper trays

Inkjet printer:

- one-off printouts, such as photos
- small batch printing jobs
- can print on various types of paper and other media (for example, can print on T-shirts)
- where small paper trays and small capacity ink cartridges are not important for small print jobs

2 Order: C, E, G, A, I, B, F, H, D

3 a In an LCD–LED screen, each pixel is made from liquid crystals that are backlit using LEDs. In pure LED screens, the pixels are made up of the LEDs themselves.

b LEDs:

- use less power
- they do not need to warm up to reach maximum brightness
- produce a 'whiter' light
- produce a brighter light
- last for much longer

- #### 4
- Filipe uses an application on his computer that can play music files.
 - The application finds the files on the backing store (HDD or SSD) and moves it into RAM.
 - The data in the file is read by the application and then sent via a device driver to loudspeakers which are connected to the computer.
 - The device driver software translates the digital data into a series of analogue electrical signal using a digital-to-analogue converter (DAC).
 - The electrical signals make the loudspeaker cones move, through the use of a solenoid.
 - If the file is a stereo recording then two different signals are sent to each loudspeaker.

- The movement of the loudspeaker cone moves molecules of air, which Filipe's ears detect, and means he can hear music.

3.5

1 a Control:

- sensors used to collect data to send to a microprocessor
- sensor data is compared to stored/pre-set data
- the output affects the next set of inputs
- microprocessor sends signals to actuators to operate valves, motors and switches

Monitoring:

- sensors used to collect data to send to a microprocessor
- sensor data is compared to stored/pre-set data
- no actual changes to the process are made
- results of data comparison are shown on a screen/printed out or an alarm is activated

- b**
- temperature and humidity sensors are used
 - both sensors constantly send data to a microprocessor
 - the data is converted to digital, if necessary, using an ADC
 - the temperature levels are set by the user ...
 - ... humidity levels are set at the factory
 - if the temperature sensor reading $>$ set temperature, the microprocessor will send a signal to an actuator ...
 - ... to switch on the air conditioning unit
 - if the temperature sensor reading $\leq$ the set temperature, the microprocessor will send a signal to an actuator ...
 - ... to switch off the air conditioning unit
 - if the humidity sensor reading $>$ factory set value, the microprocessor will send a signal to an actuator ...
 - ... to switch on the air conditioning unit
 - if the humidity sensor reading $\leq$ the factory setting AND temperature sensor reading $<$ the set temperature ...
 - ... the microprocessor will send a signal to an actuator to switch off the air conditioning unit
 - the process is continuous

- 2**
- pH sensors send data back to microprocessor
 - data changed to digital if necessary (using ADC)
 - if $\text{pH} < \text{stored value}$ then microprocessor sends signals to actuators ...
 - ... to open valve to add alkaline solution to soil
 - if $\text{pH} > \text{stored value}$ then microprocessor sends signals to actuators ...
 - ... to open valve to add acidic solution to soil
 - if $\text{pH} = \text{stored value}$ then microprocessor sends signals to actuators ...

- ... to close all valves
- the process is continuous
- 3**
 - readings taken from pressure and temperature sensors
 - readings changed to digital if necessary (using ADC)
 - readings are sent to microprocessor
 - if temperature > stored value, microprocessor sends signal to actuators ...
 - ... to open water valves
 - if temperature ≤ stored value, microprocessor sends signals to actuators ...
 - ... to close water valves
 - if pressure ≥ stored value, microprocessor sends signals to actuators ...
 - ... to close CO² gas valves
 - if pressure < stored value, microprocessor sends signals to actuators ...
 - ... open CO² gas valves
 - monitoring/control is continuous
- 4**
 - pressure sensors in all four roads at junction used to detect vehicles approaching lights in all directions
 - data is sent back to a microprocessor
 - number of vehicles approaching each set of traffic lights is calculated by the microprocessor based on sensor data
 - generated data is compared to a data model stored in the microprocessor system
 - signals sent by the microprocessor to alter light sequences as required
 - monitoring of roads is a continuous process

3.6

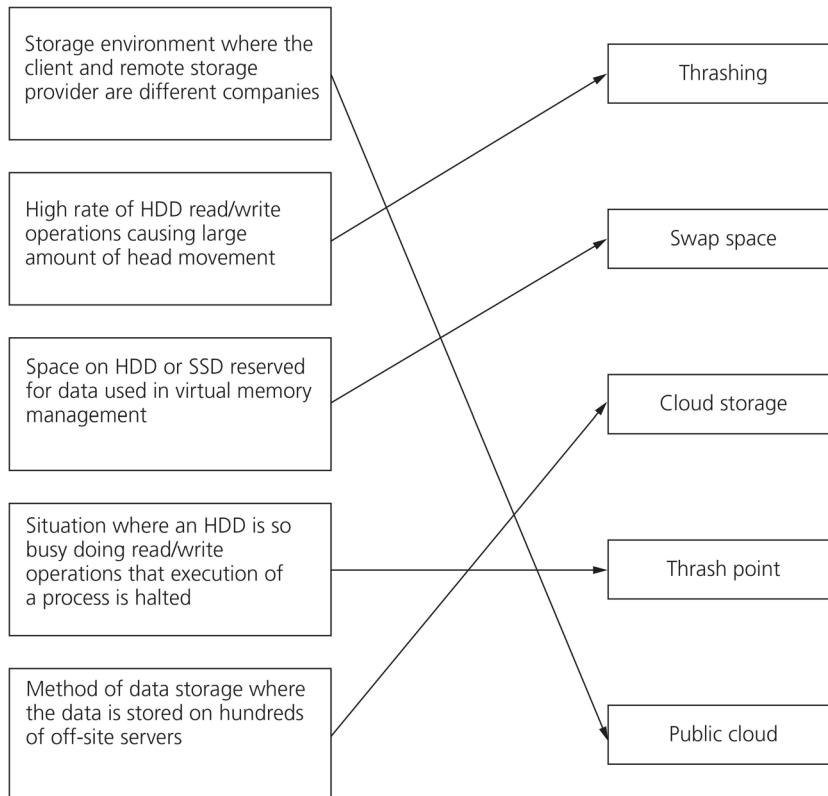
- 1**
 - a** RAM – settings (time, weight, defrost, ...)
ROM – microwave wavelength, pre-programmed cooking times, safety systems
 - b** RAM – temperature settings
ROM – control program to monitor fridge
 - c** RAM – user settings, bespoke programs
ROM – wavelength of remote signals, pre-programmed flights, ...

3.7

- 1** RAM:
 - used by both OS and applications software
 - when phone switched on, it pulls OS and app data from solid state memory to allow it to function; user's data
 Solid state:
 - used to store apps and operating system so that it is always there even when phone switched off
 - part of solid-state memory not accessible to users (for example, the OS)
 - stores user's photos, music files, and so on
- 2**
 - a**
 - uses movement of data between RAM and HDD/SSD

- RAM is physical memory and virtual memory is RAM + swap space on HDD/SSD
- to execute a program, data is loaded into RAM from HDD/SSD
- gives the illusion that RAM is much larger than it really is
- b**
 - when P5 wants access to RAM we use virtual memory since RAM is full
 - assuming P1 is oldest data, P1 data (10 GiB) is moved out of RAM into HDD/SSD
 - P5 now points to memory space in RAM once occupied by P1
 - process continues to allow all five programs to run in RAM

3



4

- a**
- RAM:**
- temporary memory device
 - volatile memory
 - can be written to and read from
 - used to store data, files, programs, part of OS **currently** in use
 - can be increased in size to improve operational speed of a computer
 - a computer has far more RAM memory than ROM memory
- ROM:**
- permanent memory device
 - non-volatile memory device
 - read-only
 - used to store the boot-up instructions
 - an increase in the amount of ROM has no effect on computer performance
- b** For example:
- RAM – used by a PC to store the contents of an application when the computer is running
 - ROM – used by a PC to find out what it should to do when the power is switched on

- 5 a** electrons
NAND
matrix
intersection
control gate
floating gate
electrons
control gate
- b** Compared to HDDs, SSDs:
- have faster data access times
 - are lighter
 - consume less power
 - don't get as hot
 - are thinner
 - are more reliable
- c** More expensive – questions over how long they can store data for

3.8

- 1 a**
- A network interface card (NIC) is needed to allow a device to connect to a network (such as the internet).
 - It is usually part of the device hardware and contains the Media Access Control (MAC) address generated at the manufacturing stage.
- b**
- Media Access Control is an address which refers to a number which uniquely identifies a device on the internet.
 - It takes the form: NN-NN-NN-DD-DD-DD.
- c**
- Internet Protocol identified either as IPv4 or IPv6
 - It gives a unique address to each device connected to the internet which identifies their location.
- d**
- These enable data packets to be routed between the different networks, for example, to join a LAN to a WAN onto the internet.
- e**
- A server that automatically provides and assigns an IP address.
- 2 a**
- changes each time a device connects to the internet
 - allows for greater privacy
 - can be less reliable for some applications (e.g. VoIP)
- b**
- IP address is associated with a particular device and does not change each time the device connects to the internet
 - allows faster upload and download speeds
 - less privacy as a specific device can be identified by its IP address
- c** Dynamic IP addresses are needed for the following reasons:
- they give greater privacy since they change each time a user logs on

- dynamic IP addresses can be an issue when using, for example, VoIP since this type of addressing is less reliable as it can disconnect and change the IP address causing the VoIP connection to fail

Static IP addresses are needed for the following reasons:

- since static IP addresses don't change, they allow each device to be fully traceable
- allow for faster upload and download speeds

3 MAC:

- Permanent address associated with the piece of hardware
- Unique
- 48-bit address
- Used to find the particular device within a network

IP:

- An address assigned to hardware by an ISP – even a static IP address is not physically associated with the hardware
- Not necessarily unique
- 32 (IPv4) or 128 (IPv6) bit address
- Used to route data to the correct network

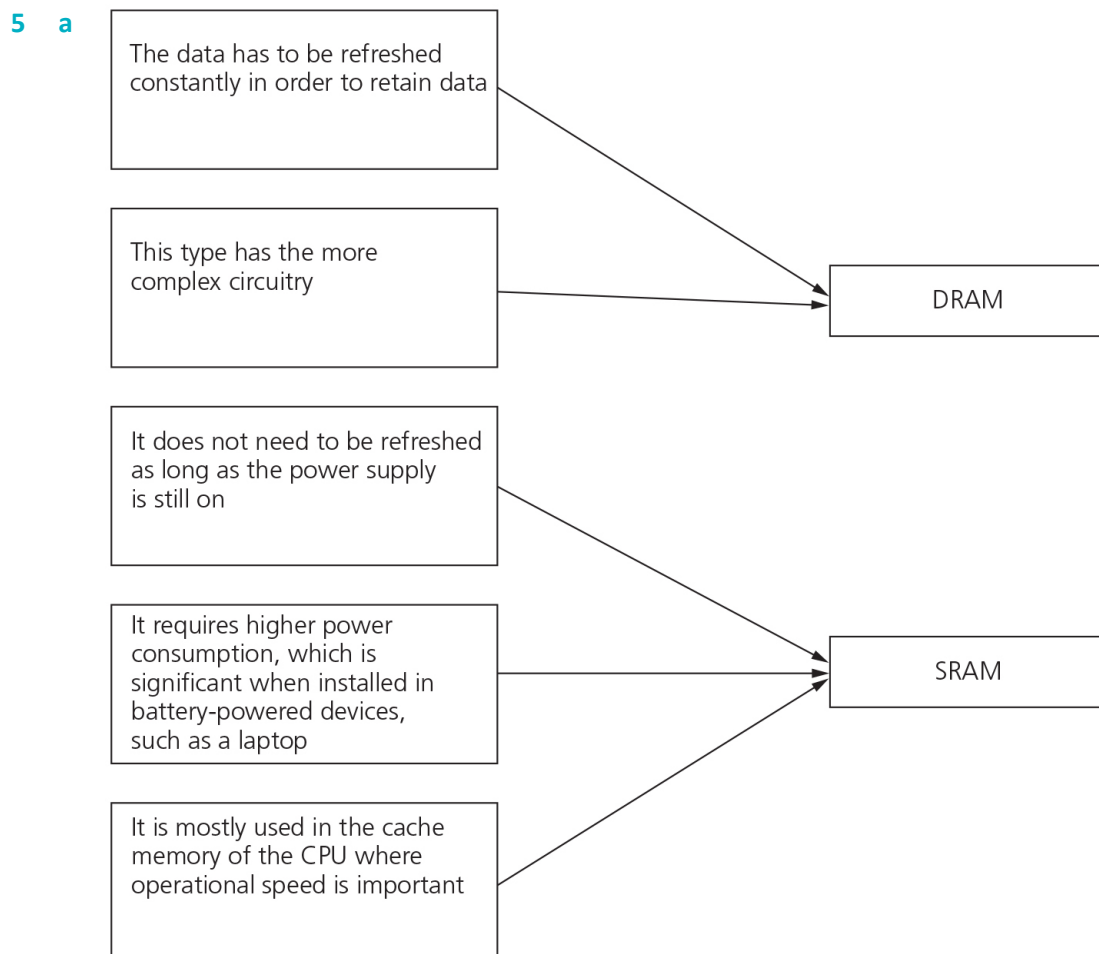
Answers to exam-style questions

Answers to questions 7–9 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

- 1 a**
- The plastic, organic layers of an OLED are thinner, lighter and more flexible than the crystal structures used in LEDs or LCDs.
 - The light-emitting layers of an OLED are lighter; OLED layers can be made from plastic rather than the glass used in LED and LCD screens.
 - OLEDs give a brighter light than LEDs.
 - OLEDs do not require backlighting like LCD screens – OLEDs generate their own light.
 - Since OLEDs require no backlighting, they use much less power than LCD screens (most of the LCD power is used to do the backlighting); this is very important in battery-operated devices such as mobile phones.
 - Since OLEDs are essentially plastics, they can be made into large, thin sheets (this means they could be used on large advertising boards in airports, subways, and so on).
 - OLEDs have a very large field of view, about 170 degrees, which makes them ideal for use in televisions and for advertising screens.
- b**
- many colours are possible if each primary colour (RGB) uses several bits to represent its colour intensity ...
 - ... for example, 32 bits per colour gives $2^{32} \times 2^{32} \times 2^{32}$ possible colour combinations
 - however, the human eye cannot detect more than a few thousand different colour shades ...
 - ... therefore, it is impossible to experience the 1 million different colours as claimed by the manufacturer

- television contrast (etc.) needs to be set up perfectly to show all colours accurately
- 2 a i** Two from:
- cannot be changed by the user
 - more difficult to copy the software
 - immediately available as soon as device powered up
 - programs are permanently stored
- ii** RAM allows the user to store their own data, such as game status
- only the part of the software needed at each stage in the game would be in RAM at any time
- iii** USB ports allow other games stored on memory sticks, for example, to be plugged into the games console
- can plug the console into external speakers for better sound effects
 - can plug in keyboard and/or mouse to navigate menu systems
 - can plug in VR headsets for consoles with this capability
- b**
- the design may change several times ...
 - ... therefore, it is cheaper and quicker to make a prototype using a 3D printer
 - the 3D solid object lets the development team see how well it will work in real life before building the final version
- 3 a**
- both use sensors to gather data
 - in control, the output affects the input and actuators are used to change the process (for example, open valve, operate pump)
 - in monitoring, no changes to the process are made; the system simply shows the status on a screen or prints out data or sounds an alarm
- b**
- temperature and humidity sensors are used
 - both sensors constantly send data to a microprocessor
 - the data is converted to digital, if necessary, using an ADC
 - the temperature levels are set by the user ...
 - ... humidity levels are set at the factory
 - if the temperature sensor reading is $>$ set temperature, the microprocessor will send a signal to an actuator ...
 - ... to switch on the air conditioning unit
 - if the temperature sensor reading $\leq$ the set temperature, the microprocessor will send a signal to an actuator ...
 - ... to switch off the air conditioning unit
 - if the humidity sensor reading $>$ factory set value, the microprocessor will send a signal to an actuator ...
 - ... to switch on the air conditioning unit
 - if the humidity sensor reading $\leq$ factory setting AND temperature sensor reading $<$ set temperature ...
 - ... the microprocessor will send a signal to an actuator to switch off the air conditioning unit

- 4 a
- the process is continuous
 - static IP address doesn't change
 - dynamic IP address is a temporary IP address assigned to a device each time it logs onto the internet
 - dynamic IP addresses make use of a Dynamic Host Configuration Protocol (DHCP) server which assigns the IP addresses whereas static IP addresses are assigned by the ISP
 - static IP addresses assigned to servers which host websites or to FTP servers; this gives the device a consistent location and also means less bandwidth is used reducing network traffic (and takes less time to retrieve data)
 - dynamic IP addresses afford greater privacy since they change each time a device logs onto the internet, therefore it makes the device more difficult to trace
- b
- MAC address is a media access control address
 - MAC is used to uniquely identify a device
 - uses the format: NN – NN – NN – DD – DD – DD
 - where NN – NN – NN refers to the manufacturer and DD – DD – DD refers to the serial number of the device
 - there are two types referred to as UAA (set at factory) or LAA (can be changed by user)



b RAM:

- temporary memory device
- volatile memory
- can be written to and read from
- used to store data, files, part of OS currently in use
- can be increased in size to improve the performance of computer

ROM:

- permanent memory device
- non-volatile memory device
- data cannot be altered
- sometimes used to store the BIOS and other start-up procedures

c DVD:

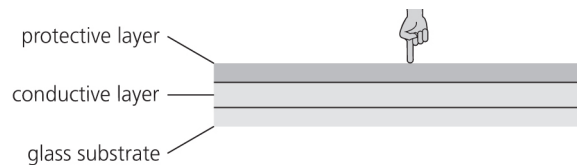
- uses red laser technology
- storage capacity is ~ 4.7 GB max
- doesn't allow great interactivity
- relatively easy to copy
- data transfer rate is ~ 10 Mbps
- always comes in dual-layer format

Blu-ray:

- uses blue laser technology
- storage capacity can be up to ~ 50 GB
- allows much greater interactivity
- has built-in data encryption
- data transfer rate is ~ 36 Mbps
- can come in single-layer, dual-layer ... up to 6-layer formats

6 a Capacitive touch screens:

- capacitive touch screens are composed of a layer of glass (protective layer), a transparent electrode (conductive) layer and a glass substrate
- since human skin is a conductor of electricity, when bare fingers (or special stylus) touch the screen, the electrostatic field of the conductive layer is changed
- the installed microcontroller is able to calculate where this change took place and hence determine the coordinates of the point of touching



- with surface capacitive screens, sensors are placed at the corners of a screen
- small voltages are also applied at the corners of the screen creating an electric field
- a finger touching the screen surface will draw current from each corner reducing the capacitance
- a microcontroller measures the decrease in capacitance and hence determines the point where the finger touched the screen
- this system only works with a bare finger or stylus
- with projective capacitive screens, the transparent conductive layer is in the form of an X–Y matrix pattern
- this creates a three dimensional (3D) electrostatic field
- when a finger touches the screen it disturbs the 3D electrostatic field allowing a microcontroller to determine the coordinates of the point of contact
- this system works with bare fingers, stylus and thin surgical or cotton gloves; it also allows multi-touch facility (for example, pinching or sliding)

Infrared touch screens:

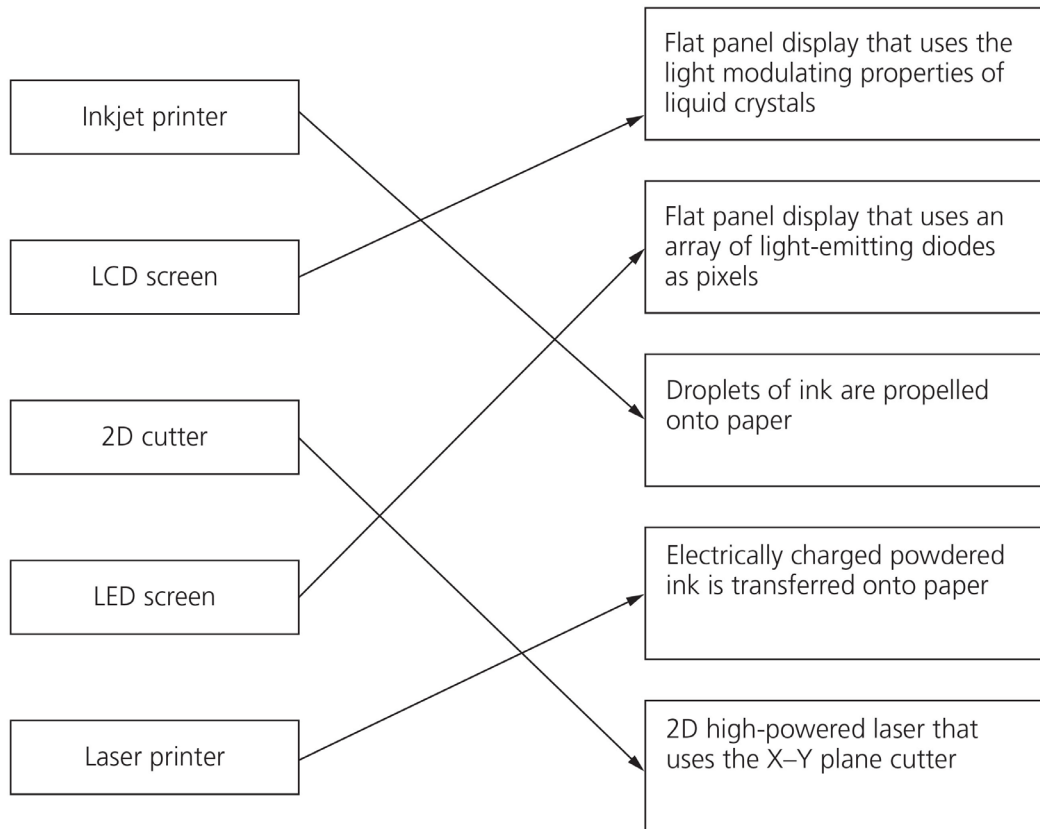
- infrared touch screens use a glass screen with an array of sensors and infrared transmitters
- the sensors detect the infrared radiation; if any of the infrared beams are broken (for example, with a finger touching the screen), the infrared radiation reaching the sensors is reduced
- the sensor readings are sent to a microcontroller which calculates where the screen was touched

Resistive touch screens:

- these screens are made up of two layers of electrically resistive material with a voltage applied across them
 - the upper layer is made of flexible polyethylene (a type of polymer) with a resistive coating on one side
 - the bottom layer is made of glass also with a resistive coating (usually indium tin oxide) on one side
 - these two layers are separated by air or an inert gas (such as argon)
 - when the top polyethylene surface is touched, the two layers make contact
 - since both layers are coated in a resistive material, a circuit is now completed which results in a flow of electricity
 - the point of contact is detected where there was a change in voltage
 - a microcontroller converts the voltage (created when the two resistive layers touch) to digital data which it then sends to the microprocessor
- b i** Uses organic layers which emit their own light and therefore there is no need for any backlighting
- ii**
- The plastic, organic layers of an OLED are thinner, lighter and more flexible than the crystal structures used in LEDs or LCDs.
 - The light-emitting layers of an OLED are lighter; OLED layers can be made from plastic rather than the glass used in LED and LCD screens.
 - OLEDs give a brighter light than LEDs.
 - OLEDs do not require backlighting like LCD screens – OLEDs generate their own light.
 - Since OLEDs require no backlighting, they use much less power than LCD screens (most of the LCD power is used to do the backlighting); this is very important in battery-operated devices such as mobile phones.
 - Since OLEDs are essentially plastics, they can be made into large, thin sheets (this means they could be used on large advertising boards in airports, subways, and so on).
 - OLEDs have a very large field of view, about 170 degrees, which makes them ideal for use in televisions and for advertising screens.

- 7 a touchscreen/concept keyboard/microphone/trackerball/touchpad
b headphones/speakers
c (removable) hard disk drive/SSD/USB drive/Blu-ray disk
d
- QR code scanned using a camera on smartphone or tablet
 - QR code is decoded/read using the app on the smartphone/tablet
 - customer is provided with text about the animal
 - customer is directed to a website about the animal
 - QR code can be saved for future use
- 8 a input device which measures/takes readings from the surroundings/environment
b
- moisture sensor:
 - measures water content of the soil
 - alerts Anna when soil is too dry/too wet
 - maintains correct water level in the soil
 - pH sensor:
 - measures how acidic/alkaline the soil is
 - alerts Anna when there may be something polluting the soil
 - maintains correct pH level of the soil
 - light sensor:
 - measures brightness of the surroundings
 - alerts Anna when fruit has too little/too much light
 - temperature sensor:
 - measures the temperature of the surroundings
 - alerts Anna when it is too hot/too cold for optimum growing conditions
 - gas/O₂/CO₂ sensor:
 - measures amount of CO₂ in the air
 - alerts Anna if CO₂ levels are too high/too low
 - humidity sensor:
 - measures the water content of the air
 - alerts Anna if the air is too dry/too damp
 - infrared sensor/motion sensor:
 - measures infrared/microwaves
 - detects intruders/wild animals
 - pressure sensor:
 - measures the weight of fruit fallen to the ground ...
 - ... to alert Anna about how much fruit has fallen from the trees

9



Answers to Student's Book

Exam-style questions and sample answers have been written by the authors. References to assessment and/or assessment preparation are the publisher's interpretation of the syllabus requirements and may not fully reflect the approach of Cambridge Assessment International Education. Cambridge International recommends that teachers consider using a range of teaching and learning resources in preparing learners for assessment, based on their own professional judgement of their students' needs.

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4 Software

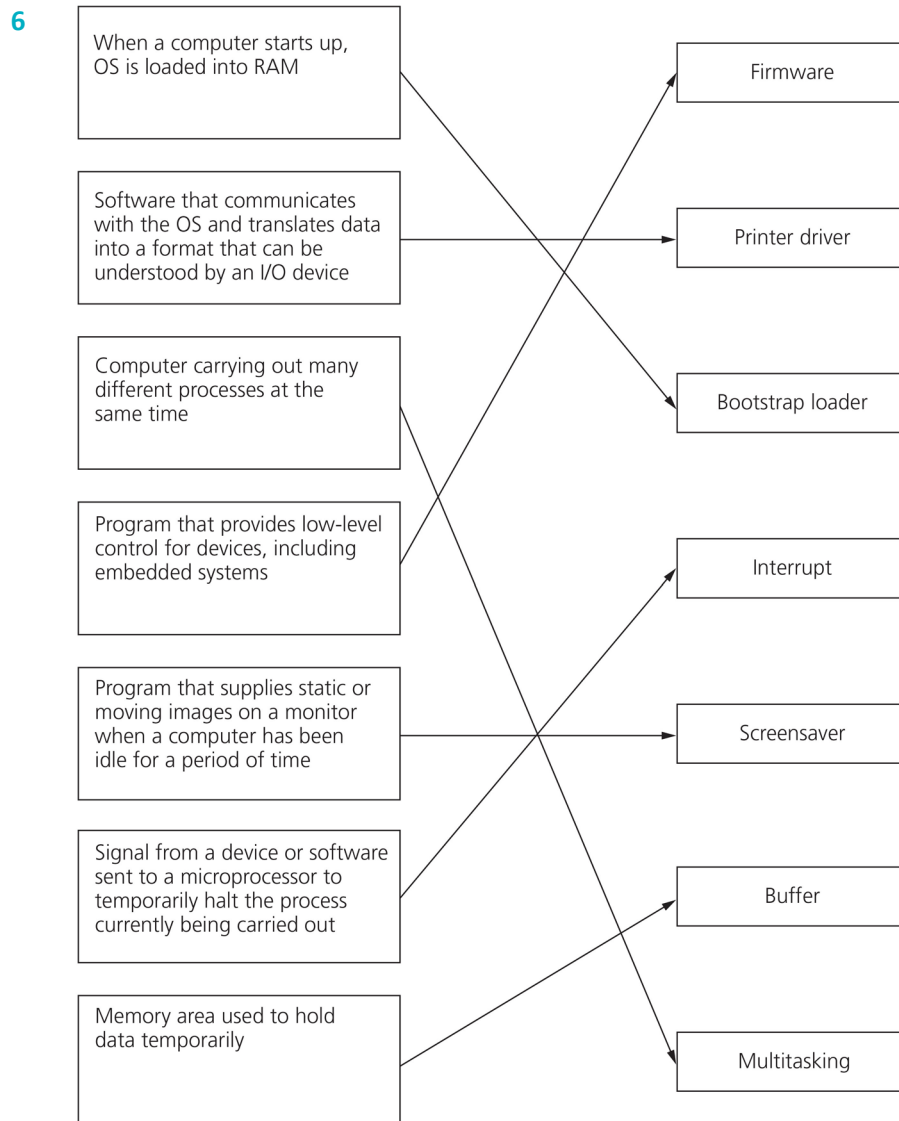
Answers to activities

4.1

1	Software	System software	Application software
	Screensaver	✓	
	Anti-virus software	✓	
	Control and measurement software		✓
	Printer driver	✓	
	Video editing software		✓
	Compiler	✓	
	QR code reader		✓
	On-screen calculator		✓
	Operating system software	✓	

- 2 a** buffer
- b**
- data sent to printer buffer
 - meanwhile, data from download/video buffer is sending data to video player
 - once printer buffer is full, attention returned to video download and video buffer starts to fill again
 - document starts to print
 - when printer buffer is empty, an interrupt is sent to microprocessor requesting more data
 - microprocessor temporarily halts video download process to service printer
 - more data is sent to the printer buffer
 - meanwhile video player plays movie
 - if video buffer is at minimum value, an interrupt is sent to stop other processes to allow video buffer to start filling again
 - process goes on until whole document is printed out
- c** interrupt sent to CPU indicating an error and message appears on screen; printing process suspended until new black ink cartridge is installed
- 3**
- | | |
|----------------------------|--|
| Human computer interface | can be CLI or GUI; permits user to interact with computer |
| Multitasking | allows computer to carry out several tasks at the same time (allocating resources to each task) |
| Platform (to run software) | all apps come under the control of OS; manages movement of data into and out of RAM to allow software to run with required resources |
| User account management | allows more than one user to log onto a computer; each user is allocated separate parts of memory; administrator has overall control |
| Managing files | carries out specific tasks (for example, create, open, close, delete, rename, copy, etc.); maintains directory structures; monitors access control mechanisms and ensures they are maintained; loads files into memory as required |
| Hardware management | involves management of all I/O devices; communicates with devices via device drivers; ensures allocation of resources according to priority; manages queues and buffers |
| Memory management | manages RAM and keeps track of which apps are using memory locations; prevents competing apps from using the same memory locations |
| Interrupt handling | handles interrupts from hardware and software to ensure they are serviced in correct priority and that all devices have access to the CPU and resources as required |
| System security | ensures the integrity, confidentiality and availability of data is maintained; use of passwords, maintaining access rights, and so on |

- 4
- A descriptor is a collection of information about devices.
 - Descriptors include the vendor ID, product ID and serial number of a device.
 - If a device doesn't have a unique serial number associated with it, the OS will treat the device as new every time it is plugged into a USB port.
 - When a device is plugged in, descriptors are used to recognise the device and load the device driver automatically.
- 5 a
- when the computer is powered up, part of the OS needs to be loaded into RAM ...
 - ... this is called the bootstrap (loader)
 - start-up is handled by the BIOS ...
 - ... which tells the computer where the storage device that hold the OS can be located
 - BIOS is stored on firmware, such as EEPROM
- b
- BIOS settings are stored on a CMOS chip ...
 - ... which is powered at all times by a battery on the motherboard
 - user can change BIOS settings if required but not the BIOS software



4.2

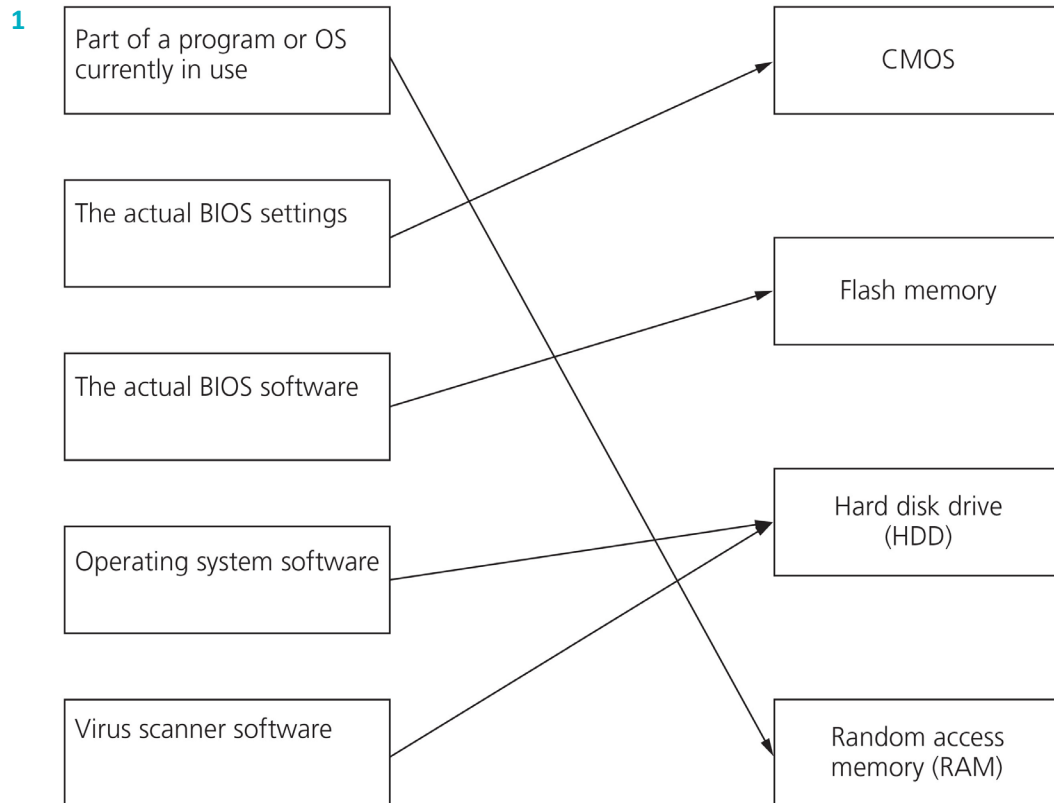
1

	True	False
An assembler translates a high-level language program.		✓
It is more difficult to write a program in a low-level language.	✓	
Java is an assembly language.		✓
It is quicker to develop a program using a high-level language.	✓	
You always need a compiler to run a compiled program.		✓
A program that is interpreted takes a longer time to run than a compiled program.	✓	
Low-level languages are machine dependent.	✓	

- 2 a
- Auto-completion means that an IDE can automatically show the reserved words or variable names that are available for use, so that the correct one can be picked from a list.
 - Prettyprinting – code editors can colour code parts of a program, for example reserved words or strings and automatically lay out the program in a meaningful way.
 - Translators are supplied as part of the IDE; these can be compilers or interpreters or both.
- b A compiler translates the whole program written in a high-level language into machine code program. An interpreter executes a program written in a high-level language by translating then executing each line of the program.
- A compiler can produce a file containing the translated machine-code program that can be executed without the compiler, whereas an interpreter will always need to be present during the execution of an interpreted program.
- c Assembly language programs can be smaller in size and execute tasks more quickly as fewer instructions are used, programs can also control specialist hardware using appropriate simple commands.

Answers to exam-style questions

Answers to questions 6 and 7 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.



- 2**
- a** virus scanner
 - b** defragmentation
 - c** security software
 - d** screen saver
 - e** device driver
- 3**
- a** provide a platform for the execution of apps and also provides an HCI (human-computer interface), memory management, security management and management of user accounts, hardware management (e.g. keyboards and monitors), file management, handles interrupts, manages multitasking (i.e. running more than one application at a time)
 - b** memory management system which gives the impression that the RAM is larger than it is; involves moving data into and out of RAM and HDD/SSD
 - c** this occurs when large volumes of data are being moved in and out of HDD; involves a large number of head movements
 - d** program within the operating system that allows a user to run several applications at the same time; but only one app is actually 'active' at any given time
 - e** signal sent from a device or software to a processor to temporarily halt the process currently being run

4 a Graphical user interface (GUI):

- uses pictures or symbols (icons) on screen to allow selection of options
- selection is done via pointing devices or using touchscreen technology
- common system is WIMP when pointing devices used
- Windows manager looks after the interaction between Windows, applications and pointing devices/cursor position for touchscreen

Command line interface (CLI):

- user has to type in instructions/commands to choose options or launch apps
- user is in direct communication with the computer
- interface/input is via a keyboard and command box
- user needs to learn a number of commands

b and c

Interface	Advantages	Disadvantages
Graphical user interface (GUI)	The user doesn't need to learn any commands. It is more user-friendly; icons are used to represent applications. A pointing device (such as a mouse) is used to click on an icon to launch the application – this is simpler than typing in commands OR a touch screen can be used where applications are chosen by simply touching the icon on the screen.	This type of interface uses up considerably more computer memory than a CLI interface. The user is limited to the icons provided on the screen. It needs an operating system, such as Windows, to operate, which uses up considerable memory.
Command line interface (CLI)	The user is in direct communication with the computer. The user is not restricted to a number of pre-determined options. It is possible to alter computer configuration settings. It uses a small amount of computer memory.	The user needs to learn a number of commands to carry out basic operations. All commands need to be typed in which takes time and can be error-prone. Each command must be typed in using the correct format, spelling, and so on.

- 5
- a C – an area in memory that temporarily holds data
 - b B – software that communicates with the operating system and translates data into a format that can be printed out
 - c C – allocating memory to competing applications running on a computer system
 - d D – multitasking
 - e A – command line interface (CLI)
 - f C – a GUI interface makes it easier to use the computer
 - g C – interrupt
 - h D – CMOS
 - i A – any device plugged into the USB port of a computer must have a unique serial number
 - j B – post-WIMP interfaces make use of the more modern optical mouse to select icons
- 6
- a
 - A compiler produces a file containing a translated machine-code program; an interpreter does not.
 - An interpreter executes a program written in a high-level language by translating then executing each line of the program, provided there are no errors in that line.
 - A compiler does not execute the program.
 - b Advantage – compiled programs are smaller in size; source code not included in compiled file. Disadvantage – longer development time if compilers are used whilst developing code.
 - c Advantage – easier to edit programs during development. Disadvantage – programs cannot run without the interpreter.
- 7
- A compiler translates a high-level language into machine code, whilst an assembler translates a low-level assembly language into machine code. One assembly language instruction usually translates to one machine code instruction, whereas one high-level language instruction usually translates to many machine code instructions.
- 8
- a An Integrated Development Environment (IDE) is software that is used by programmers to aid the writing, editing and development of programs.
 - b Auto-completion means that an IDE can automatically include matching keywords when the first keyword is typed. For example, when `IF` is written in Visual Basic, the IDE automatically adds the keywords: `End If`
 Auto-correction means that an IDE can automatically correct errors, for example, underlining the places where syntax errors occur and offering the corrected statement to use.
 Prettyprinting means colour-coding words and adjusting the layout of a program for readability, for example, always showing reserved words in orange and strings in red.
 - c
 - A debugger to test a program in a controlled environment. For example, a separate report window can be used to display the values of each variable as the program is running.
 - A code editor to type the program and correct errors means that there is no need to exit the IDE for every alteration during development.

- 9 As the program has already been written it can be translated using a compiler as the development has already been done. This means that there would be an executable program available in binary that can be run on his friend's computer without the use of any additional software. Pedro knows that his friend's internet connection is very slow, and the compiled program is smaller than the original program so there will be less data to transmit, which means that it will take less time for the program to arrive and the transmission is more likely to be successful.

Answers to Student's Book

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5 The internet and its uses

Answers to activities

5.1

1 a urlexample

b .co

c sample_page

d http

2 a Session cookie:

- temporary cookie
- ceases to exist once browser is closed or website session terminated
- doesn't actually collect any information from user's computer

Persistent cookie:

- stored on hard drive of user's computer ...
 - ... until expiry date reached or user deletes cookie
 - remains in operation on user's computer even when browser closed or website session terminated
- b
- allows a website to remember a user's email address and invoice details so these don't have to be typed in each time a user visits a website, which allows a faster log in
 - serves as a memory, enabling a website to recognise users every time they visit the website
 - saves user's items in a virtual shopping basket
 - tracks internet habits and user's website history/favourites (bookmarks)/recommendations made
 - stores customer payment details (in secure format)
 - stores user preferences
 - allows progress in online games/quizzes to be stored

- allows social networking sites to recognise certain preferences and browsing history
- allows different languages to be used on web pages automatically/personalised web pages stored
- indicates if customer is new or returning

3

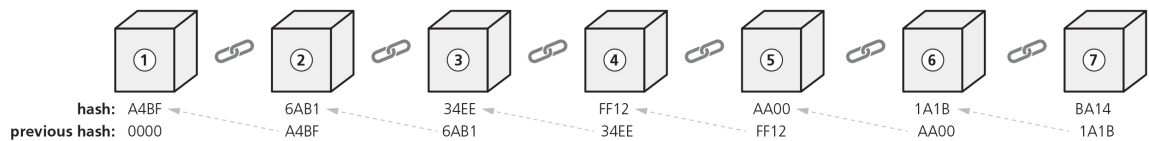
Feature	Internet	World Wide Web
It is possible to send and receive emails	✓	
Makes use of http protocols		✓
Uses URLs to specify the locations of websites and web pages		✓
Resources can be accessed by using web browsers		✓
Makes use of TCP and IP	✓	

4

- they record data about the user ...
- ... such as browsing activity, purchases and preferences ...
- ... IP addresses and geographical location
- they store marketing data which allows adverts to target user

5.2

1 a



- b If block 4 is hacked and any data is changed, then the hash value for the block will change; this will result in the chain being broken since block 5 can no longer point to block 4.

2

- cryptocurrency uses cryptography to track transactions
- cryptocurrency is decentralised unlike digital currencies which are centralised
- traditional digital currencies are regulated by central banks and governments
- unlike normal digital currencies, cryptocurrency transactions are publicly available and can be tracked/money monitored
- cryptocurrency uses blockchaining

5.3

1

a	b	c
Three from:		
brute force attacks	to crack passwords	make passwords difficult to crack and change them often
data interception	tapping into networks to obtain information	use private lines/networks and encrypt all data
denial of service attack	'... which cause it to crash	use a malware checker, firewalls and email filters
hacking	illegal access to a computer system	firewalls and passwords (including biometrics)
malware (worms, viruses, spyware, ...)	deletion or altering of data	use up-to-date anti-malware software
phishing	legitimate-looking emails which contain links to fake websites	don't open links in emails, use up-to-date browsers, look out for https, check if url looks legitimate
pharming	code on HDD/SSD or server which redirects browser to fake website without user's knowledge	use anti-virus software to detect website tampering, check spelling of websites
social engineering	cybercriminal creates social situation that can lead a victim to dropping their guard using: instant messaging, scareware, phishing, baiting or phone calls	be aware of scam text messages, phone calls, memory sticks left lying around on purpose, and so on

- 2
- worm: type of stand-alone virus that can replicate itself with the intention of spreading to other computers/networks; often uses networks to search out computers with weak security
 - ransomware: programs that encrypt data on a user's computer; a decryption key is sent back to the user once they pay a sum of money (ransom); they are often sent via a Trojan horse or by social engineering
 - Trojan horse: malicious programs often disguised as legitimate-looking software; they replace all or part of the legitimate software with the intent of carrying out some harm to the user's computer system

- 3 a**
- i** a date of birth – relatively easy to find or guess
 - ii** too close to name of a pet – easy to crack
 - iii** uses his own name in the password – makes it relatively easy to guess
- b** ensure at least eight characters are used, made up of upper-case and lower-case letters, numbers and other keyboard characters; passwords should be changed on a regular basis
- c** if the computer can be accessed by other people, then it is not a good idea to store passwords on the computer itself since these could be found by another user

5.4

- 1 a** 25-May-2000 – weak password because it is just a date
- b** Pas5word – one of the ten most used passwords!
- c** ChapTer@06 – strong password with letters, numbers and other characters
- d** AbC*N55! – strong password with letters, numbers and other characters
- e** 12345X – weak password since it is easy to guess/crack
- 2 a** make use of levels of access; for example, level 1: passenger information, level 2: airport staff, level 3: senior staff only – use passwords and user IDs
- b** make use of:
- strong passwords and user IDs
 - firewalls
 - scan for malware at all times
 - don't allow plug-in devices at information desks
 - don't allow internet access from information desks

5.5

- 1 a**
- as soon as motion detector recognises a person, a camera automatically photographs the face of the person
 - software scans face for key features and creates an electronic image/file
 - the file is compared to the database containing face scans of authorised people
 - if key features match database data then entry is allowed ...
 - ... otherwise entry is denied
- b** finger/thumb print, voice recognition using a microphone
- 2**
- | | |
|---|---|
| <p>Benefits:</p> <ul style="list-style-type: none"> • more secure system since only certain people can be recognised as authorised drivers • safer since driver doesn't need to look at screens/dials to select options or look at key data • relatively inexpensive technology | <p>Drawbacks:</p> <ul style="list-style-type: none"> • a person's voice can be recorded to overcome the security system • low accuracy; doesn't understand verbal commands • change of voice (e.g. due to a cold) can make recognition 'hit and miss' • sometimes there is a need to remember certain commands to make system work |
|---|---|

5.6

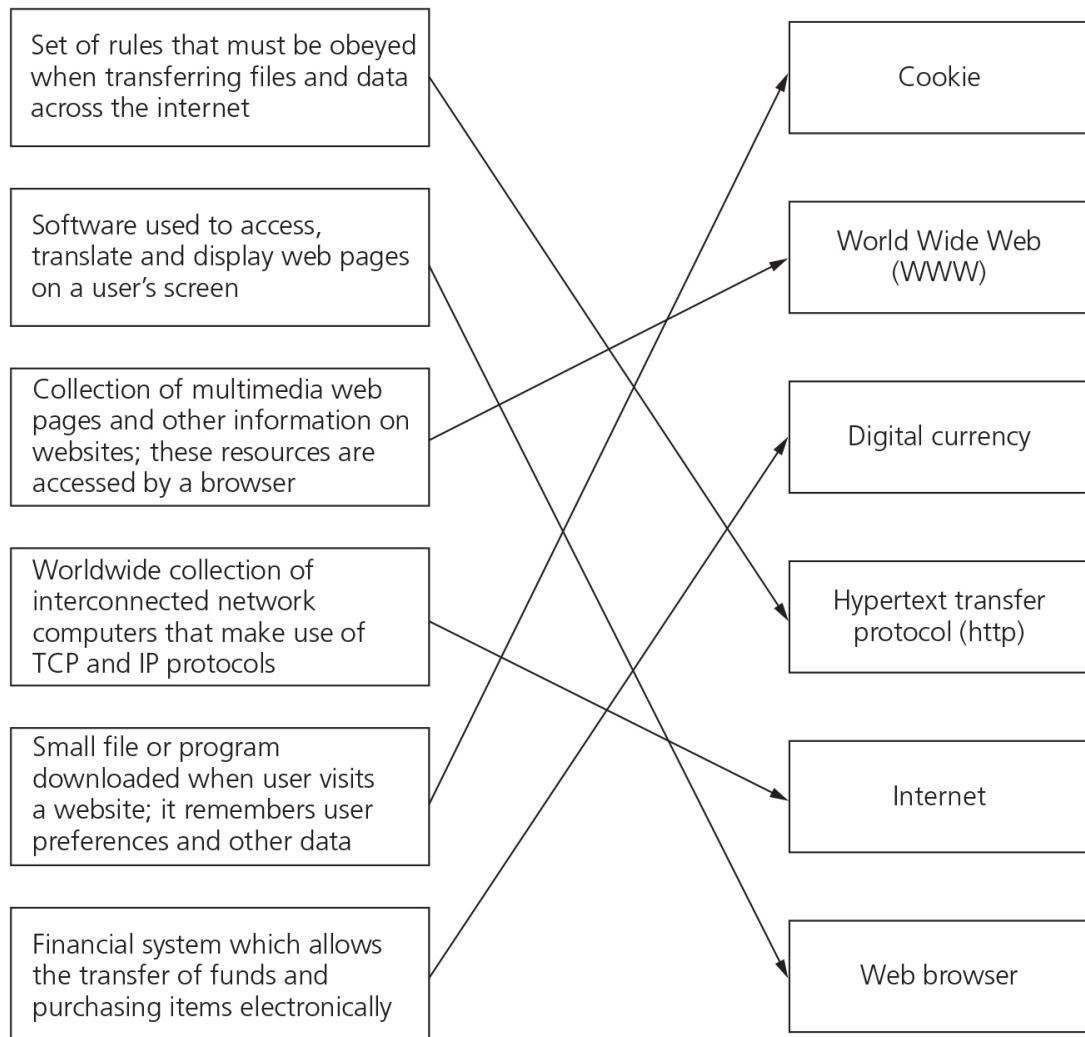
- 1
 - a two-step verification
 - b proxy server
 - c privacy settings
 - d SSL
 - e firewall
 - f DNS
 - g biometrics
 - h authentication
 - i social engineering
 - j pharming
- 2 A browser requests an SSL/TLS digital certificate from the website it is trying to access. This is sent from the web server to the user's browser, containing a public key which can be used to authenticate the website. Once the digital certificate is authenticated, the browser lets the web server know, and secure communication can begin.
- 3
 - a Three from:
 - Are there spelling or grammar errors?
 - Does the email FROM address look like it is real?
 - Is the domain name in the email address spelt correctly, or does it exactly match the real company?
 - Is the email addressed to you?
 - Is there a clickable link? If so, what address does the link point to when you hover over it? (DO NOT CLICK THE LINK!) It may not match the expected website.
 - If it claims to be from a real company, log into your account with the company without using the link.
 - b Three from:
 - poor spelling and grammar
 - not addressed to the user
 - WW and Co does not look like a real company
 - the email address which this is from does not look like a real or reputable company
 - there is no further information in the email, such as details about the company itself
 - reputable companies would not expect a customer to click on a link in an email to pay an account
 - does not look professional

Answers to exam-style questions

- 1 a i** A cookie is a small file stored on a user's computer; they are sent by a website to a browser when a user visits the site.
- ii** Session cookies are stored in a temporary memory on the computer; they don't collect any information from the user's computer and don't identify the user; they cease to exist when the browser is closed or the user terminates the session with the website. Persistent cookies are stored on the hard drive of the user's computer; they remain stored until an expiry date is reached or the user deletes the cookie; they remain in operation on the user's computer even when the browser is closed or the website session has been terminated; they remove the need to type in login details every time a website is visited.
- iii** Three from:
- allows a website to remember a user's email address and invoice details so these don't have to be typed in each time a user visits a website, which allows a faster log in
 - serves as a memory, enabling a website to recognise users every time they visit the website
 - saves users items in a virtual shopping basket
 - tracks internet habits and user's website history/favourites (bookmarks)/recommendations made
 - stores user preferences
 - allows progress in online games/quizzes to be stored
 - allows social networking sites to recognise certain preferences and browsing history
 - allows different languages to be used on web pages automatically/personalised web pages stored
 - indicates if customer is new or returning
- 2 a •** Spyware is software that gathers information by monitoring activity on a user's computer. It can gather information by finding information such as bank account details and passwords stored on the computer or within web browser cookies. Spyware called keyloggers can record what a user types and which websites they visit. The gathered information is then sent back to the person who created or sent the software, which can be used for fraudulent purposes.

- b** Two step verification – extra security that means that even if passwords are stolen criminals cannot gain access to the bank account. It requires some biometric identification – this would limit access to specified employees only.
- c**
 - checks for any unusual network traffic and blocks access to the local network
 - can be used to only allow specified external connections, so that all traffic is blocked apart from these specified sites
 - any known undesirable sites can be added to a list of blocked network sites
- d**
 - passwords/biometrics
 - levels of access
 - proxy servers
 - physical security methods – e.g. PCS in locked rooms, etc.

3



4 a Order: C, E, A, B, D

- b
- student places finger on scanner
 - map of fingerprints taken and converted to digital image
 - data sent to a computer
 - fingerprint map compared to those stored
 - if recognised, the database is updated to show student present (including time and date of entry)
 - on leaving, the same process is carried out and the database is again updated to show the date/time student left the school premises

5 a Four from:

- instant messaging – links embedded in messages to fake websites
 - scareware – pop-up message claiming, for example, user's computer is infected with a virus and user downloads the fake virus checker
 - emails/phishing – apparently genuine emails sent out which contain links to fake websites
 - baiting – cybercriminal leaves malware-infected memory stick so that it can be easily 'found'; once inserted into a computer, the malware is transferred
 - phone calls – so-called IT professional calls a user on their mobile phone claiming their device has been compromised and they are advised to download software which lets the cybercriminal take over their device
- b
- Stage 1 – victim is identified and necessary information is gathered
 - Stage 2 – victim is targeted by the chosen method
 - Stage 3 – attack on victim takes place and information is gathered or disruption to their system done
 - Stage 4 – when cybercriminal has achieved their goal, all traces of the attack are removed
- c
- i digital currency exists purely in a digital format; there is no physical money (fiat currency); digital currency involves electronic transfer of funds
- ii if a hacker accessed a block and changed anything (for example, an amount of money) ...
- ... the hash value of the block would change
 - this means that all blocks beyond the hacked block would now be invalid because the chain would be broken
 - use of 'proof-of-work' is necessary before a block can be added to the chain ...
 - ... thus preventing a hacker quickly creating a new block to complete the chain

6 a Used for layout of content – this structure is defined by HTML tags, e.g. for each heading and paragraph text.

b browser

Internet Protocol address (IP address)

web server

Hypertext Markup Language (HTML)

7 a SSL is a security protocol:

- it encrypts any data that is sent
- it uses/sends digital certificates ...
- ... which are sent to the user's browser ...
- ... that contains the gallery's public key ...
- ... that can be used to authenticate the gallery
- once the digital certificate is authenticated, the transaction can begin

b

Statement	True (✓)	False (✓)
Firewalls are only available as hardware devices		✓
Firewalls allow a user to set rules for network traffic	✓	
Firewalls will automatically stop all malicious traffic		✓
Firewalls only examine traffic entering a network		✓
Firewalls encrypt all data that is transmitted around a network		✓
Firewalls can be used to block access to certain websites	✓	

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6 Automated and emerging technologies

Answers to activities

6.1

- a**
- sensors to detect the presence of a bottle are essential
 - level sensors in mixing tank to ensure required volume maintained
 - may be other sensors such as temperature and pH
 - bottle arrives at bottling station
 - sensor sends data back to central computer continuously
 - if bottle is present, computer sends signal to actuator to open valve to allow liquid to fill bottle
 - level sensor used to check correct amount of liquid added
 - once level sensor reading indicates bottle is full, a signal is sent to computer which then sends a signal to an actuator to close the valve
 - computer then sends signal to actuator to open valve and add measured amount of carbon dioxide gas
 - after two seconds, the gas valve is closed
 - next empty bottle arrives and process continues
 - bottle now moves to capping station
 - again sensor detects bottle and computer sends signal to robot arm to place cap on bottle and tighten it
 - then robot arm retracts
 - bottle moves to labelling station
 - sensor detects bottle and computer sends signal to actuator to rotate table through 360° as label is glued to bottle
 - whole process is continuous
- b** Advantages:
- faster than a human operator to take any necessary action

- more consistent – each bottle will have almost exactly the same amount of each ingredient
- less expensive over time as it can replace most of the workforce who would need to monitor the process 24 hours a day
- less wastage and more efficient use of materials
- higher productivity

Disadvantages:

- expensive to set up
- needs considerable testing
- possible for a set of conditions to occur that were never considered during testing
- automated systems always need enhanced maintenance which can be expensive
- any computerised system is subject to cyberattacks

6.2

- needs camera to pick up parallel lines in parking spot
- extra data needs to be width and length of gap (infrared and ultrasonic sensors can be used to find these values)
- data needs to indicate that no vehicle is already in the parking space
- parallel parking will use cameras which will search out white (parallel) lines and ensure that the distance between vehicle and white lines is consistent

6.3

- the computer takes readings from sensors in the car's front bumper ...
- ... to calculate the distance between the car and the vehicle in front
- the computer takes additional sensor readings to determine the speed of the car
- the computer calculates required safe braking distance
- data from the (given) table is then used and the computer determines if the calculated braking distance is greater than the data in the table
- ... if braking distance is greater than safe distance (in table) then brakes are applied and accelerator/throttle also reduced and warning signal/light produced
- if distance between vehicles is OK then no action taken

6.4

Students should produce a table showing all the advantages and disadvantages of using an automated weather station. This will require research using the internet or a local weather station (if it exists). No answer has been supplied because this answer will vary from country to country and students will need to make their own judgements.

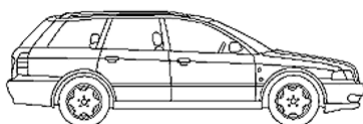
6.5

- 1 • data gathered automatically from various locations around the world ...
- ... could all be collated over time and patterns of global weather changes automatically generated
- automatic projection of key weather data into the future to predict weather could then be checked for its accuracy (by comparison of predicted weather with ACTUAL weather)

- results of this comparison would allow the climate change model to be modified
- 2**
- three sensors needed: temperature, light and moisture
 - all three sensors send data continuously to a microprocessor
 - data would need to be converted to digital (using ADC) if necessary
 - if temperature readings $<$ pre-set value, the microprocessor sends a signal to the actuators ...
 - ... to switch on the heating system/close the windows
 - if temperature readings $\geq$ pre-set value, the microprocessor sends a signal to the actuators ...
 - ... to operate motors to open windows (and switch off heaters)
 - if light level readings $<$ pre-set value, the microprocessor sends a signal to the actuators ...
 - ... to operate motors to open blinds (and turn on artificial lighting)
 - if light level readings $\geq$ pre-set value, the microprocessor sends a signal to the actuators ...
 - ... to operate motors to close blinds (and turn off artificial lighting)
 - if moisture level readings $<$ pre-set value, the microprocessor sends a signal to the actuators ...
 - ... to switch on water pump to irrigate soil
 - if moisture level readings $\geq$ pre-set value, the microprocessor sends signals to the actuators ...
 - ... to switch off water pumps
 - process is continuous
- 3**
- cameras read road signs (60 kph) and software converts the images into a text file
 - ... which allows the microprocessor to interpret road signs
 - if the car's speed $>$ 60 kph, the microprocessor sends a signal to the actuators ...
 - ... to operate motors to apply the brakes and/or reduce throttle position
 - if the car's speed $<$ 60 kph, the microprocessor sends a signal to the actuators ...
 - ... to operate motors to increase throttle position
 - sensors in the car's bumpers/fenders will ensure a safe distance between car and vehicle in front is maintained (by applying brakes and altering throttle as necessary)
 - the Sat Nav calculates the car's position on a map
 - data is sent to a central computer which checks optimum driving conditions
 - if present driving is not the same as stored driving parameters, then computer sends signals to actuators to change gears and apply brakes as necessary

4 Parts **i** and **ii**

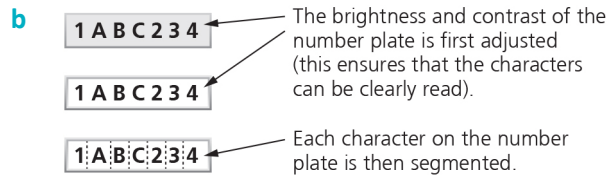
Step 1



A sensor detects a vehicle and sends a signal to instruct a camera to capture an image of the front of the vehicle (often an infrared camera is used to give a clearer image and for use at night).

Step 2

- a** An algorithm is used to locate and isolate the number plate from the image taken by the camera. This algorithm also takes into account the size of the number plate and any damage or orientation.



- c** Each character is then recognised using optical character recognition (OCR) software. The characters are converted into a string of editable text by the software.
- d** This text string is then stored on a database.

Step 3

Once all of this has happened, the car park barrier is raised and the motorist is issued with a ticket. The ticket shows the date and time of entering the car park.

Step 4

When the motorist returns to the car park, they insert their ticket into a machine which calculates the car park charges. The payment is registered on the database.

The motorist then drives to the exit barrier and the ANPR system again reads the number plate and checks its database. If the number plate is recognised (and payment made), the exit barrier is raised.

iii Advantages:

- no need to employ car park security guards which saves money
- much faster system than having to check a ticket at the exit; car parks can issue tickets on entry, but this ticket is simply used for payment purposes by the motorist before leaving the car park and isn't used at the exit since payment will now be linked to the number plate on the car
- can be used to automatically control the entry and exit to a car park or private roads
- can be used as a security system; preventing illegal parking and preventing unauthorised access to private car parks

Disadvantages:

- there is a lack of manned security car park surveillance which could lead to vandalism (and other crimes) since nobody is checking on a regular basis; CCTV is often used, but this is often just used 'after the event'
- invasion of privacy issues due to the recording of driver's number plates
- damaged or very dirty number plates won't be recognised by the system
- number plate cloning; the ANPR system only recognises the number plate and not the car, so it is possible for a car to be fitted with a cloned number plate thus by-passing car park security, for example

6.6

Each student should produce a presentation in the form of a table which highlights the advantages and disadvantages of immersive games consoles. The table should include research into which sensors are used (for example, accelerometers, proximity, magnetic field and pressure), together with a description of the function of each sensor identified.

6.7

- 1 a i flow sensors, temperature sensors
 ii light sensors, infrared/motion sensors
 iii infrared/motion sensors, cameras, pressure sensors

b i Advantages, two from:

- faster response to non-optimum conditions
- more precise monitoring
- greater consistency of product(s)
- 24/7 monitoring

Disadvantages:

- expensive to set up initially and to maintain
- may not respond well to unusual conditions

ii Advantages, two from:

- saves money on unnecessary lighting
- completely automated system
- better control of lighting

Disadvantages, two from:

- expensive to set up
- needs expertise to program the lights exactly as required
- may require specialist equipment

iii Advantages, two from:

- much safer
- fewer staff needed
- fewer delays caused by people pulling emergency cord if someone has an item stuck in the door

Disadvantages, two from:

- still requires human monitoring
- expensive to set up and implement
- requires extensive testing to make sure it works

2

Statement	True	False
Automated systems lead to less consistent results or less consistent products		✓
Automated systems are more expensive to set up than traditional manual systems	✓	
Automated systems would be quickly overwhelmed by the amount of data presented to it		✓
Automated systems are inherently less safe than manual systems		✓
Automated systems generally require enhanced maintenance when compared to manual systems	✓	
Automated systems allow processes to run at optimum conditions at all times	✓	
Software failures, due to unforeseen conditions, are unlikely to impact on an automated system		✓
Automated systems will react more quickly to unusual process conditions	✓	

6.8

Students to produce a table showing all the advantages and disadvantages of the use of robots in agriculture. This may involve research using:

- the chapters in this book
- the companion Information and Communication Technology (ICT) book
- the internet.

Typical advantages might include:

- more accurate
- much faster
- less waste (i.e. picking fruit before it is ripe)
- can work 24 hours a day

Typical disadvantages might include:

- expensive to implement
- will take lots of time to train or program to complete tasks successfully
- different conditions or plants may require more retraining, casual workers will lose their work

6.9

1 Examples:

- harvesting/picking of vegetables and fruit
- weed control
- phenotyping (plant growth and health)
- seed-planting and fertiliser distribution
- autonomous labour-saving devices

Harvesting and picking:

- robots have been designed to do this labour-intensive work; they are more accurate (only picking ripe fruit, for example) and much faster at harvesting
- for the reasons above, this leads to higher yields and reduces waste – for example, Vegebot (Cambridge University) uses cameras to scan, for example, a lettuce and decide whether or not it is ready to be harvested
- a second camera in Vegebot (near the cutting blades) guides an arm to remove the lettuce from its stalk with no damage.

Weed control:

- weed management robots can distinguish between a weed and crop using AI
- examples of weed control robots are being used in France (by Mouton Rothschild) to remove weeds between grape vines in their vineyards; this saves considerably on labour costs and improves vine growth
- weed control robots use GPS tracking to stay on course to move along the rows of vines and remove the weeds; a weed removal blade is operated by an actuator under the control of the controller (microprocessor) in the robot
- very often a drone (flying robot) is used first to do an aerial view of the vineyard, so that a programmed course of action can be produced, which is then sent to the weed control robot's memory.

Phenotyping:

- phenotyping is the process of observing physical characteristics of a plant in order to assess its health and growth
- robots designed to do phenotyping are equipped with sensors (including spectral sensors and thermal cameras) that can create a 3D image/model of the plant, thus allowing it to be monitored for health and growth
- machine learning is used to recognise any issues with leaves (for example, if they have a blight or have the wrong colour) so that the robot can convey this back to the farmer
- these robots are much more accurate and faster at predicting problems than when done manually.

Seed-planting and fertiliser distribution:

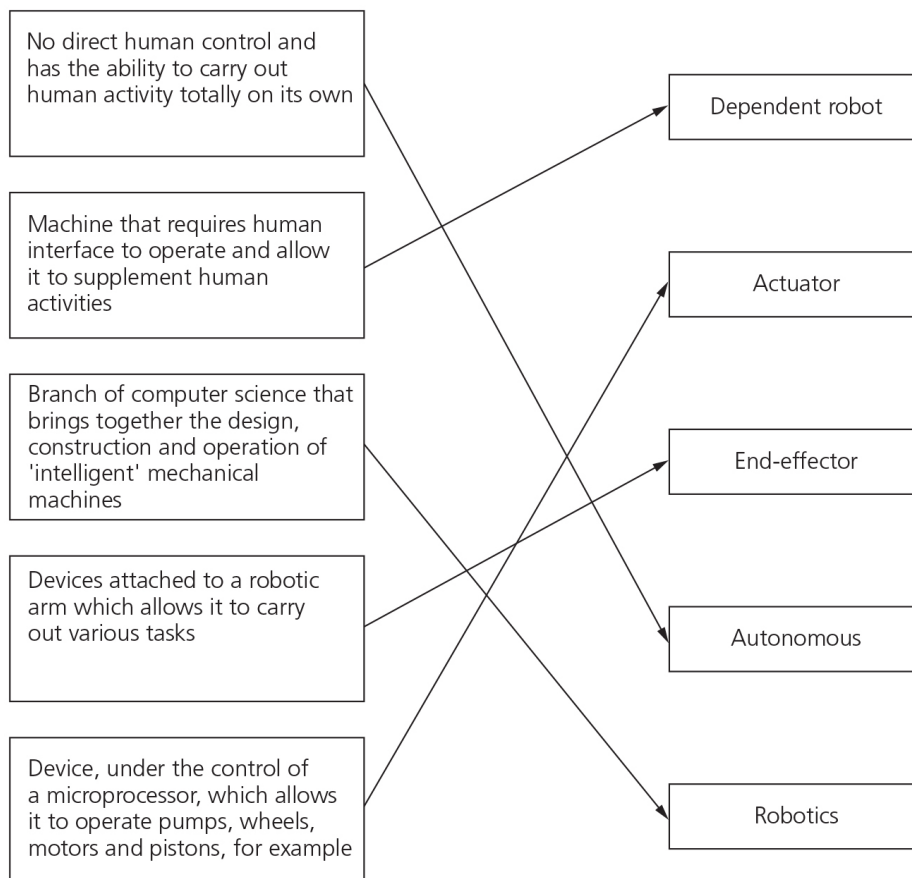
- drones (flying robots) can produce an aerial image of a farm sending back a 'bird's eye view' of the crops and land
- they allow seed-planting to be done far more accurately
- they also allow for more efficient fertiliser-spreading to reduce waste and improve coverage (this is much more efficient than conventional crop spraying)
- drones can also be used in cloud seeding where the drone can add silver iodide crystals to a cloud forcing it to give up its rainwater
- the drones use a very complex camera system to target seeding and allow fertiliser spraying

Autonomous labour-saving devices:

Several of the devices described above could be referred to as autonomous. The following list summarises some of the devices which can work independently of humans:

- grass mowers/cutters
- weeding, pruning and harvesting robots
- seeding robots
- fertiliser spraying
- All of these devices use sensors and cameras to go around obstacles, or they can even be programmed to 'go to sleep' if the weather turns bad.

2



6.10

Essay produced by student on robotics in entertainment including their advantages and disadvantages.

6.11

1

Advantages	Disadvantages
- robots are capable of working in conditions which may be hazardous to humans - robots work 24/7 without the need to stop - robots are less expensive in the long run (since there will be fewer salaries to pay) - robots are more productive than humans (higher productivity) - although not necessarily more accurate, robots are more consistent - robots are better suited to boring, repetitive tasks than humans (therefore less likely to make mistakes) - there will be less cost in heating and lighting (robots don't need good light or warmth)	- robots can find it difficult to do 'non-standard' tasks (for example, windscreen being fitted to a car is cracked) - robots can lead to higher unemployment amongst manual labourers - there is a risk of deskilling when robots take over certain tasks (for example, welding and paint spraying) - factories can now be moved to anywhere in the world where operation costs are lower (leading again to unemployment in some countries) - robots are expensive to buy and set up in the first place

2 Ability to sense its surroundings – two from:

- sensors gather data from surroundings
- allowing robot to recognise its immediate surroundings ...
- ... giving it the ability to determine size, shape, weight, etc.

Ability to move in some way – two from:

- make use of wheels, cogs, pistons, gears, etc. to carry out turning, twisting, moving, gripping, lifting, etc.
- mechanical structures made up of motors, hydraulic pipes, actuators and circuit boards
- makes use of electronic components
- uses end-effectors to carry out different tasks

A perceived intelligence:

- have a controller which determines the action to be taken to perform a task
- controllers can be programmed to allow the robot to do certain tasks

3 autonomous

sensors

cameras

actuators

light detecting and ranging/radar and ultrasonics

lasers

4 sensors

microprocessor

environment

end-effectors

controller

programs

repetitive

adaptive

- 5 a**
- sensors can use radar/sonar to detect objects under the sea
 - cameras take images which allow device to be recognised
 - if ship recognised, then microprocessor sends signals to actuators to operate digital cameras
 - actuators also used to operate electric motors in undersea robots to allow them to move around the shipwreck
- b i**
- sensors can be used to recognise samples
 - data sent to microprocessor after digital conversion (using ADC)
 - if rock samples recognised, microprocessor sends signals to actuators to ...
 - ... stop robot's motors
 - ... to operate 'claws' to take samples of rock
- ii** camera used to recognise terrain, recognise potential rock samples and to allow manoeuvring safely
- c** Advantages include:
- harsh environment which is dangerous to humans
 - more precise than humans
 - can operate non-stop
 - no need for any humans to be present (useful on long journeys, such as visit to Mars which takes several years)
- Disadvantages include:
- expensive devices to design, test and manufacture
 - difficult to repair such devices remotely if they break down
 - as with any robot, they are limited to the functions programmed in (although some ability to reprogram will exist – they may not have the necessary attachments to carry out a newly required task)
- d** Could be used in defusing bombs, clean-up operations in nuclear plants (for example, Chernobyl), etc.

6.12

- a** knowledge base
- b** explanation system
- c** expert system shell
- d** robotics
- e** rules base

6.13

1 artificial intelligence

cognitive
expert systems
explanation system
inference engine
search engine
inference rules

- 2 a
- Is there anything showing on the screen?
 - Can sound be heard through internal loud speakers?
 - If external speakers are plugged into another computer, do they work?
 - Are the connections between the speakers and the computer OK?
(or any series of logical questions)

- b
- oil and mineral prospecting
 - diagnosis of patient's illness
 - tax and financial calculations
 - strategy games, such as chess
 - logistics (parcel deliveries)
 - identification of plants, animals and chemicals (for example)

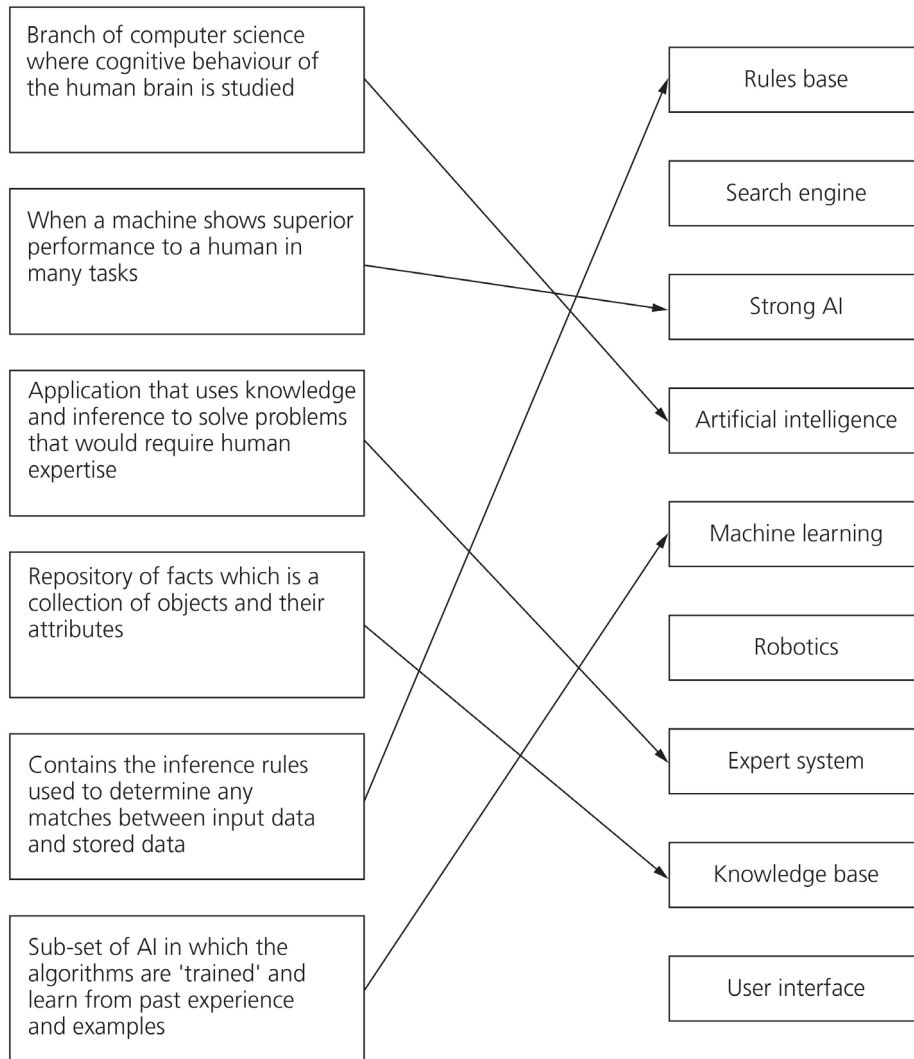
- 3 a
- sub-set of AI
 - algorithms are trained and learn from past experiences and examples
 - offer fast and accurate outcomes due to very powerful processing capability
 - ability to manage and analyse considerable volumes of complex data
 - machines learn through data acquisition to solve problems
 - can make decisions without being programmed to do so

b

AI	Machine learning
represents simulated intelligence in machines	this is the practice of getting a machine to make decisions without being programmed to do so
the aim is to build machines which are capable of thinking like humans	the aim is to make machines that learn through data acquisition, so that they can solve new problems

- c
- use web scraping to find key data in websites (search bots)
 - redundant data (collaboration filtering) removed
 - relationship between key words typed in and key words following collaboration filtering is used to train the model
 - these searches are evaluated to further refine the search engine model

4



Answers to exam-style questions

1 a Advantages:

- robots don't need good lighting or working temperatures
- robots can work 24/7
- lower wage bill in the long term
- higher productivity
- fewer errors made
- leaves humans to do more skilled work

Disadvantages:

- expensive to maintain and repair
- expensive to set up initially
- need to pay redundancy money to workers

b

- sensors (proximity) to detect presence of humans
- on sensing a human, the robot will automatically stop its activity
- verbal warning made by robot to indicate its presence

- 2
- accelerometers
 - to measure and respond to tilting of device
 - to respond to forward/backward and side-to-side movement
 - proximity sensors
 - used in smart touch pads ...
 - ... to detect hand/finger positions
 - embedded accelerometers and other sensors in game consoles
 - allow increased human interaction with the game ...
 - ... giving a more immersive games experience
- 3 a
- data from proximity sensor compared to pre-set value
 - pre-set value shows minimum distance
 - if sensor value is the same/lower
 - then the microprocessor sends a signal to actuators to stop the robot
 - if sensor value higher
 - then no action is taken
 - whole process is continuous
- b i
- light sensors in offices
 - light sensors and infrared/motion sensors in offices and corridors
 - if a person is detected entering an office or corridor, a signal is sent by microprocessor to turn on the lights
 - if light sensor value < stored value for office the microprocessor sends signal to turn on lights
- ii Advantages:
- less expensive, because lights only on when office or corridor occupied
 - longer bulb life because switched on for less time
 - possible to program light displays if required
- Disadvantages:
- expensive to set up initially
 - will require more maintenance than a manual system

4 a

Task	Control	Monitor	Expert system
automatic control of the atmosphere in a large greenhouse (to ensure ideal growing conditions)	✓		
self-parking system in a car	✓		
playing a strategy game, such as chess			✓
an automatic weather station sending data to pilots flying near an airport		✓	
a 24/7 check on patients in an intensive care unit of a hospital		✓	
automatic system used to diagnose the illness of a patient			✓
automatic closing and opening of the doors on a train	✓		
fault identification in electronic devices together with suggested actions to rectify the fault			✓

b Four from:

- explanation system informs user of the expert system's reasoning behind its conclusions and recommendations
- inference engine is a type of search engine used in an expert system that examines the knowledge base for information that matches the queries
- knowledge base is a repository of facts which contain all the knowledge of the expert system; it is composed of a collection of objects and attributes
- rules base is a collection of inference rules used to draw conclusions, based on inputs
- user interface is a method whereby the expert system interacts with the user; the interaction can be through dialogue boxes or command prompts; questions are often simple Yes/No responses and are based on responses to previous questions

5 a

- sensors (infrared, pressure) indicate the presence of a person trying to leave/enter the train
- cameras can also be used to check that nobody is trapped in the doors
- sensors send data back to a microprocessor ...
- ... after being converted to digital (using an ADC)
- camera data is put through an algorithm which determines if a person is trapped in a door
- if camera or sensor data indicate a trapped person ...
- ... a signal is sent to actuators to operate motors to open doors/halt the closing of doors
- if sensors and camera detect no presence of passenger in door area, ...
- ... signals are sent to actuators to operate motors to close doors
- the train is then able to move

- b**
 - can be fully automatic/no need to manually check every door
 - it is safer since no one can get trapped and train can't move off ...
 - ... until all sensors indicate no passenger presence
 - faster response to 'dangerous' situations
- c**
 - allows driver to make a manual check on whether it is safe to move off (a double safety check)
 - gives passengers confidence if they know there is a manual back-up safety system
- d**
 - i** autonomous means driverless, no human operators

ii Sensors:

- cameras to recognise signals on the track
- radar and ultrasonics to sense the environment and build up a 3D picture of surroundings
- use of LiDaR, laser systems and GPS to track position

Actuators:

- to apply brakes to stop train
- to operate accelerator to control speed of train
- to open and close doors

iii

Advantages of autonomous trains	Disadvantages of autonomous trains
improves the punctuality of the trains	ever-present fear of hacking into the vehicle's control system
reduced running costs (fewer staff are required)	system doesn't work well with very busy services (at the moment)
improves safety because human error is removed	high capital costs and operational costs initially (that is, buying the trains, expensive signalling and control equipment and the need to train staff)
minimises energy consumption since there is better control of speed and an optimum service requires less energy (trains stuck in stations still use energy)	ensuring passenger behaviour is acceptable particularly during busy times (for example, jamming doors open on trains, standing too near the edge of platforms, and so on)
possible to increase the frequency of trains (automated systems allow for shorter times between trains)	passenger reluctance of the new technology
easier to change train scheduling (for example, more trains during busier times)	no drivers mean there will be a need for CCTV to monitor railway stations

6 a Ability to sense their surroundings:

- this is done via sensors (such as light, pressure, temperature, acoustic, and so on)
- sensors allow a robot to recognise its immediate environment and gives it the ability to determine things like size, shape or weight of an object, detect if something is hot or cold, and so on; all sensor data is sent to a microprocessor or computer

Have a degree of movement:

- they can make use of wheels, cogs, pistons, gears (etc.) to carry out functions such as turning, twisting, moving backwards/forwards, gripping or lifting
- they are mechanical structures made up of many parts (for example, motors, hydraulic pipes, actuators and circuit boards)
- they contain many electrical components to allow them to function
- can make use of end effectors (different attachments to allow them to carry out specific tasks such as welding, spraying, cutting or lifting)

Apparent intelligence:

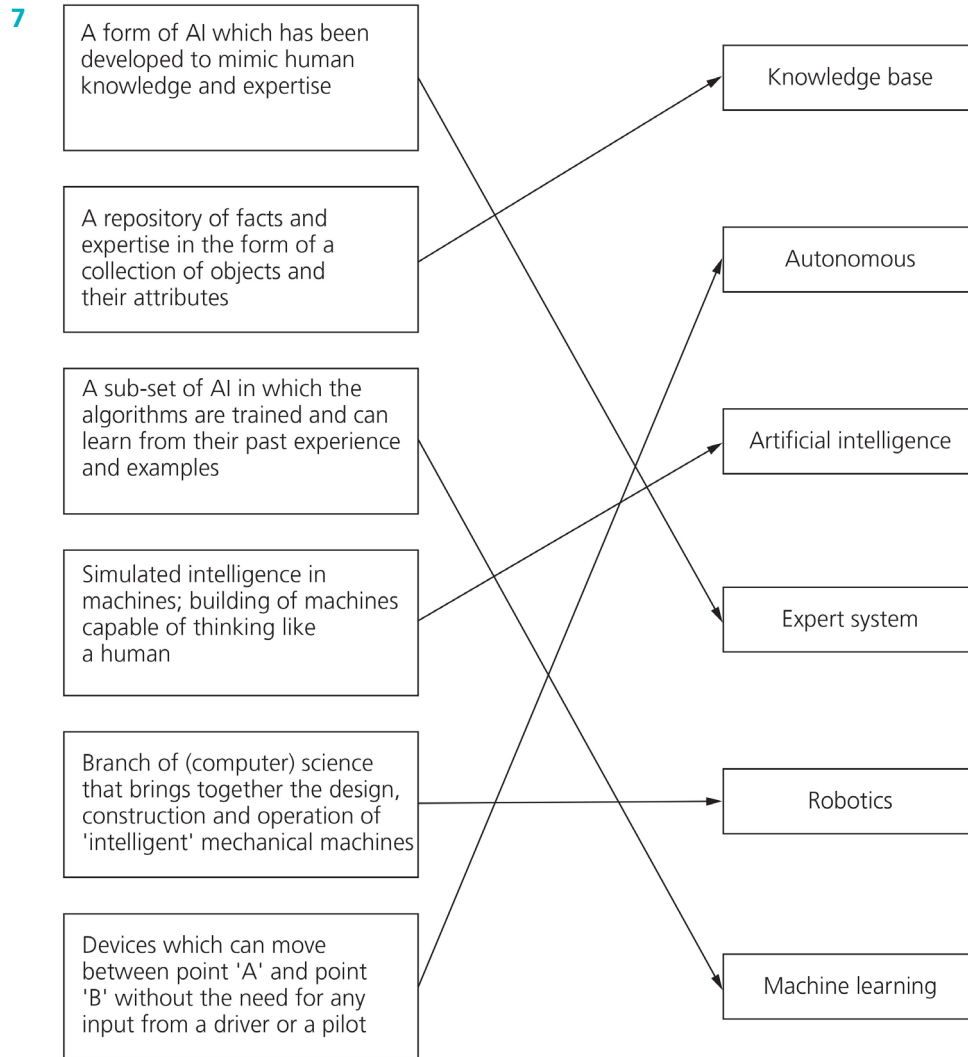
- they have a 'brain' known as a controller which determines the action to be taken to perform a certain task (the controller relies on data sent from sensors or cameras, for example)

b Independent robots:

- have no direct human control (they are said to be autonomous; for example, an autonomous vehicle)
- can replace the human activity totally (no human interaction is required for the robot to function fully)

Dependent robots:

- have a human interfacing directly with the robot (the human interface may be a computer or a control panel)
- can supplement, rather than totally replace, the human activity (for example, in a car assembly plant where both humans and robots work together to produce a car)



- 8
- a end effector
 - b inference engine
 - c autonomous
 - d machine learning
 - e automated systems
 - f artificial intelligence
 - g chatbots
 - h search bot/web crawler
 - i robotics
 - j cognitive
 - k attributes
 - l knowledge base
 - m drone
 - n controller
 - o inference rules

Answers to Student's Book

Exam-style questions and sample answers have been written by the authors. References to assessment and/or assessment preparation are the publisher's interpretation of the syllabus requirements and may not fully reflect the approach of Cambridge Assessment International Education. Cambridge International recommends that teachers consider using a range of teaching and learning resources in preparing learners for assessment, based on their own professional judgement of their students' needs.

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7 Algorithm design and problem solving

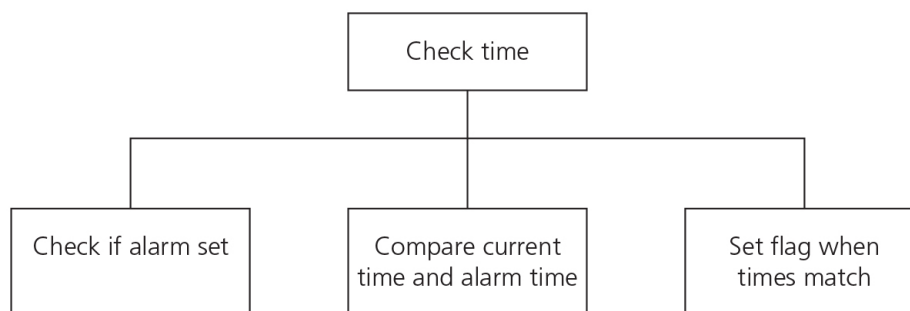
Answers to activities

7.1

For example, for a step tracking app:

- Input a 3-axis accelerometer senses my steps
- Process totals the steps for each day
- Output the total number of steps taken so far
- Storage the number of steps taken each day

7.2



7.3

```

Amount  100
TotalPrice  350
Discount  0.2
FinalPrice  280
Name  "Nikki"
Message  "Hello Nikki"
  
```

7.4

```

OUTPUT "Please enter a mark "
INPUT Mark
  
```

```

IF Mark < 0 OR Mark > 20
    THEN
        OUTPUT "Invalid Mark"
    ELSE
        IF Mark > 9
            THEN
                OUTPUT "Pass"
            ELSE
                OUTPUT "Fail"
        ENDIF
    ENDIF
ENDIF

```

7.5

```

CASE OF Day
    1 : OUTPUT "Sunday"
    2 : OUTPUT "Monday"
    3 : OUTPUT "Tuesday"
    4 : OUTPUT "Wednesday"
    5 : OUTPUT "Thursday"
    6 : OUTPUT "Friday"
    7 : OUTPUT "Saturday"
    OTHERWISE OUTPUT "Error in day number"
ENDCASE

```

7.6

Purpose: Display the largest number

Processes:

- Input two numbers
- Compare the two numbers
- Display the largest number with a suitable message

Output if 7 and 18 input: 18 is the largest

7.7

- 1 Telephone number – presence check (to ensure something is entered), length check (to check the number is not too long or short), or format check (to make sure number matches the style of national telephone numbers).

Pupil's name – presence check (to ensure something is entered) and length check (i.e. between 2 and 30 letters, to check the name is of a reasonable length).

Part number – presence check (to ensure something is entered) and format check (to make sure data matches the character/number style XXX999).

- 2 OUTPUT "Please enter age of child "

```
INPUT ChildAge
```

```
OUTPUT "Please enter height of child in centimetres"
```

```
INPUT ChildHeight
```



```

IF ChildAge < 8 OR ChildAge > 11 OR ChildHeight < 110 OR
  ChildHeight > 150
THEN
  IF ChildAge < 8
    THEN
      OUTPUT "Too young"
    ENDIF
  IF ChildAge > 11
    THEN
      OUTPUT "Too old"
    ENDIF
  IF ChildHeight < 110
    THEN
      OUTPUT "Too small"
    ENDIF
  IF ChildHeight > 150
    THEN
      OUTPUT "Too tall"
    ENDIF
  ELSE
    OUTPUT "Child can ride"
  ENDIF
3 OUTPUT "Please enter your password "
REPEAT
  INPUT Password
  IF LENGTH(Password) > 12 OR LENGTH(Password) < 8
    THEN
      IF LENGTH(Password) > 12
        THEN
          OUTPUT "Password is too long, please re-enter "
        ELSE
          OUTPUT "Password is too short, please re-enter "
        ENDIF
      ENDIF
    ENDIF
  UNTIL LENGTH(Password) <= 12 AND LENGTH(Password) >= 8

```

7.8

Telephone number – the number needs to be the one that is required as well as having the necessary characteristics.

Pupil's name – the name needs to be the one that is associated with the pupil.

Part number – a part number could have the required format but not be for the part intended.

7.9

For example:

Normal test data: 7, 34, 55, 67, 74, 83, 21, 45, 72, 64; Expected result: 54.2

(Other values acceptable.)

7.10

For example:

Erroneous test data: -7, 1001; Expected result: both values rejected

(Other values acceptable.)

7.11

1 Boundary test data for 100 is: 100, 101; Expected result: 100 accepted, 101 rejected

2 Normal test data: 2, 14, 15, 6, 7, 13, 20, 4, 10, 9; Expected result: 10

Normal test data: 17, 13, 15, 15, 14, 16, 2, 8, 7, 6; Expected result: 11.3

Erroneous test data: -7, 23; Expected result: both values rejected

Boundary test data: -1, 0 and 20, 21

Expected results: -1 rejected, 0 accepted and 20 accepted, 21 rejected

7.12

Answer in chapter (page 285).

7.13

A	B	C	X	OUTPUT
0	0	100		Enter your ten values
1	4	4	4	
2	8		8	
3	19		19	
4			17	
5		3	3	
6			11	
7			6	
8		1	1	
9			13	
10			9	19 1

7.14

First	Last	Index	Swap	Temp	Temperature							
					[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
1	8	1	FALSE		35	31	32	36	39	37	42	38
			TRUE	35	31	35						
		2	TRUE	35		32	35					
		3										
		4										
		5	39	39					37	39		
		6										
		7		42							38	42
	7	1	FALSE									
		2										
		3										
		4										
		5										
		6	TRUE	39						38	39	
	6	1	FALSE									
		2										
		3										
		4										
		5										
	5											

7.15

Sample data: -1, -7, -11, -19, -3, -20, -9, -4, -6, -8

A	B	C	X	OUTPUT
0	0	100		
1			-1	
2		-7	-7	
3		-11	-11	
4		-19	-19	
5			-3	
6		-20	-20	
7			-9	
8			-4	
9			-6	
10	0		-3	
				0 -20

The highest value is -1 not zero.

7.16

-1000001(or any number less than this) and 1000001(or any number greater than this) will fail.

7.17

```

A ← 0
OUTPUT "Enter your ten values"
INPUT X
B ← X
C ← X
REPEAT
  INPUT X
  IF X > B
    THEN
      B ← X
  ENDIF
  IF X < C
    THEN
      C ← X
  ENDIF
  A ← A + 1
UNTIL A = 9
OUTPUT B, C

```

A	B	C	X	OUTPUT
0	-97	-97	-97	
1	12390		12390	
2			0	
3			77	
4			359	
5			-2	
6			-89	
7			5000	
8			21	
9			67	
				12390, -97

7.18

0, 26

Expected results rejected – erroneous data

1, 25

Expected results: Your tickets cost 20

Your tickets cost 400 – **extreme data**

9, 10

Expected results: Your tickets cost 180

Your tickets cost 180 – **normal data**

19, 20

Expected results: Your tickets cost 342

Your tickets cost 320 – **normal data**

Extra test data:

0, 1

Expected results: 0 rejected

1 Your tickets cost 20 – **boundary data lower bound**

26, 25

Expected results: 26 rejected

25 Your tickets cost 400 – **boundary data upper bound**

-30, 200

Expected results rejected – erroneous data

7.19

```

1  Change FOR Test ← 1 to 4  // outer loop for the tests
   to FOR Test ← 1 to 2  // outer loop for the tests
   remove 3 : SubjectName ← "English" and 4 : SubjectName ← "IT"
   change FOR StudentNumber ← 1 TO 600
   to FOR StudentNumber ← 1 TO 5

   change SubjectAverage ← SubjectTotal / 600
   to SubjectAverage ← SubjectTotal / 5
   change OverallAverage ← OverallTotal / 2400
   to OverallAverage ← OverallTotal / 10

```

The code should now look like this:

```

// initialisation of overall counters
OverallHighest ← 0
OverallLowest ← 100
OverallTotal ← 0
FOR Test ← 1 to 2  // outer loop for the tests
// initialisation of subject counters
  SubjectHighest ← 0
  SubjectLowest ← 100
  SubjectTotal ← 0
  CASE OF Test
    1 : SubjectName ← "Maths"
    2 : SubjectName ← "Science"
  ENDCASE
  FOR StudentNumber ← 1 TO 5  // inner loop for the students
    REPEAT
      OUTPUT "Enter Student", StudentNumber, "'s mark for ", SubjectName
      INPUT Mark
    UNTIL Mark < 101 AND Mark > -1
    IF Mark < OverallLowest THEN OverallLowest ← Mark
    IF Mark < SubjectLowest THEN SubjectLowest ← Mark
    IF Mark > OverallHighest THEN OverallHighest ← Mark
    IF Mark > SubjectHighest THEN SubjectHighest ← Mark
    OverallTotal ← OverallTotal + Mark
    SubjectTotal ← SubjectTotal + Mark
  NEXT StudentNumber
  SubjectAverage ← SubjectTotal / 5
  OUTPUT SubjectName
  OUTPUT "Average mark is ", SubjectAverage
  OUTPUT "Highest Mark is ", SubjectHighest
  OUTPUT "Lowest Mark is ", SubjectLowest
NEXT Test
OverallAverage ← OverallTotal / 10
OUTPUT "Overall Average is ", OverallAverage
OUTPUT "Overall Highest Mark is ", OverallHighest
OUTPUT "Overall Lowest Mark is ", OverallLowest

```

Test data (sample – yours might be different): 1, 99, -1, 101, 20, 80, 30, 70, 40, 60, 50, 50

2 Trace table:

Test	Student Number	Mark	Subject Name	Subject				Overall				Output
				Av	High	Low	Total	Av	High	Low	Total	
									0	100	0	
1					0	100	0					
	1	1			1	1	1		1	1	1	Enter student 1's mark for Maths
	2	99			99		100		99		100	Enter student 2's mark for Maths
	3	-1										Enter student 3's mark for Maths
		101										Enter student 3's mark for Maths
		20					120				120	Enter student 3's mark for Maths
	4	80					200				200	Enter student 4's mark for Maths
	5	30					230				230	Enter student 5's mark for Maths
				46								Maths Average mark is 46 Highest mark is 99 Lowest mark is 1
2					0	100	0					
		1	70		70	70	70				300	Enter student 1's mark for English
		2	40			40	110				340	Enter student 2's mark for English
		3	60			60	170				400	Enter student 3's mark for English
		4	50			50	220				450	Enter student 4's mark for English
		5	50			50	270				500	Enter student 5's mark for English
				54								English Average mark is 54 Highest mark is 70 Lowest mark is 40
								50				Overall average mark is 50 Overall highest mark is 99 Overall lowest mark is 1

7.20

1 Total $\leftarrow$ 0

FOR Counter $\leftarrow$ 1 TO 10

REPEAT

 OUTPUT "Enter positive number "

 INPUT Number

UNTIL Number > 0

 Total $\leftarrow$ Total + Number

NEXT Counter

Average $\leftarrow$ Total / 10

OUTPUT "Average is ", Average

OUTPUT "Total is ", Total

2 Total $\leftarrow$ 0

Counter $\leftarrow$ 0

REPEAT

 REPEAT

 OUTPUT "Enter positive number, -1 to finish "

 INPUT Number

 UNTIL Number > 0 OR Number = -1

 IF Number <> -1

 THEN

 Total $\leftarrow$ Total + Number

 Counter $\leftarrow$ Counter + 1

 ENDIF

UNTIL Number = -1

Average $\leftarrow$ Total / Counter

OUTPUT "Average is ", Average

OUTPUT "Total is ", Total

3 For 1, I chose a FOR ... NEXT loop because it has a fixed number of iterations.

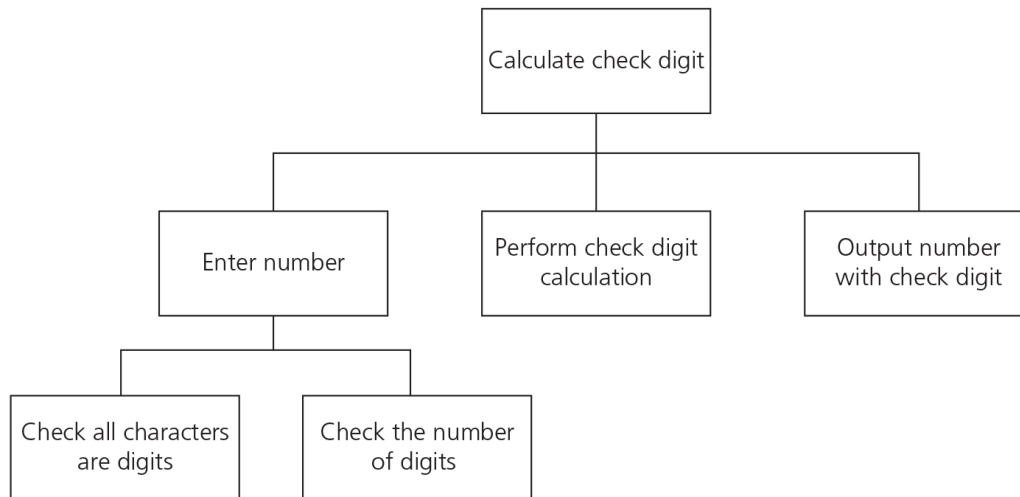
For 2, I chose a REPEAT ... UNTIL loop because there must be at least one number in the list but the number of iterations is unknown.

Answers to exam-style questions

Answers to questions 8 and 9 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

- 1
 - Inputs – the data used by the system that needs to be entered while the system is active
 - Processes – the tasks that need to be performed using the input data and any other previously stored data
 - Outputs – information that needs to be displayed or printed for the users of the system
 - Storage – data that needs to be stored in files on an appropriate media for use in the future

2



3 a Diners – range check 2 to 12 inclusive, presence check and type check of integer

Bill – range check 10 to 500 inclusive, presence check and type check of real

b Test data (sample – yours might be different):

- Normal data: 5 diners and a bill of \$90.00
Expected result: \$18.00
- Normal data: 8 diners and a bill of \$69.00
Expected result: \$8.63

c Test data (sample – yours might be different):

- Abnormal data: 15 diners and a bill of \$5.00
Expected result: both values rejected

d • Boundary data for lower bounds 1 diner and 2 diners and bills of \$9.99 and \$10.00
Expected result: 1 and \$9.99 both values rejected, 2 and \$10.00 \$5.00

- Boundary data for upper bounds 12 diners and 13 diners and bills of \$500.00 and \$500.01
Expected result: 13 and \$500.01 both values rejected, 12 and \$500.00 \$41.67

4 Validation – the automated checking by a program that data is reasonable before it is accepted into a computer system

Verification – checking that data has been accurately copied from another source and input into a computer or transferred from one part of a computer system to another

5 **Name:**

Validation checks:

- presence to ensure name is entered
- length check to ensure there are not too many letters

Verification check – yes to ensure that the name entered is the one intended and there are no typing errors

Date of birth:

Validation checks:

- presence to ensure date of birth is entered
- format check to ensure that all the components of the data are in range
- range check to ensure that the value for the date is realistic

Verification check – yes to ensure that the date entered is the one intended and there are no typing errors

Password:

Validation checks:

- presence to ensure password is entered
- length check to ensure there are the appropriate number of characters
- format check for any special rules, for example at least one digit

Verification check – yes to ensure that the password entered is the one intended and there are no typing errors

Phone number:

Validation checks:

- presence to ensure phone number is entered
- length check to ensure there are the appropriate number of digits/characters
- format check for any special rules, for example must start with +44

Verification check – yes to ensure that the phone number entered is the one intended and there are no typing errors

6 a

Counter	Accept	Reject	Length	Breadth	Size	OUTPUT
0	0	0				
1	1		15	10	150	
2	2		20	17	340	
3		1	32	10		
4		2	30	35		
5	3		30	15	450	
6		3	30	28	840	
7		4	25	25	625	
8	4		20	15	300	
9		5	40	20	800	
10	5		12	10	120	5 5

- b**
- input length and breadth of 10 parcels
 - check length and breadth less than 30 for each parcel
 - calculate size and check size less than 600 for each parcel
 - increment the number of parcels accepted or the number of parcels rejected
 - output number of parcels accepted and number of parcels rejected
- c** Length and breadth must be less than or equal to 30. Size must be less than or equal to 600.
- 7 a**
- Error 1: Counter $\leftarrow$ 1 should be Total $\leftarrow$ 0
 - Error 2: UNTIL Number < 0 should be UNTIL Number > 0
 - Error 3: Total $\leftarrow$ Total + Counter should be Total $\leftarrow$ Total + Number
 - Error 4: Counter $\leftarrow$ Counter + 1 is not required
 - Error 5: OUTPUT Total should be after the end of the loop
 - Error 6: NEXT Number should be NEXT Counter

b

```

Total  $\leftarrow$  0
FOR Counter  $\leftarrow$  1 TO 10
    REPEAT
        OUTPUT "Enter a positive whole number "
        INPUT Number
    UNTIL Number > 0
    Total  $\leftarrow$  Total + Number
NEXT Counter
OUTPUT Total

```

- c** Test data: 1, 0, 67, 10, 22, -10, 3, 7, 2, 8, 1, 9 (sample – yours might be different).

Counter	Number	Total	OUTPUT
		0	
1	1	1	
2	0		
	67	68	
3	10	78	
4	22	100	
5	-10		
	3	103	
6	7	110	
7	2	112	
8	8	120	
9	1	121	
10	9	130	130

- d** Test data: 1, 0, 67, 10, 22, -10, 3, 7, 2, 8, 1, 9 (sample – yours might be different).
 1 extreme, 0 erroneous, 67 normal, 10 normal, 22 normal, -10 erroneous, 3 normal,
 7 normal, 2 normal, 8 normal, 1 normal and 9 normal.

8 a

Number1	Number2	Sign	Answer	OUTPUT
5	7	+	12	12
6	2	-	4	4
4	3	*	12	12
7	8	?	0	
0	0	/	(0)	

- b** CASE Sign OF
 '+' : Answer ← Number1 + Number2
 '-' : Answer ← Number1 - Number2
 '*' : Answer ← Number1 * Number2
 '/' : Answer ← Number1 / Number2
 OTHERWISE Answer ← 0
 ENDCASE

- 9 a** To ensure no changes are made on input, because the details do not have fixed values and there is no clear set of rules that can be used for validation.
 The programmer could ask the contributor to type in each detail twice and then check that both values are equal. If they are not equal, then the input should be rejected.
- b** Email – check for @ to ensure that it is an email address. Password – length check to ensure sufficient characters have been entered.

Answers to Student's Book

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8 Programming

Answers to activities

8.1

Python

```
print("Hello World")
```

Visual Basic

```
Module Module1
    Sub Main()
        Console.WriteLine("Hello World")
        Console.ReadKey()
    End Sub
End Module
```

Java

```
public class HelloWorld
{
    public static void main(String[] args)
    {
        System.out.println("Hello World");
    }
}
```

8.2

Pseudocode

```
DECLARE radius, length : REAL
CONSTANT PI = 3.142
```

Python

```
# Python does not require separate declarations...
# ...values are simply assigned as required
# Python does not differentiate between constants and variables
radius = 10.0
length = 10.0
PI = 3.142
```

Visual Basic

```
Dim radius As Decimal
Dim length As Decimal
Const PI As Decimal = 3.142
```

Java

```
final double PI = 3.142;
double radius;
```

```
double length;
```

8.2a

Python

```
StoreInteger = int(input())  
StoreDecimal = float(input())  
StoreLetter = input()  
StoreText = input()  
StoreBoolean = bool(input())
```

Visual Basic

```
StoreInteger = Integer.parse(Console.ReadLine())  
StoreDecimal = Decimal.parse(Console.ReadLine())  
StoreLetter = Console.ReadLine()  
StoreText = Console.ReadLine()  
StoreBoolean = Bool.parse(Console.ReadLine())
```

Java

```
int StoreInteger = myObj.nextInt();  
double StoreDecimal = myObj.nextDouble();  
char StoreLetter = myObj.next();  
String StoreText = myObj.next();  
Boolean StoreBoolean = myObj.nextBoolean();
```

8.3

Python

```

ConstPi = 3.142
radius = float(input("Please enter the radius of the cylinder "))
length = float(input("Please enter the length of the cylinder "))
volume = radius * radius * length * ConstPi
print("Volume of the cylinder is ", volume)

```

Visual Basic

```

Module Module1
    Public Sub Main()
        Dim radius As Decimal
        Dim length As Decimal
        Dim volume As Decimal
        Const PI As Decimal = 3.142
        Console.Write("Please enter the radius of the cylinder ")
        radius = Decimal.Parse(Console.ReadLine())
        Console.Write("Please enter the length of the cylinder ")
        length = Decimal.Parse(Console.ReadLine())
        volume = radius * radius * length * PI
        Console.WriteLine("Volume of cylinder is " & volume)
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Cylinder
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        final double PI = 3.142;
        double radius;
        double length;

        System.out.println("Please enter the radius of the cylinder ");
        radius = myObj.nextDouble();
        System.out.println("Please enter the length of the cylinder ");
        length = myObj.nextDouble();
        double volume = radius * radius * length * PI;
        System.out.println("Volume of cylinder is " + volume);
    }
}

```


8.4

Python

```
num = int(input("Please enter a number "))
if num >= 100:
    print ("100 or over")
else:
    print ("less than 100")
```

Visual Basic

```
Module Module1
    Sub Main()
        Dim num As Integer
        Console.Write("Please enter a number ")
        num = Integer.Parse(Console.ReadLine())
        If num >= 100 Then
            Console.WriteLine("100 or over")
        Else
            Console.WriteLine("less than 100")
        End If
        Console.ReadKey()
    End Sub
End Module
```

Java

```
import java.util.Scanner;
class Activity_8_4
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        int num;

        System.out.println("Please enter a number ");
        num = myObj.nextInt();
        if (num >= 100){
            System.out.println("100 or over");
        } else {
            System.out.println("Less than 100");
        }
    }
}
```

8.5

Python

```

num = int(input("Please enter a number "))
if num == 1:
    print ("Number equals 1")
elif num == 2:
    print ("Number equals 2")
elif num == 3:
    print ("Number equals 3")
elif num == 4:
    print ("Number equals 4")
else: print ("Invalid number")

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim num As Integer
        Console.Write("Please enter a number ")
        num = Integer.Parse(Console.ReadLine())
        Select Case num
            Case 1
                Console.WriteLine("Number equals 1")
            Case 2
                Console.WriteLine("Number equals 2")
            Case 3
                Console.WriteLine("Number equals 3")
            Case 4
                Console.WriteLine("Number equals 4")
            Case Else
                Console.WriteLine("Invalid number")
        End Select
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_5_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        int num;

        System.out.println("Please enter a number ");
        num = myObj.nextInt();
        switch (num){
            case 1:
                System.out.println("Number equals 1");
                break;
            case 2:
                System.out.println("Number equals 2");
                break;
            case 3:
                System.out.println("Number equals 3");
                break;
            case 4:
                System.out.println("Number equals 4");
                break;
            default:
                System.out.println("Invalid number");
        }
    }
}

```

8.6

Python

```
#count controlled loop
for Counter in range (1, 30, 3):
    print(Counter)
#condition controlled loop
Counter = 1
while Counter < 30:
    print(Counter)
    Counter = Counter + 3
```

Visual Basic

```
Module Module1
    Sub Main()
        Dim Counter As Integer
        'count controlled loop
        For Counter = 1 To 30 Step 3
            Console.WriteLine(Counter)
        Next
        'condition controlled loop
        Counter = 1
        Do
            Console.WriteLine(Counter)
            Counter = Counter + 3
        Loop Until Counter >= 30
        Console.ReadKey()
    End Sub
End Module
```

Java

```
class Activity_8_6_Java
{
    public static void main(String args[])
    {
        int Counter;
        //count controlled loop
        for (Counter = 1; Counter <= 30; Counter = Counter + 3){
            System.out.println(Counter);
        }

        //condition controlled loop
        Counter = 1;
        do {
            System.out.println(Counter);
            Counter = Counter + 3;
        }
        while (Counter <=30);
    }
}
```

8.7

Python

```

Count = 0
TotalSackWeight = 0
SackWeight = float(input("Enter weight of next sack, -1 to finish "))
while SackWeight != -1:
    Count = Count + 1
    TotalSackWeight = TotalSackWeight + SackWeight
    SackWeight = float(input("Enter weight of next sack, -1 to finish "))
print ("Total number of sacks is ", Count)
print ("Total weight of sacks is ", TotalSackWeight)

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim Count As Integer
        Dim SackWeight As Decimal
        Dim TotalSackWeight As Decimal
        Count = 0
        TotalSackWeight = 0
        Console.Write("Please enter weight of next sack ")
        SackWeight = Decimal.Parse(Console.ReadLine())
        While SackWeight <> -1
            Count = Count + 1
            TotalSackWeight = TotalSackWeight + SackWeight
            Console.Write("Please enter weight of next sack ")
            SackWeight = Decimal.Parse(Console.ReadLine())
        End While
        Console.WriteLine("Total number of sacks is " & Count)
        Console.WriteLine("Total weight of sacks is " & TotalSackWeight)
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_7_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        int Count = 0;
        double SackWeight;
        double TotalSackWeight = 0;

        System.out.println("Please enter weight of next sack ");
        SackWeight = myObj.nextDouble();
        while (SackWeight != -1) {
            Count ++;
            TotalSackWeight = TotalSackWeight + SackWeight;
            System.out.println("Please enter weight of next sack ");
            SackWeight = myObj.nextDouble();
        }
        System.out.println("Total number of sacks is " + Count);
        System.out.println("Total weight of sacks is " + TotalSackWeight);
    }
}

```

8.8

Python

```

MyName = input("Please enter your full name ")
print ("There are ", len(MyName), " characters in your name")
print ("The first three characters are are ", MyName[0:3])
print ("Upper Case ", MyName.upper())
print ("Lower Case ", MyName.lower())

```

Visual Basic

```

Module Module1
    Public Sub Main()
        Dim MyName As String
        Console.Write("Please enter your full name ")
        MyName = Console.ReadLine()
        Console.WriteLine("There are " & MyName.Length() & " characters in your
name")
        Console.WriteLine("The first three characters are are " &
MyName.Substring(0, 3))
        Console.WriteLine("Upper case " & UCase(MyName))
        Console.WriteLine("Lower case " & LCase(MyName))
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_8_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Please enter your full name ");
        String MyName = myObj.next();
        System.out.println("There are " + MyName.length() + " characters in
your name");
        System.out.println("The first three characters are " +
MyName.substring(0,3));
        System.out.println("Upper case " + MyName.toUpperCase());
        System.out.println("Lower case " + MyName.toLowerCase());
    }
}

```

8.9

Python

```

Number1 = float(input ("Enter first number "))
Number2 = float(input ("Enter second number "))
print("Add ", Number1 + Number2)
print("Subtract ", Number1 - Number2)
print("Multiply ", Number1 * Number2)
print("Divide ", Number1 / Number2)
print("Raise to the power of ", Number1 ** Number2)
Number3 = int(input ("Enter third number "))
print("Number3 * (Number1 + Number2) ", Number3 * (Number1 + Number2))
print("Number3 * (Number1 - Number2) ", Number3 * (Number1 - Number2))
print("(Number1 + Number2) ^ Number3 ", (Number1 + Number2) ** Number3)

```

Visual Basic

```

Module Module1
    Public Sub Main()
        Dim Number1 As Decimal
        Dim Number2 As Decimal
        Dim Number3 As Integer
        Console.Write("Please enter the first number ")
        Number1 = Decimal.Parse(Console.ReadLine())
        Console.Write("Please enter the second number ")
        Number2 = Decimal.Parse(Console.ReadLine())
        Console.WriteLine("Add " & (Number1 + Number2))
    End Sub
End Module

```

```

        Console.WriteLine("Subtract " & (Number1 - Number2))
        Console.WriteLine("Multiply " & (Number1 * Number2))
        Console.WriteLine("Divide " & (Number1 / Number2))
        Console.WriteLine("Raise to the power of " & (Number1 ^ Number2))
        Console.Write("Please enter the third number ")
        Number3 = Integer.Parse(Console.ReadLine())
        Console.WriteLine("Number3 * (Number1 + Number2) " & (Number3 *
(Number1 + Number2)))
        Console.WriteLine("Number3 * (Number1 - Number2) " & (Number3 *
(Number1 - Number2)))
        Console.WriteLine("(Number1 + Number2) ^ Number3 " & ((Number1 +
Number2) ^ Number3))
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
import java.lang.Math;
class Activity_8_9_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Please enter first number ");
        double Number1 = myObj.nextDouble();
        System.out.println("Please enter second number ");
        double Number2 = myObj.nextDouble();
        System.out.println("Add " + (Number1 + Number2));
        System.out.println("Subtract " + (Number1 - Number2));
        System.out.println("Multiply " + (Number1 * Number2));
        System.out.println("Divide " + (Number1 / Number2));
        System.out.println("Raise to the power of " +
(Math.pow(Number1,Number2)));
        System.out.println("Please enter third number ");
        int Number3 = myObj.nextInt();
        System.out.println("Number3 * (Number1 + Number2) " + (Number3 * (Number1
+ Number2)));
        System.out.println("Number3 * (Number1 - Number2) " + (Number3 * (Number1
- Number2)));
        System.out.println("(Number1 + Number2) ^ Number3 " +
(Math.pow((Number1 + Number2),
            Number3)));
    }
}

```

8.10

Python

```

Number1 = float(input ("Enter first number "))
Number2 = float(input ("Enter second number "))
if (Number1 == Number2):
    print("Both numbers are equal ")
elif (Number1 > Number2):
    print("Numbers are not equal, Number 1 is the largest and Number 2 is the
smallest")
else:
    print("Numbers are not equal, Number 2 is the largest and Number 1 is the
smallest")
Number3 = float(input ("Enter third number "))
if (Number1 == Number2) and (Number2 == Number3):
    print("All numbers are equal ")

```

```

elif Number1 == Number2:
    if Number1 > Number3:
        print("Numbers 1 and 2 are equal largest, Number 3 is smallest")
    else:
        print("Numbers 1 and 2 are equal smallest, Number 3 is largest")
elif Number2 == Number3:
    if Number2 > Number1:
        print("Numbers 2 and 3 are equal largest, Number 1 is smallest")
    else:
        print("Numbers 2 and 3 are equal smallest, Number 1 is largest")
elif Number3 == Number1 :
    if Number3 > Number2:
        print("Numbers 1 and 3 are equal largest, Number 2 is smallest")
    else:
        print("Numbers 1 and 3 are equal smallest, Number 2 is largest")

else:
    print("All numbers are not equal")
    if (Number1 > Number2 and Number2 > Number3):
        print("Number 1 is the largest and Number 3 is the smallest")
    if (Number1 > Number3 and Number3 > Number2):
        print("Number 1 is the largest and Number 3 is the smallest")
    if (Number2 > Number1 and Number1 > Number3):
        print("Number2 is the largest and Number3 is the smallest")
    if (Number2 > Number3 and Number3 > Number1):
        print("Number2 is the largest and Number1 is the smallest")
    if (Number3 > Number2 and Number2 > Number1):
        print("Number3 is the largest and Number1 is the smallest")
    if (Number3 > Number1 and Number1 > Number2):
        print("Number3 is the largest and Number2 is the smallest")

```

Visual Basic

```

Module Module1
    Public Sub Main()
        Dim Number1 As Decimal
        Dim Number2 As Decimal
        Dim Number3 As Decimal

        Console.WriteLine("Please enter the first number ")
        Number1 = Decimal.Parse(Console.ReadLine())
        Console.WriteLine("Please enter the second number ")
        Number2 = Decimal.Parse(Console.ReadLine())
        If (Number1 = Number2) Then
            Console.WriteLine("Both numbers are equal ")
        Else
            Console.WriteLine("Both numbers are not equal")
            If (Number1 > Number2) Then
                Console.WriteLine("Number1 is largest and Number2 is
smallest")
            Else
                Console.WriteLine("Number2 is largest and Number1 is
smallest")
            End If
        End If
        Console.WriteLine("Please enter the third number ")
        Number3 = Decimal.Parse(Console.ReadLine())
        If Number1 = Number2 And Number2 = Number3 Then
            Console.WriteLine("All numbers are equal ")
        ElseIf Number1 = Number2 Then
            If Number1 > Number3 Then
                Console.WriteLine("Numbers 1 and 2 are equal largest, Number 3
is smallest")
            Else
                Console.WriteLine("Numbers 1 And 2 are equal smallest, Number
3 is largest")
            End If
        End If
    End Sub
End Module

```

```

        ElseIf Number2 = Number3 Then
            If Number2 > Number1 Then
                Console.WriteLine("Numbers 2 and 3 are equal largest, Number 1
is smallest")
            Else
                Console.WriteLine("Numbers 2 And 3 are equal smallest, Number
1 is largest")
            End If

            ElseIf Number3 = Number1 Then
                If Number3 > Number2 Then
                    Console.WriteLine("Numbers 1 and 3 are equal largest, Number 2
is smallest")
                Else
                    Console.WriteLine("Numbers 1 And 3 are equal smallest, Number
2 is largest")
                End If

            Else
                If Number1 > Number2 And Number2 > Number3 Then
                    Console.WriteLine("Number1 is largest and Number3 is
smallest")
                End If
                If Number1 > Number2 And Number3 > Number2 And Number1 > Number3
Then
                    Console.WriteLine("Number1 is largest and Number2 is
smallest")
                End If
                If Number2 > Number3 And Number3 > Number1 Then
                    Console.WriteLine("Number2 is largest and Number1 is
smallest")
                End If
                If Number2 > Number1 And Number1 > Number3 And Number1 > Number3
Then
                    Console.WriteLine("Number2 is largest and Number3 is
smallest")
                End If
                If Number3 > Number2 And Number2 > Number1 Then
                    Console.WriteLine("Number3 is largest and Number1 is
smallest")
                End If
                If Number3 > Number1 And Number1 > Number2 And Number3 > Number2
Then
                    Console.WriteLine("Number3 is largest and Number2 is
smallest")
                End If
            End If
        End If
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_10_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Please enter first number ");
        double Number1 = myObj.nextDouble();
        System.out.println("Please enter second number ");
        double Number2 = myObj.nextDouble();
        if (Number1 == Number2) {
            System.out.println("Both numbers are equal");
        } else {
            System.out.println("Both numbers are not equal");
            if (Number1 > Number2) {

```



```

        System.out.println("Number1 is the largest and Number2 is the
smallest");
    } else {
        System.out.println("Number2 is the largest and Number1 is the
smallest");
    }
}
System.out.println("Please enter third number ");
double Number3 = myObj.nextDouble();
if(Number1 == Number2 && Number2 == Number3) {
    System.out.println("All numbers are equal");
}
else if(Number1 == Number2) {
    if (Number1 > Number3) {
        System.out.println("Numbers 1 and 2 are equal largest, Number
3 is smallest");
    } else {
        System.out.println("Numbers 1 and 2 are equal smallest, Number
3 is largest");
    }
}

else if(Number2 == Number3) {
    if (Number2 > Number1) {
        System.out.println("Numbers 2 and 3 are equal largest, Number
1 is smallest");
    } else {
        System.out.println("Numbers 2 and 3 are equal smallest, Number
1 is largest");
    }
}

else if(Number3 == Number1) {
    if (Number3 > Number2) {
        System.out.println("Numbers 1 and 3 are equal largest, Number
2 is smallest");
    } else {
        System.out.println("Numbers 1 and 3 are equal smallest, Number
2 is largest");
    }
}

else {
    if (Number1 > Number2 && Number2 > Number3) {
        System.out.println("Number1 is the largest and Number3 is the
smallest");
    }
    if (Number1 > Number3 && Number3 > Number2) {
        System.out.println("Number1 is the largest and Number2 is the
smallest");
    }
    if (Number2 > Number1 && Number1 > Number3) {
        System.out.println("Number2 is the largest and Number3 is the
smallest");
    }
    if (Number2 > Number3 && Number3 > Number1) {
        System.out.println("Number2 is the largest and Number1 is the
smallest");
    }
    if (Number3 > Number2 && Number2 > Number1) {
        System.out.println("Number3 is the largest and Number1 is the
smallest");
    }
    if (Number3 > Number1 && Number1 > Number2) {
        System.out.println("Number3 is the largest and Number2 is the
smallest");
    }
}
}
}

```

8.11, 8.12 and 8.13**Python**

```

ClassHigh = 0
ClassLow = 100
ClassTotal = 0
NumberOfTests = 5
NumberOfSubjects = 6
ClassSize = 20
for Student in range (1, ClassSize + 1):
    StudentHigh = 0
    StudentLow = 100
    StudentTotal = 0
    for Subject in range (1, NumberOfSubjects + 1):
        SubjectHigh = 0
        SubjectLow = 100
        SubjectTotal = 0
        for Test in range (1, NumberOfTests + 1):
            Mark = int(input("Please enter mark "))
            if (Mark > SubjectHigh):
                SubjectHigh = Mark
            if (Mark < SubjectLow):
                SubjectLow = Mark
            SubjectTotal = SubjectTotal + Mark
        SubjectAverage = SubjectTotal / NumberOfTests
        print ("Average mark for Subject ", Subject, " is ", SubjectAverage)
        print ("Highest mark for Subject ", Subject, " is ", SubjectHigh)
        print ("Lowest mark for Subject ", Subject, " is ", SubjectLow)
        if SubjectHigh > StudentHigh:
            StudentHigh = SubjectHigh
        if SubjectLow < StudentLow:
            StudentLow = SubjectLow
        StudentTotal = StudentTotal + SubjectTotal
    StudentAverage = StudentTotal / (NumberOfTests * NumberOfSubjects)
    print ("Average mark for Student ", Student, " is ", StudentAverage)
    print ("Highest mark for Student ", Student, " is ", StudentHigh)
    print ("Lowest mark for Student ", Student, " is ", StudentLow)

    if StudentHigh > ClassHigh:
        ClassHigh = StudentHigh
    if StudentLow < ClassLow:
        ClassLow = StudentLow
    ClassTotal = ClassTotal + StudentTotal
ClassAverage = ClassTotal / (NumberOfTests * NumberOfSubjects * ClassSize)
print ("Average mark for Class is ", ClassAverage)
print ("Highest mark for Class is ", ClassHigh)
print ("Lowest mark for Class is ", ClassLow)

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim ClassAverage, StudentAverage, SubjectAverage As Decimal
        Dim Student, Subject, Test, Mark As Integer
        Dim ClassHigh, ClassLow, ClassTotal As Integer
        Dim StudentHigh, StudentLow, StudentTotal As Integer
        Dim SubjectHigh, SubjectLow, SubjectTotal As Integer
        ClassHigh = 0
        ClassLow = 100
        ClassTotal = 0
        'Use of constants enables you to easily change the values for testing
        Const NumberOfTests As Integer = 5
        Const NumberOfSubjects As Integer = 6
        Const ClassSize As Integer = 20
        For Student = 1 To ClassSize
            StudentHigh = 0
            StudentLow = 100
            StudentTotal = 0
            For Subject = 1 To NumberOfSubjects

```

```

        SubjectHigh = 0
        SubjectLow = 100
        SubjectTotal = 0
        For Test = 1 To NumberOfTests
            Console.Write("Please enter mark ")
            Mark = Decimal.Parse(Console.ReadLine())
            If Mark > SubjectHigh Then
                SubjectHigh = Mark
            End If
            If Mark < SubjectLow Then
                SubjectLow = Mark
            End If
            SubjectTotal = SubjectTotal + Mark
        Next Test
        SubjectAverage = SubjectTotal / NumberOfTests
        Console.WriteLine("Average mark for Subject " & Subject & " is "
& SubjectAverage)
        Console.WriteLine("Highest mark for Subject " & Subject & " is
" & SubjectHigh)
        Console.WriteLine("Lowest mark for Subject " & Subject & " is
" & SubjectLow)
        If SubjectHigh > StudentHigh Then
            StudentHigh = SubjectHigh
        End If
        If SubjectLow < StudentLow Then
            StudentLow = SubjectLow
        End If
        StudentTotal = StudentTotal + SubjectTotal
    Next Subject
    StudentAverage = StudentTotal / (NumberOfTests * NumberOfSubjects)
    Console.WriteLine("Average mark for Student " & Student & " is " &
StudentAverage)
    Console.WriteLine("Highest mark for Student " & Student & " is " &
StudentHigh)
    Console.WriteLine("Lowest mark for Student " & Student & " is " &
StudentLow)
    If StudentHigh > ClassHigh Then
        ClassHigh = StudentHigh
    End If
    If StudentLow < ClassLow Then
        ClassLow = StudentLow
    End If
    ClassTotal = ClassTotal + StudentTotal
Next Student
ClassAverage = ClassTotal / (NumberOfTests * NumberOfSubjects *
ClassSize)
Console.WriteLine("Average mark for Class is " & ClassAverage)
Console.WriteLine("Highest mark for Class is " & ClassHigh)
Console.WriteLine("Lowest mark for Class is " & ClassLow)
Console.ReadKey()
End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_13_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        double ClassAverage, StudentAverage, SubjectAverage;
        int Student, Subject, Test, Mark;
        int ClassHigh, ClassLow, ClassTotal;
        int StudentHigh, StudentLow, StudentTotal;
        int SubjectHigh, SubjectLow, SubjectTotal;
        ClassHigh = 0;
        ClassLow = 100;
        ClassTotal = 0;
    }
}

```

```

final int NumberOfTests = 5;
final int NumberOfSubjects = 6;
final int ClassSize = 20;

for (Student = 1; Student <= ClassSize ; Student ++ ) {
    StudentHigh = 0;
    StudentLow = 100;
    StudentTotal = 0;
    for (Subject = 1; Subject <= NumberOfSubjects; Subject ++ ) {
        SubjectHigh = 0;
        SubjectLow = 100;
        SubjectTotal = 0;
        for (Test = 1; Test <= NumberOfTests; Test ++ ) {
            System.out.println("Please enter mark ");
            Mark = myObj.nextInt();
            if (Mark > SubjectHigh) {
                SubjectHigh = Mark;
            }
            if (Mark < SubjectLow) {
                SubjectLow = Mark;
            }
            SubjectTotal = SubjectTotal + Mark ;
        }
        SubjectAverage = SubjectTotal / NumberOfTests ;
        System.out.println("Average mark for subject " + Subject + "
is " + SubjectAverage );
        System.out.println("Highest mark for subject " + Subject + "
is " + SubjectHigh);
        System.out.println("Lowest mark for subject " + Subject + " is
" + SubjectLow);
        if (SubjectHigh > StudentHigh) {
            StudentHigh = SubjectHigh;
        }
        if (SubjectLow < StudentLow) {
            StudentLow = SubjectLow;
        }
        StudentTotal = StudentTotal + SubjectTotal;
    }
    StudentAverage = StudentTotal / (NumberOfTests * NumberOfSubjects);
    System.out.println("Average mark for student " + Subject + " is " +
StudentAverage);
    System.out.println("Highest mark for student " + Subject + " is " +
StudentHigh);
    System.out.println("Lowest mark for student " + Subject + " is " +
StudentLow);
    if (StudentHigh > ClassHigh) {
        ClassHigh = StudentHigh;
    }
    if (StudentLow < ClassLow) {
        ClassLow = StudentLow;
    }
    ClassTotal = ClassTotal + StudentTotal ;
}
ClassAverage = ClassTotal / (NumberOfTests * NumberOfSubjects *
ClassSize);
System.out.println("Average mark for class is " + ClassAverage );
System.out.println("Highest mark for class is " + ClassHigh);
System.out.println("Lowest mark for class is " + ClassLow);
}
}

```

8.14**Python**

```
def Stars():
    print("*****")
    print("*****")
    print("*****")
Stars()
```

Visual Basic

```
Module Module1
    Sub Main()
        Stars()
        Console.ReadKey()
    End Sub
    Sub Stars()
        Console.WriteLine("*****")
        Console.WriteLine("*****")
        Console.WriteLine("*****")
    End Sub
End Module
```

Java

```
import java.util.Scanner;
class Activity_8_14_Java
{
    static void Stars()
    {
        System.out.println("*****");
        System.out.println("*****");
        System.out.println("*****");
    }
    public static void main(String args[])
    {
        Stars();
    }
}
```

8.15**Python**

```
def Stars(Number):
    for Counter in range(Number):
        print("*****\n")

Stars(7)
```

Visual Basic

```
Module Module1
    Sub Main()
        Stars(7)
        Console.ReadKey()
    End Sub
    Sub Stars(Number As Integer)
        Dim Counter As Integer
        For Counter = 1 To Number
            Console.WriteLine("*****")
        Next
    End Sub
End Module
```

Java

```
import java.util.Scanner;
class Activity_8_14_Java
{
    static void Stars(int Number)
    {
        for (int Counter = 1; Counter <= Number; Counter++)
        {
            System.out.println("*****");
        }
    }
}
```

```

    }
}
public static void main(String args[])
{
    Stars(7);
}
}

```

8.16

Python

```

def Fahrenheit (Temperature):
    return (1.8 * Temperature + 32)
Answer = Fahrenheit(0)
print (Answer)

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim Answer As Decimal
        Answer = Fahrenheit(0)
        Console.WriteLine(Answer)
        Console.ReadKey()
    End Sub
    Function Fahrenheit(ByVal Temperature As Decimal)
        Return 1.8 * Temperature + 32
    End Function
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_16_Java
{
    static double Fahrenheit (double Temperature)
    {
        return (Temperature * 1.8 + 32);
    }
    public static void main(String args[])
    {
        double Answer = Fahrenheit (0);
        System.out.println(Answer);
    }
}

```

8.17

Python

```

def Test():
    Number1 = 10
    Number2 = 20
    Number3 = 30
    Answer = Number1 + Number2
    print ("Number1 is now ", Number1)
    print ("Number2 is now ", Number2)
    print ("Answer is now ", Answer)

Number1 = 50
Number2 = 100
Answer = Number1 + Number2
print ("Number1 is ", Number1)
print ("Number2 is ", Number2)
print ("Answer is ", Answer)
Test()
print ("Number1 is still ", Number1)
print ("Number2 is still ", Number2)
print ("Answer is still ", Answer)
print ("Number3 is ", Number3)

```

Output from Python:

```

Number1 is 50
Number2 is 100
Answer is 150
Number1 is now 10
Number2 is now 20
Answer is now 30
Number1 is still 50
Number2 is still 100
Answer is still 150
Traceback (most recent call last):
  File "... Hodder/Chapter 8 programs/Activity_8_17_Python.py", line 24, in
<module>
    print ("Number3 is ", Number3)
    NameError: name 'Number3' is not defined

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim Number1, Number2, Answer As Integer
        Number1 = 50
        Number2 = 100
        Answer = Number1 + Number2
        Console.WriteLine("Number1 is " & Number1)
        Console.WriteLine("Number2 is " & Number2)
        Console.WriteLine("Answer is " & Answer)
        Test()
        Console.WriteLine("Number1 is still " & Number1)
        Console.WriteLine("Number2 is still " & Number2)
        Console.WriteLine("Answer is still " & Answer)
        Console.WriteLine("Number3 is " & Number3)
        Console.ReadKey()
    End Sub
    Sub Test()
        Dim Number1, Number2, Number3, Answer As Integer
        Number1 = 10
        Number2 = 20
        Number3 = 30
        Answer = Number1 + Number2
        Console.WriteLine("Number1 is now " & Number1)
        Console.WriteLine("Number2 is now" & Number2)
        Console.WriteLine("Answer is now " & Answer)
    End Sub
End Module

```

Note – the program will not run, try defining Number3 in the main program instead.



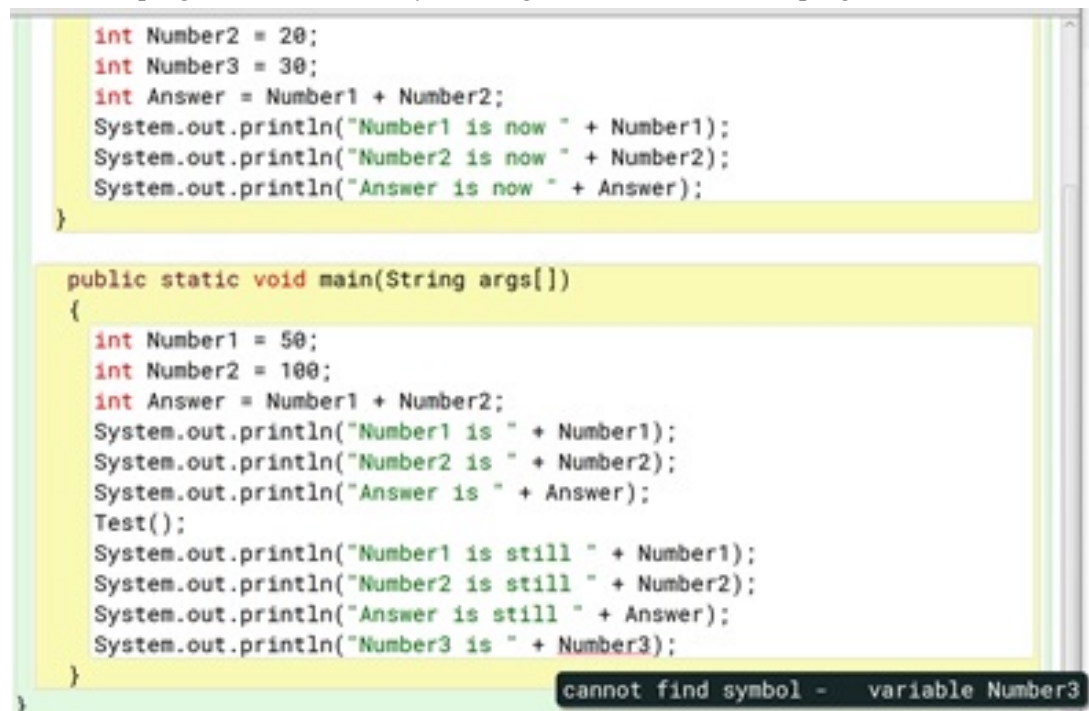
Java

```

import java.util.Scanner;
class Activity_8_17_Java
{
    static void Test()
    {
        int Number1 = 10;
        int Number2 = 20;
        int Number3 = 30;
        int Answer = Number1 + Number2;
        System.out.println("Number1 is now " + Number1);
        System.out.println("Number2 is now " + Number2);
        System.out.println("Answer is now " + Answer);
    }
    public static void main(String args[])
    {
        int Number1 = 50;
        int Number2 = 100;
        int Answer = Number1 + Number2;
        System.out.println("Number1 is " + Number1);
        System.out.println("Number2 is " + Number2);
        System.out.println("Answer is " + Answer);
        Test();
        System.out.println("Number1 is still " + Number1);
        System.out.println("Number2 is still " + Number2);
        System.out.println("Answer is still " + Answer);
        System.out.println("Number3 is " + Number3);
    }
}

```

Note – the program will not run, try defining Number3 in the main program instead.

**8.18**

Subject Procedure local variables:

Test, Mark, SubjectAverage, SubjectTotal

Student Procedure local variables:

Subject, StudentAverage, StudentTotal, SubjectHigh, SubjectLow,

8.19

Value1 is 1
 Value2 is 3
 Value3 is 6.97

8.20**Python**

```
import random
a = int(input ("Enter first integer "))
b = int(input ("Enter second integer "))
AnswerMod = a%b
AnswerDiv = a//b
print(a, "MOD", b, "is", AnswerMod)
print(a, "DIV", b, "is", AnswerDiv)
c = int(random.random() * 200 + 100)
print(c, "is a random number between 100 and 300")
```

Visual Basic

```
Module Module1
    Sub Main()
        Dim a, b, c As Integer
        Dim AnswerMod, AnswerDiv As Integer
        Console.WriteLine("Please enter the first integer ")
        a = Integer.Parse(Console.ReadLine())
        Console.WriteLine("Please enter the second number ")
        b = Integer.Parse(Console.ReadLine())
        AnswerMod = a Mod b
        AnswerDiv = a \ b
        Console.WriteLine(a & " MOD " & b & " is " & AnswerMod)
        Console.WriteLine(a & " DIV " & b & " is " & AnswerDiv)
        c = CInt(Rnd() * 200 + 100)
        Console.WriteLine(c & " is a random integer between 100 and 300")
        Console.ReadKey()
    End Sub
End Module
```

Java

```
import java.util.Scanner;
import java.lang.Math;
import java.util.Random;
class Activity_8_20_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        System.out.println("Please enter first integer ");
        int a = myObj.nextInt();
        System.out.println("Please enter second integer ");
        int b = myObj.nextInt();

        int AnswerMod = a % b;
        int AnswerDiv = a / b;

        System.out.println(a + " MOD " + b + " is " + AnswerMod);
        System.out.println(a + " DIV " + b + " is " + AnswerDiv);

        Random rand = new Random();
        int c = rand.nextInt(200) + 100;

        System.out.println(c + " is a random integer between 100 and 300");
    }
}
```

8.21

Use the list in 8.1.8 to check for any features not used.

8.22

Python

```

MyList=[]
for counter in range(10):
    MyList.append (int(input("Enter next value")))
print (MyList)

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim MyList(9), Counter As Integer
        For Counter = 0 To 9
            Console.Write("Enter next value ")
            MyList(Counter) = Integer.Parse(Console.ReadLine())
        Next
        For Counter = 0 To 9
            Console.Write(MyList(Counter) & " ")
        Next
        Console.ReadKey()
    End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_22_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        int MyList[] = new int[10];
        int Counter;
        for (Counter = 0; Counter <= 9; Counter++) {
            System.out.print("Enter next value ");
            MyList[Counter] = myObj.nextInt();
        }
        for (Counter = 0; Counter <= 9; Counter++) {
            System.out.print(MyList[Counter] + " ");
        }
    }
}

```

8.23

Python

```

MyTable =[]
for ColumnCounter in range(3):
    Column =[]
    for RowCounter in range(10):
        Column.append(int(input("Enter next value ")))
    MyTable.append(Column)
print (MyTable)

```

Visual Basic

```

Module Module1
    Sub Main()
        Dim MyTable(9, 2), ColumnCounter, RowCounter As Integer
        For ColumnCounter = 0 To 2
            For RowCounter = 0 To 9
                Console.Write("Enter next value for Column " & ColumnCounter)
                MyTable(RowCounter, ColumnCounter) =
Integer.Parse(Console.ReadLine())
            Next
        Next
        For RowCounter = 0 To 9
            For ColumnCounter = 0 To 2
                Console.Write(MyTable(RowCounter, ColumnCounter) & " ")
            Next
        Next
    End Sub
End Module

```

```

        Console.WriteLine()
    Next
    Console.ReadKey()
End Sub
End Module

```

Java

```

import java.util.Scanner;
class Activity_8_23_Java
{
    public static void main(String args[])
    {
        Scanner myObj = new Scanner(System.in);
        int MyTable[][] = new int[10][3];
        int ColumnCounter, RowCounter;
        for (ColumnCounter = 0; ColumnCounter <= 2; ColumnCounter++) {
            for (RowCounter = 0; RowCounter <= 9; RowCounter++) {
                System.out.print("Enter next value for column " +
ColumnCounter);
                MyTable[RowCounter][ColumnCounter] = myObj.nextInt();
            }
        }
        for (RowCounter = 0; RowCounter <= 9; RowCounter++) {
            for (ColumnCounter = 0; ColumnCounter <= 2; ColumnCounter++) {
                System.out.print( MyTable[RowCounter][ColumnCounter]+ " ");
            }
            System.out.println();
        }
    }
}

```

8.24

Python

```

# writing to and reading a line of text from a file
MyFile = open ("MyText.txt","w") # open file for writing
TextLine = input("Please enter a line of text ") # enter line of text from
keyboard
MyFile.write(TextLine) # write line of text to file
print("The file contains this line of text")
MyFile = open ("MyText.txt","r")# open file for reading
TextLine = MyFile.read()# read line of text from file
print(TextLine) # output line of text from file

```

Visual Basic

```

' writing to and reading from a text file
Imports System.IO
Module Module1
    Sub Main()
        Dim textLn As String
        Dim objMyFileWrite As StreamWriter
        Dim objMyFileRead As StreamReader
        objMyFileWrite = New StreamWriter("textFile.txt") ' setup file for
writing
        Console.Write("Please enter a line of text ")
        textLn = Console.ReadLine() ' enter line of text from keyboard
        objMyFileWrite.WriteLine(textLn) ' write line of text to file
        objMyFileWrite.Close() ' close file after writing line of text
        Console.WriteLine("The line of text is ")
        objMyFileRead = New StreamReader("textFile.txt") ' setup file for
reading
        textLn = objMyFileRead.ReadLine ' read line of text from file
        Console.WriteLine(textLn) ' output line of text from file
        objMyFileRead.Close() ' close file after reading line of text
        Console.ReadLine()
    End Sub
End Module

```

Java

```
//writing to and reading from a text file
import java.util.Scanner;
import java.io.BufferedReader; // library routines for file handling
import java.io.PrintWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

class TextFile {

    public static void main(String[] args) {
        Scanner myObj = new Scanner(System.in);
        String textLn;
        try {
            FileWriter myFileWriter = new FileWriter("textFile.txt", false);
            PrintWriter myPrintWriter = new PrintWriter(myFileWriter);
            // setup file for writing

            System.out.println("Please enter a line of text ");
            textLn = myObj.next(); //enter line of text from keyboard
            myPrintWriter.printf("%s" + "%n", textLn); // write line of text
to file
            myPrintWriter.close(); // close file after writing line of text
        } catch (IOException e) {
            e.printStackTrace();
        }

        try { // exception handling
            FileReader myFileReader = new FileReader("textFile.txt");
            BufferedReader myBufferedReader = new BufferedReader(myFileReader);
            // setup file for reading

            textLn = myBufferedReader.readLine(); // read line of text from file
            System.out.println(textLn); // output line of text from file

            myFileReader.close(); // close file after reading line of text

        } catch (IOException e) { // exception handling
            e.printStackTrace();
        }
    }
}
```

8.25

```
DECLARE TextLine : STRING // variables are declared as normal
DECLARE MyFile : STRING
MyFile ← "MyText.txt"
MyOtherFile ← "MyOtherText.txt"
// reading the line of text from the file
OUTPUT "The file contains this line of text:"
OPEN MyFile FOR READ // opens file for reading
    READFILE, TextLine // reads a line of text from the file
CLOSEFILE(MyFile) // closes the file
// writing the line of text to another file
OPEN MyOtherFile FOR WRITE // opens the file for writing
    WRITEFILE, TextLine // writes the line of text to the
file
CLOSEFILE(MyFile) // closes the file
```

Answers to exam-style questions

Answers to question 9 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

- 1 Both types of data store contain values that can be accessed by a program during execution using the name of the data store. Data stores for variables contain data that may be changed during the execution of the program. Whereas data stores for constants contain data that may not be changed during the execution of the program.

- 2
 - Item name: `ItemName` – string
 - Manufacturer: `Manufacturer` – string
 - Description: `Description` – string
 - Number in stock: `NumberInStock` – integer
 - Reorder level: `ReorderLevel` – integer
 - Whether the item is on order or not: `OnOrder` – Boolean

- 3
 - Sequence – the order that steps in a program are executed. For example, these two statements need to be in the correct sequence:

```
INPUT Value
IF Value > 10...
```

- Selection – allowing the selection of different paths through the steps of a program.

```
IF Value > 10 THEN ... ELSE ...
```

- Iteration – a section of programming code can be repeated under certain conditions.

```
FOR Value = 1 TO 10
    OUTPUT "*****"
NEXT
```

- Totalling – keeping a total that values are added to.

```
Total = Total + Value
```

- Counting – keeping a count of the number of times an action is performed.

```
Count = Count + 1
```

- 4 OUTPUT "Please enter password "

```
INPUT Password
IF (LENGTH(Password) = 10) AND (Password = UCASE(Password))
    THEN
        OUTPUT "Password meets the rules"
    ENDF
```

- 5 A global variable can be used by any part of a program – its scope covers the whole program. Any variable declared in the main program is a global variable.

A local variable can only be used by the part of the program it has been declared in – its scope is restricted to that part of the program. Any variable declared within a procedure or function is a local variable.

An error will occur if an attempt is made to access a local variable outside of its scope.

- 6 Many programmers find library routines useful because they are fully tested and ready to incorporate into a program. For example, any string handling routines such as conversion to lower or upper case, or routines for rounding a value to a fixed number of decimal places.

- 7 a** DECLARE Name : ARRAY[1:10, 1:2] OF STRING
 // Table for First name in column 1 and Family Name in column 2
 DECLARE Counter :INTEGER
 FOR Index ← 1 TO 10
 OUTPUT ("First name ", Name[Index, 1], " Family name ", Name[Index, 2])
 NEXT Index
- b** DECLARE Name : ARRAY[1:10, 1:2] OF STRING
 // Table for First name in column 1 and Family Name in column 2
 DECLARE Index, First, Last : INTEGER
 DECLARE Swap : Boolean
 DECLARE Temp1, Temp2 : STRING
 First ← 1
 Last ← 10
 REPEAT
 Swap ← FALSE
 FOR Index ← First TO Last - 1
 IF Name[Index, 2] > Name[Index + 1, 2] // Test Family name in column 2
 THEN
 Temp2 ←Name[Index, 2]
 Temp1 ←Name[Index, 1]
 Name[Index, 2] ← Name[Index + 1, 2]
 Name[Index, 1] ← Name[Index + 1, 1]
 Name[Index + 1, 2] ← Temp2
 Name[Index + 1, 1] ← Temp1
 Swap ← TRUE // swap both names
 ENDIF
 NEXT Index
 Last ← Last -1
 UNTIL (NOT Swap) OR Last = 1
 FOR Index ← 1 TO 10
 OUTPUT ("First name ", Name[Index, 1], " Family name ", Name[Index, 2])
 NEXT Index

- ```

8 DECLARE Message : STRING // variables are declared as normal
 DECLARE MessageFile : STRING
 MessageFile ← "Message.txt"
 // reading the line of text from the file
 OUTPUT "The Message is:"
 OPEN MessageFile FOR READ // opens file for reading
 READFILE, Message // reads a line of text from the file
 OUTPUT Message
 CLOSEFILE(MessageFile) // closes the file

```
- 9 a Loop with 300 repetitions, starting at 1, values input are stored in consecutive locations in the array Name at the position in the array indexed by the loop counter I
- b REPEAT ... UNTIL
- c OUTPUT "Please enter a number between 0 and 100 inclusive"
- ```

   INPUT Number
   WHILE Number < 0 OR Number > 100
       OUTPUT "Number out of range. Please enter a number
       between 0 and 100 inclusive"
       INPUT Number
   ENDWHILE

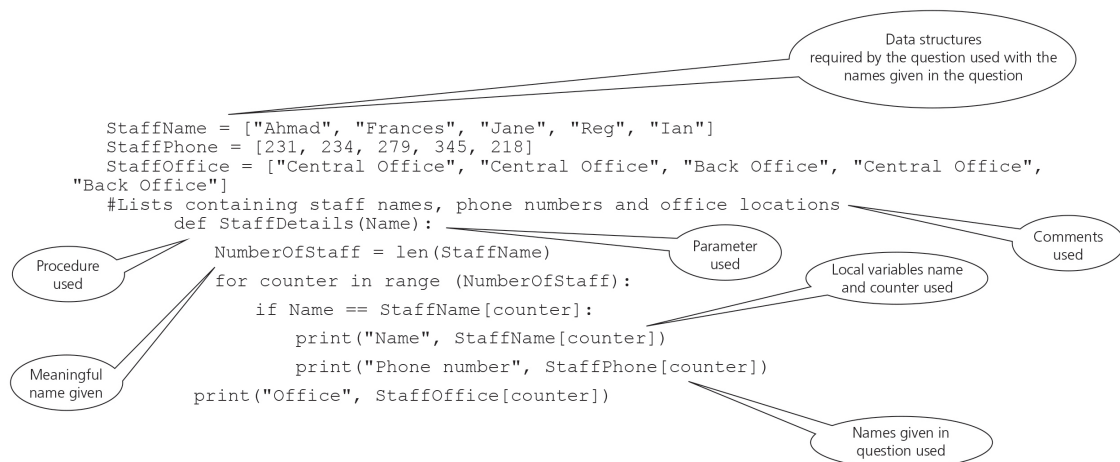
```

10–12

Important points to note when writing your program:

- There is not one 'correct' solution to this problem. Any solution is acceptable but your program must:
 - perform all the tasks set out in the question.
 - use a range of appropriate programming techniques that cover all the tasks set out in the question. These tasks must be executed by your program in a logical order.
 - use appropriate data structures, that have been given meaningful names, to store all the data mentioned in the question. If any data structures are named in the question, then those names must be used in your program.
 - be fully commented to ensure that it can be understood.

For example, in 10, see the following sample of code:



Answers to Student's Book

Exam-style questions and sample answers have been written by the authors. References to assessment and/or assessment preparation are the publisher's interpretation of the syllabus requirements and may not fully reflect the approach of Cambridge Assessment International Education. Cambridge International recommends that teachers consider using a range of teaching and learning resources in preparing learners for assessment, based on their own professional judgement of their students' needs.

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9 Databases

Answers to activities

Answers to question 3 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

9.1

Sample fields for database APPOINTMENT:

- Number of patients: e.g. PatientNumber
- Name of patient: e.g. PatientName
- Time of appointment: e.g. AppointmentTime
- Length of appointment: e.g. Duration

9.2

PATIENT					
Field Name	Data Type				
FirstName	Short Text				
FamilyName	Short Text				
DateOfAdmission	Date/Time				
Consultant	Short Text				
WardNumber	Number				
BedNumber	Number				

FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Winnie	Sing	12/10/2022	Mr Smith	6	8
Steve	Chow	23/10/2022	Miss Abebe	6	3
Chin	Wee	30/10/2022	Mr Jones	7	1
Min	Hoo	01/11/2022	Mr Smith	6	1
Peter	Patel	12/11/2022	Mr Jones	7	8
Sue	Sands	19/11/2022	Miss Abebe	6	2
Farouk	Khan	22/11/2022	Mr Jones	7	4
Ahmad	Teo	22/11/2022	Mr Jones	7	2

Field Size	Integer
Format	
Decimal Places	0
Input Mask	
Caption	
Default Value	
Validation Rule	>0<=10
Validation Text	Ward number must be between 1 and 10 inclusive

Field Size	Integer
Format	
Decimal Places	0
Input Mask	
Caption	
Default Value	
Validation Rule	>0<=8
Validation Text	Bed number must be between 1 and 8 inclusive

9.3

PATIENT							
HospitalNumber	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber	
HN000001	Winnie	Sing	12/10/2022	Mr Smith	6	8	
HN000002	Steve	Chow	23/10/2022	Miss Abebe	6	3	
HN000003	Chin	Wee	30/10/2022	Mr Jones	7	1	
HN000004	Min	Hoo	01/11/2022	Mr Smith	6	1	
HN000005	Peter	Patel	12/11/2022	Mr Jones	7	8	
HN000006	Sue	Sands	19/11/2022	Miss Abebe	6	2	
HN000007	Farouk	Khan	22/11/2022	Mr Jones	7	4	
HN000008	Ahmad	Teo	22/11/2022	Mr Jones	7	2	

9.4

1 Four from:

- Text – a number of characters, for example a person's name
- Char – a single character, for example M or F for male or female
- Boolean – True or False, for example whether an item has been sold or not
- Integer – a whole number, for example the number of items in stock
- Real – a decimal number, for example the price of an item
- Date/time – for example the date and time of an appointment

2 A single table database contains one table. Each table consists of many records. Every record has the same number of fields. Every field is given a data type. Examples of data types are char, Boolean and text. Some fields will have validation rules.

9.5

- 1
 - a `SELECT FirstName, FamilyName
FROM PATIENT
WHERE Consultant = 'Mr Jones';`
 - b `SELECT FirstName, FamilyName
FROM PATIENT
WHERE WardNumber <> 6;`
 - c `SELECT FirstName, FamilyName
FROM PATIENT
WHERE DateOfAdmission = '12/11/2022';`
 - d `SELECT FirstName, FamilyName, DateOfAdmission
FROM PATIENT
WHERE DateOfAdmission >= '12/10/2022' AND <= '30/10/2022';`

FirstName	FamilyName	BedNumber
Chin	Wee	1
Peter	Patel	8
Ahmad	Teo	2
Farouk	Khan	4

9.6

- 1 a

```
SELECT Count (WardNumber)
FROM PATIENT
WHERE WardNumber = 7;
```
- b

```
SELECT Count (WardNumber)
FROM PATIENT
WHERE WardNumber <> 7;
```

2

CubNumber	CubName	DateOfBirth	Address	Gender	School	TelephoneNt	DateJoined
C010	Suzy	12/01/2015	The House	F	WestCliffe	0779988	12/01/2022
C011	Joe	12/01/2015	The House	M	EastCliffe	0779988	12/01/2022
C001	Ali	03/11/2012	The Green	M	EastCliffe	0773366	01/01/2020
C002	Max	07/07/2012	West Green	M	EastCliffe	0774421	01/01/2020
C004	Nina	01/03/2013	East Green	F	WestCliffe	0773145	08/01/2021
C005	Maurice	27/11/2013	North Port	M	EastCliffe	0778356	08/01/2021
C006	Wendy	03/04/2014	The Field	F	WestCliffe	0779800	04/06/2021
C007	Ian	11/12/2014	Forest View	M	EastCliffe	0776535	12/01/2022
C008	Cy	09/09/2014	Garden View	M	NorthCliffe	0789355	12/01/2022
C012	Jay	30/10/2014	Sea View	M	NorthCliffe	0782121	12/01/2022

9.7

1 a

CubNumber	CubName	DateOfBirth	Address	Gender	School	TelephoneNt	DateJoined
C010	Suzy	12/01/2015	The House	F	WestCliffe	0779988	12/01/2022
C011	Joe	12/01/2015	The House	M	EastCliffe	0779988	12/01/2022
C001	Ali	03/11/2012	The Green	M	EastCliffe	0773366	01/01/2020
C002	Max	07/07/2012	West Green	M	EastCliffe	0774421	01/01/2020
C004	Nina	01/03/2013	East Green	F	WestCliffe	0773145	08/01/2021
C005	Maurice	27/11/2013	North Port	M	EastCliffe	0778356	08/01/2021
C006	Wendy	03/04/2014	The Field	F	WestCliffe	0779800	04/06/2021
C007	Ian	11/12/2014	Forest View	M	EastCliffe	0776535	12/01/2022
C008	Cy	09/09/2014	Garden View	M	NorthCliffe	0789355	12/01/2022
C012	Jay	30/10/2014	Sea View	M	NorthCliffe	0782121	12/01/2022

b

CubNumber	CubName	DateOfBirth	Address	Gender	School	TelephoneNt	DateJoined	Six
C010	Suzy	12/01/2015	The House	F	WestCliffe	0779988	12/01/2022	red
C011	Joe	12/01/2015	The House	M	EastCliffe	0779988	12/01/2022	blue
C001	Ali	03/11/2012	The Green	M	EastCliffe	0773366	01/01/2020	red
C002	Max	07/07/2012	West Green	M	EastCliffe	0774421	01/01/2020	red
C004	Nina	01/03/2013	East Green	F	WestCliffe	0773145	08/01/2021	red
C005	Maurice	27/11/2013	North Port	M	EastCliffe	0778356	08/01/2021	blue
C006	Wendy	03/04/2014	The Field	F	WestCliffe	0779800	04/06/2021	red
C007	Ian	11/12/2014	Forest View	M	EastCliffe	0776535	12/01/2022	green
C008	Cy	09/09/2014	Garden View	M	NorthCliffe	0789355	12/01/2022	green
C012	Jay	30/10/2014	Sea View	M	NorthCliffe	0782121	12/01/2022	red

- i

```
SELECT CubName
FROM CUB
WHERE Six = 'red';
```
- ii

```
SELECT CubName
FROM CUB
WHERE Six = 'red' OR Six = 'blue';
```
- iii

```
SELECT COUNT(Six)
FROM CUB
WHERE Six = 'red';
```
- c

```
SELECT Sum (Badges)
FROM CUB;
```

9.8

- 1 a
- Family name – text
 - Other names – text
 - Student ID – text
 - Date of Birth – date
 - Date of Entry to School – date
 - Current Class – date
 - Current school year/grade – integer
 - Email address – text
- b
- Family name – verification
 - Other names – verification
 - Student ID – verification and validation
 - Date of Birth – verification and validation
 - Date of Entry to School – verification and validation
 - Current Class – verification and validation
 - Current school year/grade – verification and validation
 - Email address – verification and validation
- c
- Student ID – presence check, length check and format check
 - Date of Birth – range check and format check
 - Date of Entry to School – range check and format check
 - Current Class – format check
 - Current school year/grade – format check
 - Email address – format check
- d Student ID
- e Example: five characters, S followed by four digits

Field Name	Data Type
StudentID	Short Text
FamilyName	Short Text
OtherNames	Short Text
DateOfBirth	Date/Time
DateEntrySchool	Date/Time
CurrentClass	Short Text
CurrentSchoolYear	Number
EmailAddress	Short Text

- g i** `SELECT OtherNames, FamilyName, EmailAddress`
`FROM STUDENT`
`ORDER By FamilyName;`
- ii** `SELECT OtherNames, FamilyName`
`FROM STUDENT`
`ORDER BY CurrentClass, FamilyName;`
- iii** e.g. for year 3 it would be:
`SELECT OtherNames, FamilyName, DateOfBirth`
`FROM STUDENT`
`WHERE CurrentSchoolYear = 3`
`ORDER BY CurrentClass;`
 Similar SQL queries would be used for each different school year group.

Answers to exam-style questions

- 1 a i** 10 records
- ii** 11 fields
- iii** No field is suitable to use as a primary key as none of the field values are unique.
- b** CR
 LR
- c** `SELECT *`
`FROM CAR`
`WHERE leather AND (silver OR grey);`
- 2 a i**
- model number – text – used when the field contains numbers and characters
 - description – text – suitable for characters
 - colour – text – suitable for characters
 - size of wheels – integer – wheels are measured in mm in the example so no decimal places needed, hence integer is best option
 - price of model – real – price will often finish with \$XX.99, hence real numbers with decimal required
 - still manufactured – Boolean – can only be yes or no, hence Boolean is best option
 - number in stock – integer – can only be whole numbers of stock, hence integer is used
- ii**
- model number – presence check, format check and length check as the field must contain data; the first two characters should be letters and the last five numbers, and the number of characters should be seven
 - price – range check as the value should not be below a minimum number
 - number in stock – range check as this should not fall below zero
- iii** Model number is suitable to use as a primary key as this uniquely identifies the type of bicycle.

- b** For instance, to display the stock of each model number where there are less than 20 bikes left in stock, ordered by number in stock, we would use:

```
SELECT ModelNumber, NumberInStock  
FROM CYCLE  
WHERE NumberInStock < 20  
ORDER BY NumberInStock;
```

To calculate the total number of bikes in stock we would use:

```
SELECT SUM NumberInStock  
FROM CYCLE;
```

SELECT is used to specify the fields that will be displayed.

FROM defines which table the fields are taken from.

WHERE is an optional command that can add certain conditions on the records to be displayed.

SUM is an optional command used with SELECT to specify fields whose values will be added together.

ORDER BY is an optional command that defines the order in which records will be displayed.

- 3 a**
- Fields: 5
 - Records: 8
- b**
- Validation check 1: format check
 - Validation check 2: presence check

Answers to Student's Book

Exam-style questions and sample answers have been written by the authors. References to assessment and/or assessment preparation are the publisher's interpretation of the syllabus requirements and may not fully reflect the approach of Cambridge Assessment International Education. Cambridge International recommends that teachers consider using a range of teaching and learning resources in preparing learners for assessment, based on their own professional judgement of their students' needs.

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10 Boolean logic

Answers to activities

10.1

A	B	A AND NOT B	NOT A AND B	(A AND NOT B) OR (NOT A AND B)	A OR B	NOT (A AND B)	(A OR B) AND (NOT (A AND B))
0	0	0	0	0	0	1	0
0	1	0	1	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	0	1	0	0

10.2

1 a

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

b

A	B	X
0	0	0
0	1	0
1	0	0
1	1	0

c

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

d

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

e

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

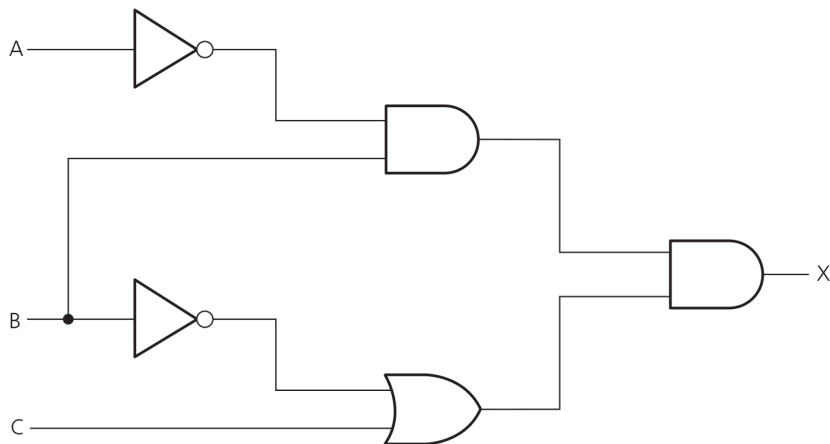
- 2 a** (A AND B) XOR (B NOR C)
b (A NAND C NAND A) NOR (B NOR C)

10.3

1 i

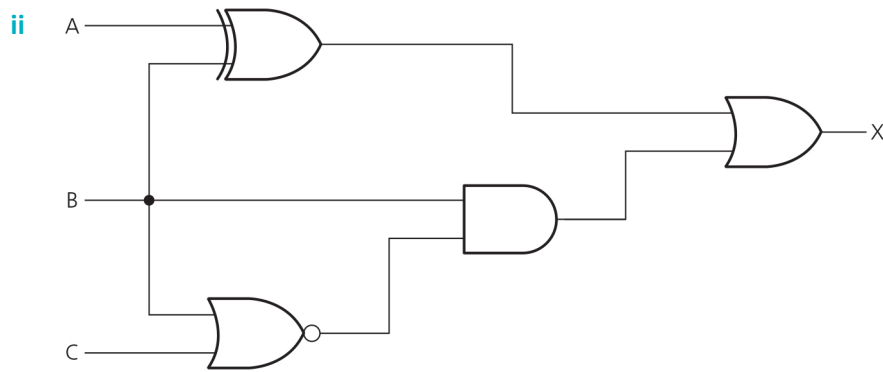
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

ii



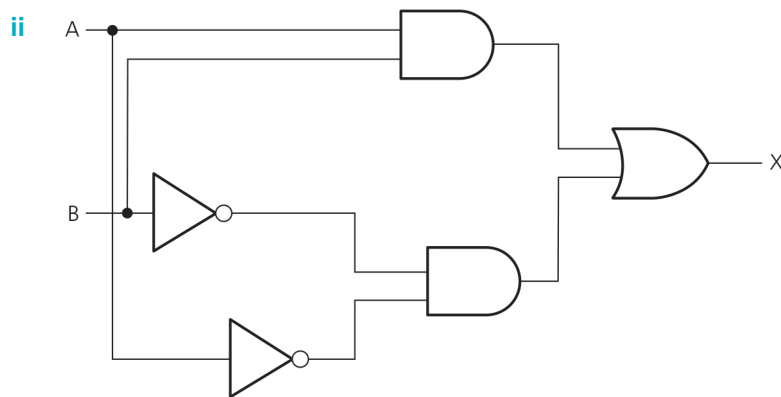
2 i

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0



3 i (NOT A AND NOT B) OR (A AND B)

[or $(\bar{a}.\bar{b}) + (a.b)$]



4 a (NOT A AND NOT B AND NOT C) OR (A AND NOT B AND NOT C) OR (A AND NOT B AND C)

or

$(\bar{a}.\bar{b}.\bar{c}) + (a.\bar{b}.\bar{c}) + (a.\bar{b}.c)$

b (NOT A AND NOT B AND C) OR (NOT A AND B AND NOT C) OR (A AND NOT B AND NOT C) OR (A AND NOT B AND C)

or

$(\bar{a}.\bar{b}.c) + (\bar{a}.b.\bar{c}) + (a.\bar{b}.\bar{c}) + (a.\bar{b}.c)$

- 5 a (NOT A AND NOT B AND NOT C) OR (NOT A AND NOT B AND C) OR (A AND NOT B AND NOT C) OR (A AND NOT B AND C)

or

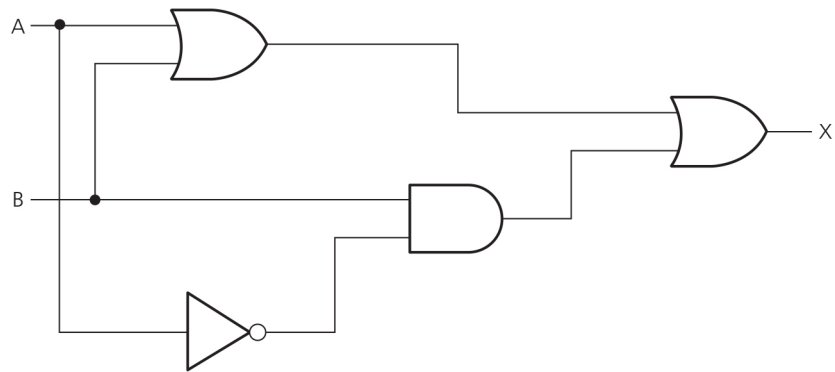
$$(\bar{a}.\bar{b}.\bar{c}) + (\bar{a}.\bar{b}.c) + (a.\bar{b}.\bar{c}) + (a.\bar{b}.c)$$

b

A	B	C	X	NOT A AND NOT B	A AND NOT B	(NOT A AND NOT B) OR (A AND NOT B)
0	0	0	1	1	0	1
0	0	1	1	1	0	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	1	0	1	1
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	1	0	0	0	0

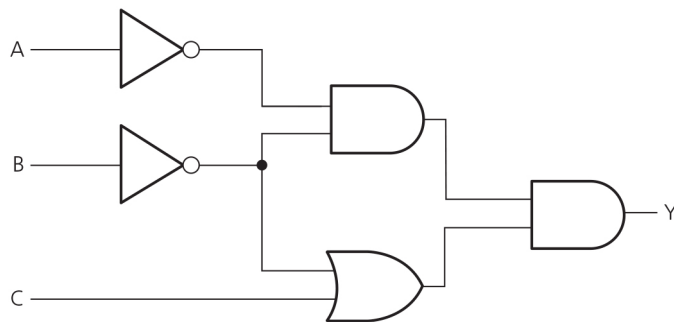
10.4

1 a



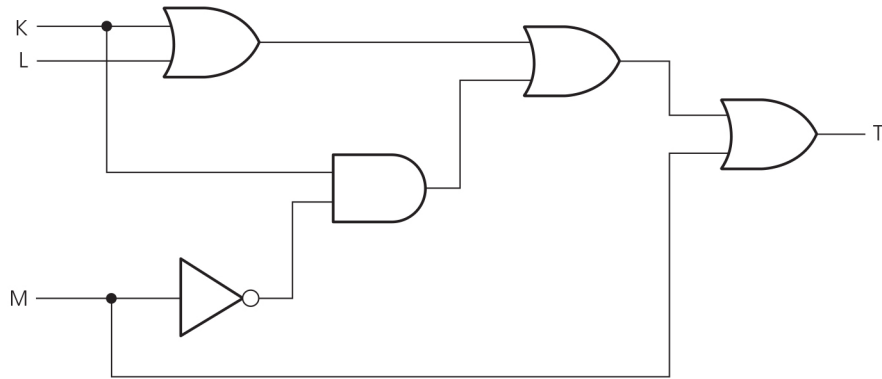
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

b



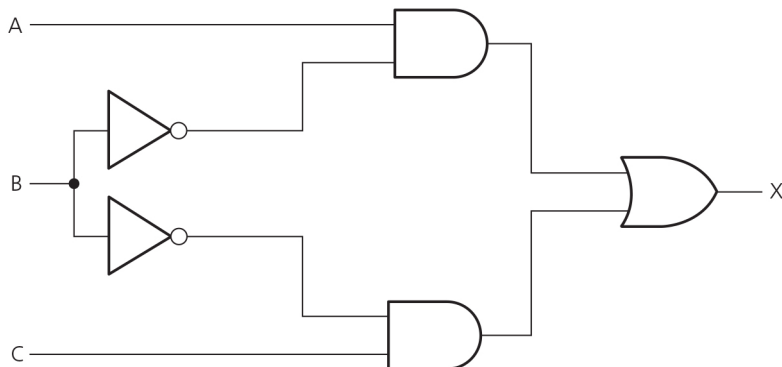
A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

c

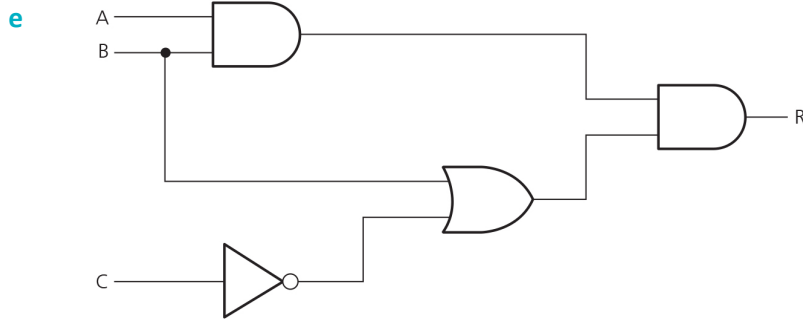


K	L	M	T
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

d



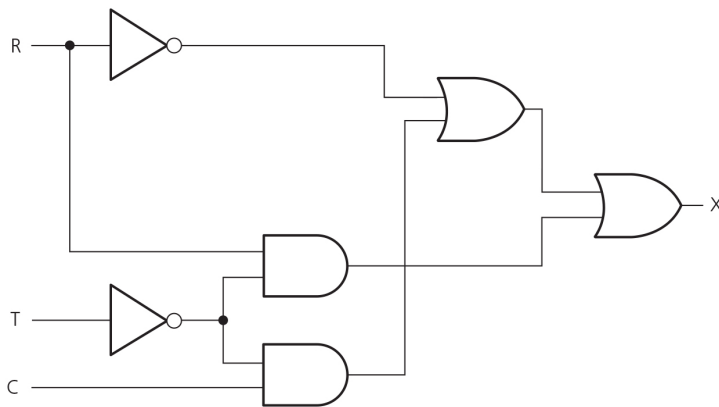
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0



A	B	C	R
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

- 2 Logic statement: (NOT R) OR (C AND NOT T) OR (R AND NOT T)

Logic circuit:

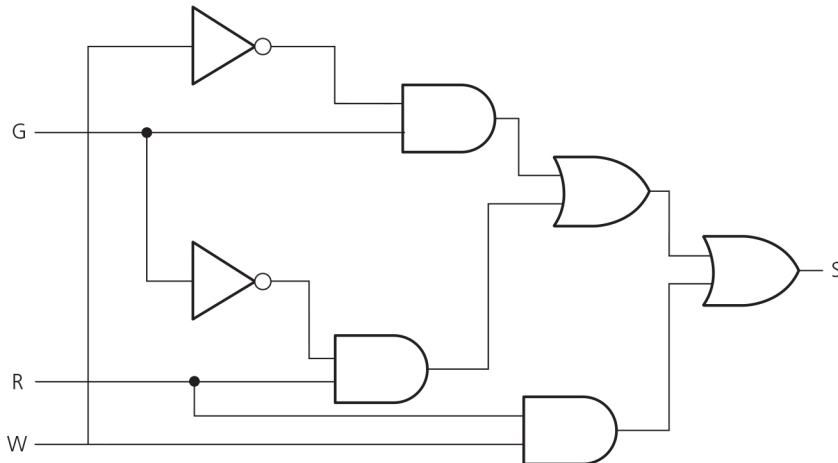


Truth table:

R	T	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

- 3 Logic statement: $(G \text{ AND NOT } W) \text{ OR } (\text{NOT } G \text{ AND } R) \text{ OR } (W \text{ AND } R)$

Logic circuit:



Truth table:

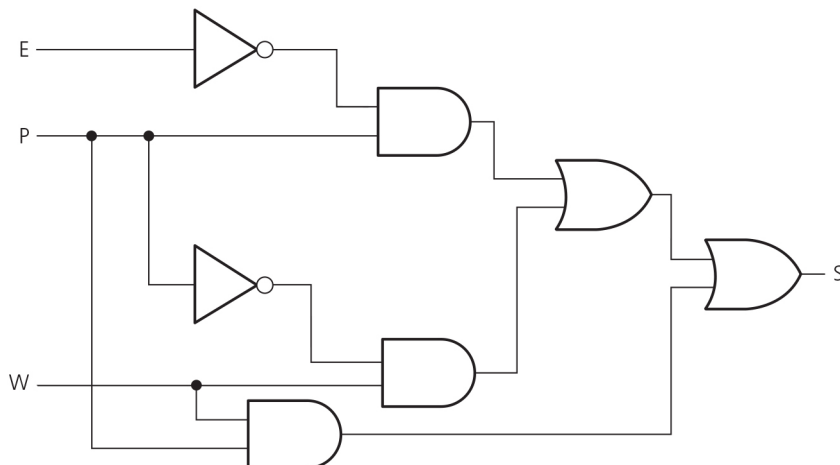
G	R	W	S
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

4 i Truth table:

E	P	W	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

ii Logic statement: $(\text{NOT } E \text{ AND } P) \text{ OR } (\text{NOT } P \text{ AND } W) \text{ OR } (P \text{ AND } W)$

iii Logic circuit:



5

A	B	A AND B	NOT A AND NOT B	(A AND B) OR (NOT A AND NOT B)	X
0	0	0	1	1	1
0	1	0	0	0	0
1	0	0	0	0	0
1	1	1	0	1	1

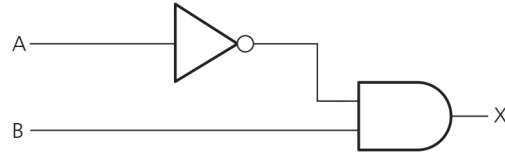
Answers to exam-style questions

Answers to question 5 in this section are example answers for past paper questions. The details of the relevant past paper can be found with the corresponding exam-style question in the Student's Book.

1 a i NOT A AND B

$(\bar{a}.b)$

ii



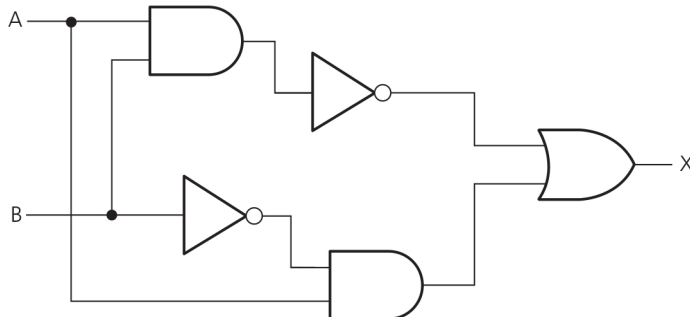
b i $(A \text{ AND } (A \text{ NAND } B)) \text{ NOR } (B \text{ NOR } C)$

ii

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

2 a $(A \text{ AND } B) \text{ NOR } (\text{NOT } A \text{ AND } \text{NOT } B)$

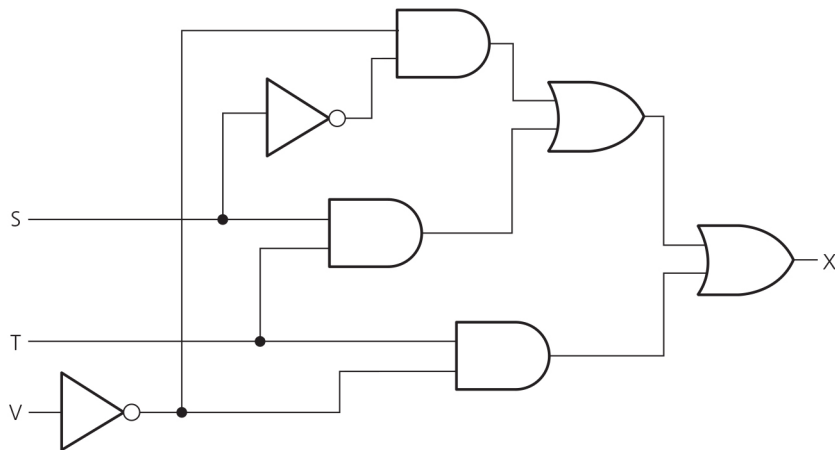
b



c

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

- 3 a** (S AND T) OR (NOT S AND NOT V) OR (T AND NOT V)



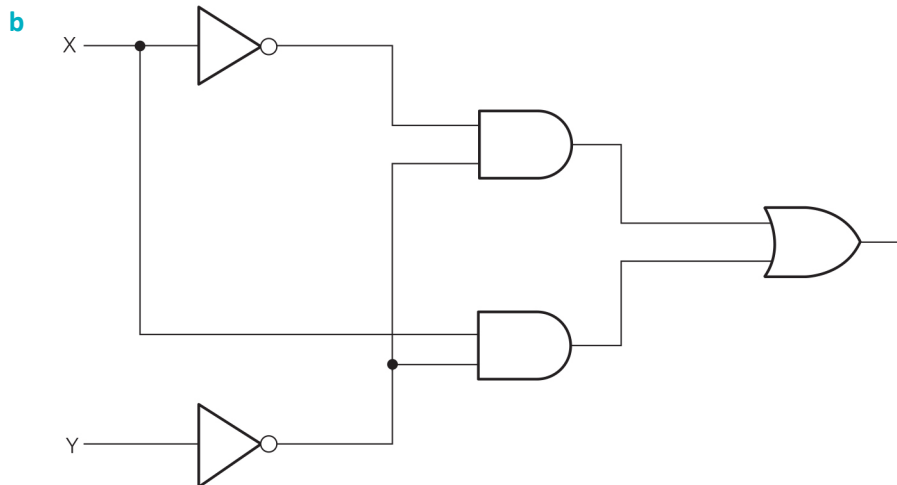
b

S	T	V	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

- 4 a** (NOT X AND NOT Y) OR (X AND NOT Y)

or in alternative notation:

$$\bar{x}.\bar{y} + x.\bar{y}$$

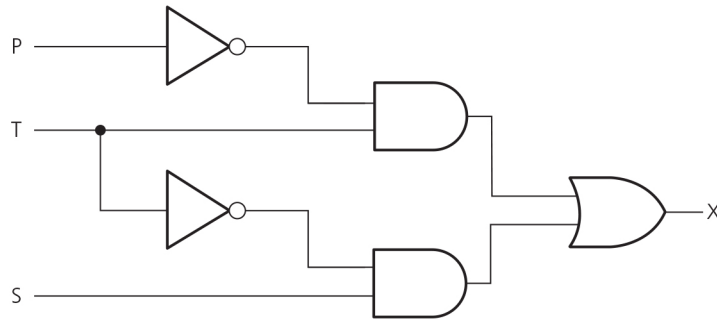


5 a

A	B	OUTPUT
0	0	0
0	1	1
1	0	1
1	1	1

b XOR gate

c (T AND NOT P) OR (S AND NOT T)

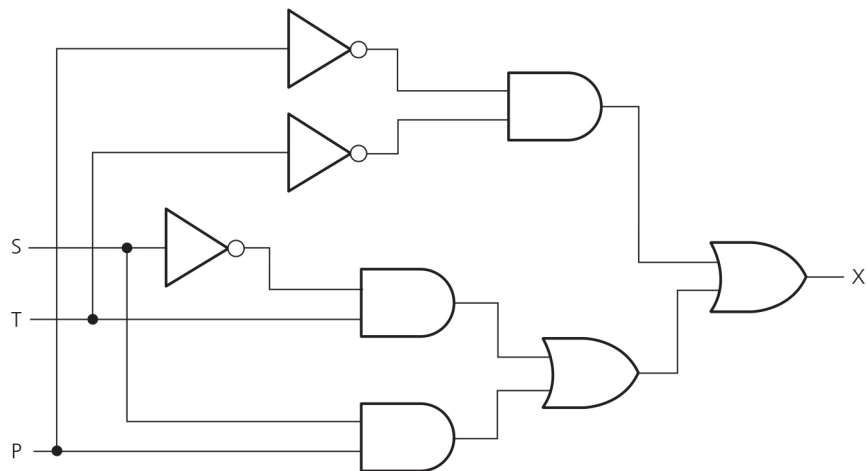


d Two from:

- continuous measurements taken
- won't forget to take readings
- more accurate readings
- safer since no human in contact with potentially dangerous process
- cost saving since no need to pay any wages

6 a (NOT S AND T) OR (S AND P) OR (NOT T AND NOT P)

b



c

S	T	P	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Chapter 10 Extension Answers

Task 1

A	B	F	X
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

This truth table is the same as an AND gate.

A	B	F	G	X
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

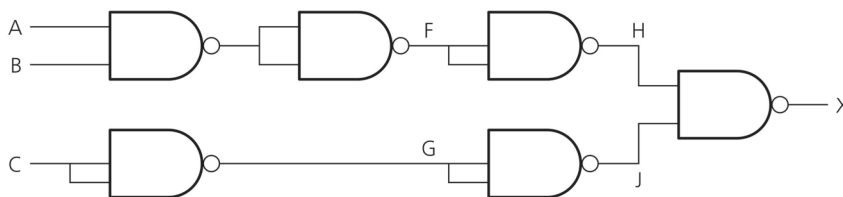
This truth table is the same as an OR gate.

A	F	X
0	0	1
1	1	0

This truth table is the same as a NOT gate.

Task 2

- 1 We simply replace each AND, OR and NOT gate with the NAND gate equivalents given in Figure 10.34:



To confirm we can write down both truth tables. First, for the circuit shown in the question:

A	B	C	J	K	X
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	1

Then, for the equivalent NAND circuit we have drawn:

A	B	C	F	G	H	J	X
0	0	0	0	1	1	0	1
0	0	1	0	0	1	1	0
0	1	0	0	1	1	0	1
0	1	1	0	0	1	1	0
1	0	0	0	1	1	0	1
1	0	1	0	0	1	1	0
1	1	0	1	1	0	0	1
1	1	1	1	0	0	1	1

The yellow highlights show the inputs and outputs for both circuits, which are the same, and therefore the circuits are equivalent.

- 2 One way to approach this is to first consider the XOR truth table, which is what we want to achieve:

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

We can see how close this is to an OR gate truth table:

A	B	F	G	X
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

The only row we need to amend from the OR gate truth table is the final one. We require F and G to both be 1 so that $X = F \text{ NAND } G$ is 0 on that final row, i.e.:

A	B	F	G	X
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	1	1	0

It is only when A and B are both 1 that we want to amend the OR gate circuit. A NAND could be used to do this, because a NAND gate has the property that only when both inputs are 1 is the output different:

A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0

How can we combine A NAND B with A in order that F has the values we require?

By combining gates as follows:

A	B	A NAND B	F
0	0	1	1
0	1	1	1
1	0	1	0
1	1	0	0

It can be seen that $F = A \text{ NAND } (A \text{ NAND } B)$.

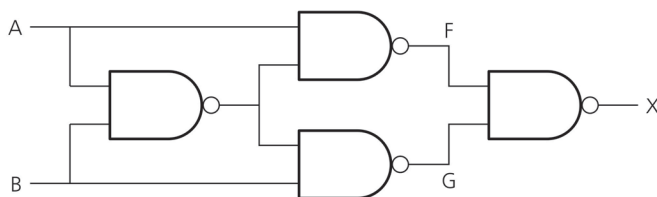
We can use the same approach for G, but this time considering the input B:

A	B	A NAND B	$F = A \text{ NAND } (A \text{ NAND } B)$	$G = B \text{ NAND } (A \text{ NAND } B)$
0	0	1	1	1
0	1	1	1	0
1	0	1	0	1
1	1	0	1	1

We now have the values of F and G that will ensure the output X (which is still $X = F \text{ NAND } G$ as per the OR circuit) behaves like an XOR gate:

A	B	$F = A \text{ NAND } (A \text{ NAND } B)$	$G = B \text{ NAND } (A \text{ NAND } B)$	X
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	1	1	0

Let's draw the circuit:



Task 3

A	B	F	G	H	X
0	0	1	1	0	1
0	1	1	0	1	0
1	0	0	1	1	0
1	1	0	0	1	0

By considering only the inputs A, B and the output X we can see this is the inverse of an OR gate, which is called an **NOR gate**.